



**GREEN
CLIMATE
FUND**

Meeting of the Board

4 – 7 March 2024

Kigali, Rwanda

Provisional agenda item 10

GCF/B.38/02/Add.07

12 February 2024

Consideration of funding proposals – Addendum VII

Funding proposal package for FP227

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme (DEFIS +)";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title:	Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme (DEFIS+)
Country(ies):	Madagascar
Accredited Entity:	International Fund for Agricultural Development (IFAD)
Date of first submission:	2020/04/24
Date of current submission	2023/11/21
Version number	V.008



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY				
A.1. Project or programme	Project	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+. <u>Not applicable</u>			
A.4. Result area(s)	Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated %age of GCF and Co-financers' contribution devoted to it. The total of the %ages when summed should be 100% for GCF and Co-financers' contribution respectively.			
		GCF contribution	Co-financers' contribution¹	
	Mitigation total	0%	0%	
	☐ Energy generation and access			
	☐ Low-emission transport			
	☐ Buildings, cities, industries and appliances			
	☐ Forestry and land use			
	Adaptation total	100%	100%	
	☒ Most vulnerable people and communities	30 %	30 %	
	☒ Health and well-being, and food and water security	25 %	20 %	
☒ Infrastructure and built environment	20 %	20 %		
☒ Ecosystems and ecosystem services	25 %	30 %		
A.5. Expected mitigation outcome	Mitigation co-benefit of 1 672 669 tCO ₂ eq	A.6. Expected adaptation outcome	3,990,920 (1,995,460 females and 1,995,460 males) ²	
			number of direct beneficiaries 447,201	number of indirect beneficiaries 3,543,719
			% of direct beneficiaries vis-à-vis total population 1.6%	% of indirect beneficiaries vis-à-vis total population 12.7%
A.7. Total financing (GCF + co-finance³)	150 847 444 USD	A.9. Project size	Medium (Up to USD 250 million)	
A.8. Total GCF funding requested	53 861 003 USD <i>For multi-country proposals, please fill out annex 17.</i>			

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² An annex is included elaborating the estimates and calculations of the beneficiaries for different project interventions.

³ Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8. GREEN CLIMATE FUND FUNDING PROPOSAL V.3.0 PAGE 2		
	☒ Grant 41 045 298 ☒ Loan 12 815 705 ☐ Guarantee		☐ Equity ☐ Results-based payment
A.11. Implementation period	6 years, 0 months.	A.12. Total lifespan	20 years (as in Economic and Financial Analysis)
A.13. Expected date of AE internal approval	March 2024	A.14. ESS category	Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category. B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1. Yes ☒ No ☐		
A.20. Executing Entity information	The Republic of Madagascar (RoM) through its Ministry of Economy and Finance (MEF) and Ministry of Agriculture and Livestock (MINAE).		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			
Madagascar has been ranked the fifth most vulnerable country to climate change in the world, and the ND-Gain index, which ranks the climate adaptation performance of 181 countries, has classified Madagascar's exposure and sensitivity to climate change as high and ability to adapt to the negative impacts of climate change as one of the least (rank 169 out of 181 countries). Madagascar belongs to the group of least developed countries (LDCs), more than 80% of the population lives below the poverty level of US\$ 2 per day and 80% of the population lives in rural zones, most of them in very remote areas with limited access to basic infrastructure. The poverty rate (at \$1.9/day) is estimated to rise in 2023; it was 74.3% in 2019, corresponding to an increase of 1.38 million people in just one year (2020). Madagascar has experienced unprecedented, excessive and consecutive droughts from 2017 to 2022, affecting crop production and food and nutrition security across the country. The southwest and the extreme south of the country are characterised by semi-desert environments with erratic and low rainfall of less than 800 mm per year. Thus, a change in average annual precipitation, ranging from -8 to +1%, is projected by 2050 and it is predicted that there will be an increase in the amplitude of inter-annual rainfall variability with greater likelihood of extreme events. This is particularly true in the project area, which is formed by six regions (Figure 4A, Annex 16) named Amoron'i Mania, Haute Matsiatra, Vatovavy, Fitovinany, Atsimo-Atsinanana, and Ihorombe in the southcentral and southeast of the country. The great variability of rainfall in the project area leads to either droughts or situations where the water balance is very much in surplus due to heavy rains and resulting floods. Projected temperature increases in the southern regions of Madagascar			

are estimated to be between +1.2°C and +2.1°C by 2050. In the next decade, under the RCP4.5 scenario, the number of days subject to extreme heat events is projected to increase from 9.2 days to 18.5 days and by 2050, the average number of days subject to extreme heat events will increase from 32.4 days to 54.7 days, posing extreme high risks to human lives.

The high occurrence of climate related disasters, limited climate investments, lack of access to formal safety nets and markets, characterise the vulnerability of rural Malagasy smallholders in the project area. The main climate-related changes include: (i) the decrease in the amount of annual precipitation, with more marked and/or more frequent periods of drought; (ii) a delay in the arrival of the rains, which requires adaptation (lag) in the agricultural calendars; (iii) the increased intensity of extreme climatic phenomena (cyclones, droughts, floods, storms and strong winds); and (iv) a very high temporal and spatial variability in rainfall, which makes farming and water supply less predictable. Additionally, smallholders are frequently exposed to pest and disease outbreaks, which cause significant crop and income losses. Since 2010, some of the regions most affected by drought events have been in the project area, including the Atsimo-Atsinanana, Vatovavy and Fitovinany regions. The GHG emissions in Madagascar come predominantly from the land-based sectors: agriculture, forestry and land use change (AFOLU) and represent 80.5% of Madagascar's national emissions. The emissions are mainly from livestock (enteric fermentation and manure) and agriculture (methane from rice cultivation, poor soil management, deforestation for agricultural purposes and fertiliser application).

The proposed project, Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme (**DEFIS+**) has the ambition of reducing the negative impact of climate change by applying sustainable and nature-based adaptation measures with mitigation co-benefits. Thus, DEFIS+ is built on positive lessons learned from past IFAD projects, and from an ongoing pro-poor agricultural investment programme, DEFIS, funded by IFAD and the OPEC Fund for International Development (OFID) and other national financiers.

The UN World Food Program has described the recent humanitarian crisis in Madagascar, as the *world's first climate related famine* and it has been exacerbated by the Covid19 pandemic. The burden of malnutrition in the country is heavy with children living in DEFIS+ regions being among the most affected⁴. According to UNICEF, the proportion of under-five Malagasy children, suffering from acute malnutrition, which has been declining from 1992 to 2018, is now increasing due to the recent famine caused by the negative impact of climate change. Small-scale farming households are among the most affected by malnutrition, driven by extreme climate events, poverty, food insecurity and insufficient access to basic services such as health, education and WASH. The effects of heavy rainfalls and floods caused by climate change are exacerbated by poorly planned and designed rural roads. Sub-optimal design leads to fast degradation during floods, which massively and repeatedly reduces the access of rural people to agricultural markets and basic services such as education, health and social services. Only 11% of the Malagasy population has access to road networks, meaning the development of agricultural businesses and livelihood resilience is hampered due to poor connectivity. Roads play a vital role in strengthening the socio-economic development and climate change resilience of isolated communities with critical connections to health, education and markets. Road infrastructure needs to be made more resilient in order to enhance the climate change resilience of local people through improved accessibility.

The project area is facing significant adaptation challenges that require urgent technical, financial and institutional support to: (i) mitigate the negative effects of climate change; (ii) improve the resilience of human population in the rural areas; (iii) improve the agricultural value chain production, food security and nutrition of smallholder farmers and their communities; (iv) make roads more flood resilient; (v) improve livelihood options and convert current production systems that lead to deforestation, land degradation and use of grey technology into nature-based and nature conserving green technologies; (vi) improve sustainable water management; and (vii) support systems and climate resilient infrastructures to facilitate access to markets and other social services to boost people's adaptive capacities and increase resilience to climate change.

The objectives of DEFIS+ are to strengthen the resilience of agricultural production systems and vulnerable marginal communities to the impacts of climate change while contributing to enhance mitigation co-benefits such as GHG emission reduction, promoting intensive/improved rice farming techniques (SRI/SRA), and implementing Climate Smart Agriculture (CSA).⁵ Thus, DEFIS+ will be organised around the following three complementary outcomes, which are strategically conceived to contribute to climate change adaptation with mitigation co-benefits:

Outcome 1: Strengthened climate resilience of agricultural production systems through enhancement and maintenance of ecosystem services by: (i) improving access to efficient water management systems for sustainable agriculture, which includes enhancing the resilience of 15,000 ha of irrigated schemes and 8,100 ha of sub-watersheds through nature-based flood protection and ecosystem-based approaches to restore 5,500 ha of landscape; (ii)

⁴ <https://dhsprogram.com/pubs/pdf/PR131/PR131.pdf>

⁵ Malagasy government: Madagascar's Intended Nationally Determined Contribution

improving access to regular and appropriate agro-climatic information for weather forecast and disaster risk reduction using automated agro-meteorological stations for flood and drought monitoring; (iii) improving the climate resilience of food production systems for 45,000 small family farms through the promotion of climate resilient farming technologies and practices, water conservation and storage techniques and integrated pest management techniques; and (iv) promoting resilient fodder production, CSA and agroforestry good practices.

Outcome 2: Increased income generation through improved market accessibility. This outcome revolves around promoting green roads for water approach, supporting the climate resilience of rural infrastructure for access to markets and utilising the water harvesting potential of roads. This includes: (i) investing in reinforcing works including well-designed cross drainage systems on 540 km of rural roads to protect the roads from damage due to climate hazards, such as heavy rainfall and floods and to harvest water for drought resilience by efficiently managing the runoff water from catchments upstream of roads with water retention and artificial recharge infrastructure associated with roads; (ii) training 2,000 trainers, construction workers and technicians on construction and maintenance standards that build upon a labour-intensive construction approach, which will be more sustainable in the long-term due to the reliance on local green technology; and (iii) improving climate resilience by introducing water harvesting options for increased water availability for farming, soil moisture improvement and aquifer recharge.

Outcome 3: Improved food and nutrition security through managing knowledge on food production systems. Local skills and knowledge will be leveraged for instance, through the development of operations manuals for Farmer Field Schools, in order to scale up innovations such as stress tolerant seed varieties and CSA practices. Outcome 3 will also promote non-timber forest products (NTFPs) and neglected and underutilised climate-tolerant food species (NUS), as a strategy to adapt to climate change, improve agricultural diversification, livelihoods and food security. The technologies that will be disseminated will not only help respond to the climate change challenge but will also contribute to reducing women's workload and time burden during farm and off-farm activities, with a positive impact on their health, education and those of their children. This outcome is inherently linked to Outcomes 1 and 2 as it benefits from the introduction of climate-smart practices and from increased income generation, both of which result in enhanced food and nutrition security and mitigate the negative impact of extreme climate events.

The DEFIS+ proposal aligns well with both the NDC priorities, and the GCF Sector Guide on climate change, agriculture and food security, especially regarding the investment pathway on CSA, which focuses on adaptation in agriculture. It is also aligned with the GCF Sector Guide on Ecosystems and Ecosystems Services through the pathway on landscape-level protection and restoration to maintain the ability to mitigate emissions (project co-benefit). Also, DEFIS+ aligns with the GCF Sector Guide on Water Security through the pathway on strengthening integrated water resources management.

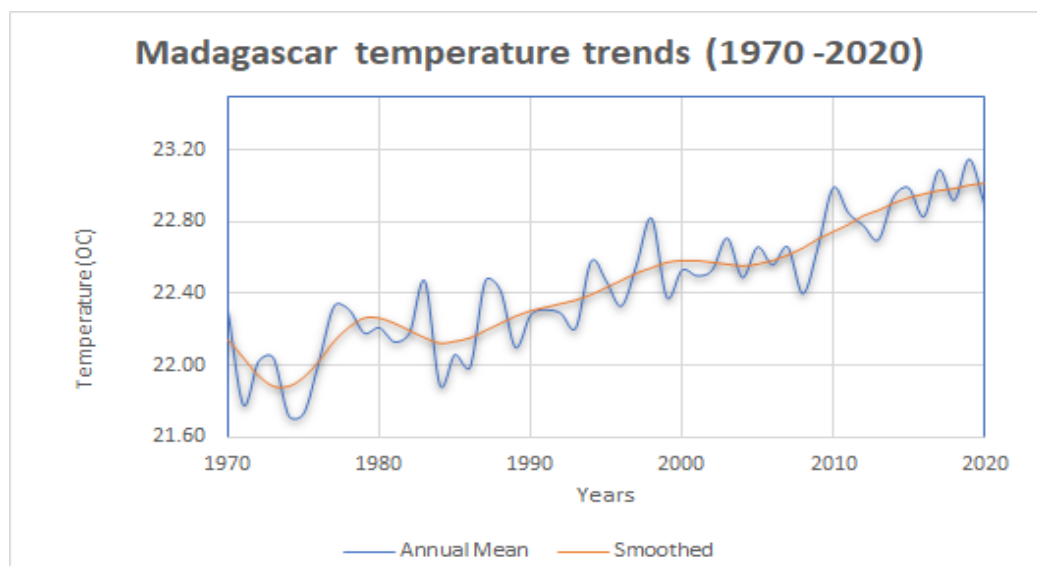
B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Climate baseline. Madagascar's geographical location, maritime influence and wind regime result in great climatic diversity. Madagascar has two main seasons: a hot, rainy season (November to April), with maximum rainfall received between December and January, and a cooler, dry season (May to October), which has less rainfall received between September and October. During the dry period, light rainfall is mostly restricted to the southern and eastern coasts.⁶ The eastern coast is hot and humid and receives between 1700mm and 3700mm of rainfall per year. The highlands enjoy a tropical climate with cold winters and average rainfall ranging from 900mm to 1500mm per year. The southwest and the extreme south are characterised by semi-arid environments with erratic and low rainfall of less than 800 mm per year. Temperatures vary significantly across the country, due to topographic differences between the west and east coasts. Average annual temperatures range from 23°C – 27°C along the coast and from 16°C – 19°C in the highlands.⁷ Temperatures are the highest on the west coast, reaching 36°C between October and November.

Temperature trends. The University of Cape Town estimates that the average temperature in Madagascar increased from 21.5°C to 22.4°C from 1961 to 2005.⁸ Another estimate by Razafitsalama et al. (2018)⁹ indicates that the average annual temperature linearly increased at a rate of about 0.2°C per decade from 1961 to 2017. Additionally, as shown in Figure 1, average temperature trends have increased from 22°C to 23°C from 1970 to 2020.¹⁰ Comparison with other timelines from the EWEMBI climate database¹¹ also indicates an increase in temperature by 0.72 - 0.84°C between 2000 and 2020 compared to the 1960-1990 period. Finally, over the period 1961-2017, minimum and maximum temperatures have increased by 0.04 and 0.05°C/year, respectively.¹² In parallel, temperature indicators show an upward trend in extreme heat waves. The highest increases of over 0.84°C are noted in central and southern Madagascar (Figure 2).

Figure 1: Madagascar observed annual temperature trends from 1970 to 2020¹³



⁶ University of Cape Town (2008). Climate Systems Analytical Group. Climate Change in Madagascar; recent past and future

⁷ <https://climateknowledgeportal.worldbank.org/country/madagascar/climate-data-historical>

⁸ University of Cape Town (2008). Climate Systems Analytical Group. Climate Change in Madagascar; recent past and future.

⁹ Razafitsalama et al (2018). Climate Change Forecasting in Ronohira, Southern Madagascar, using Liner and ARIMA Models. International Research Journal of Environmental Science. Vol. 7(12), 12-20.

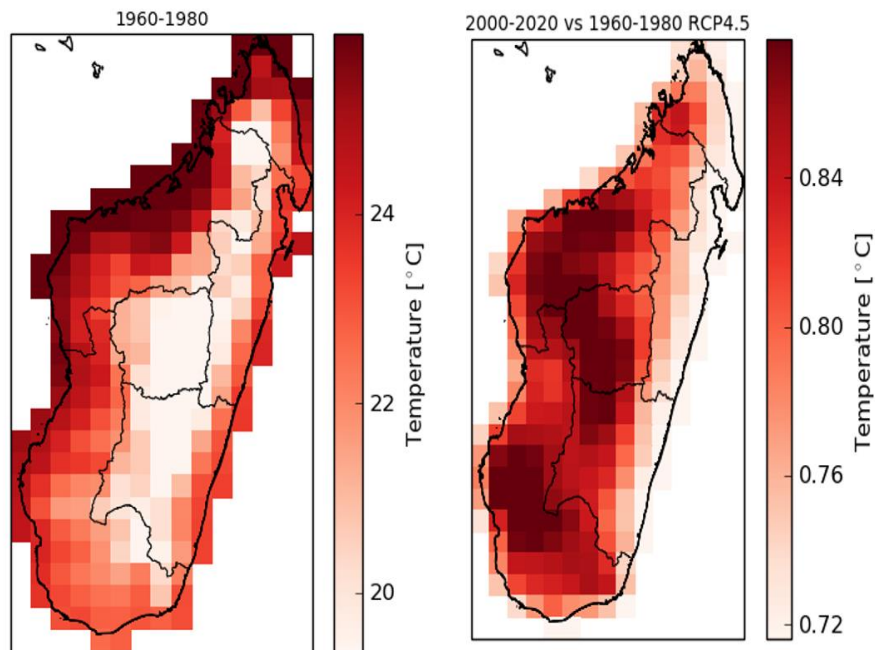
¹⁰ <https://climateknowledgeportal.worldbank.org/country/madagascar/climate-data-historical>

¹¹ Lange, Stefan (2016) Earth2Observe, WFDEI and ERA-Interim data Merged and Bias-corrected for ISIMIP (EWEMBI). GFZ Data Services. <http://doi.org/10.5880/pik.2016.004>

¹² DGM, 2019

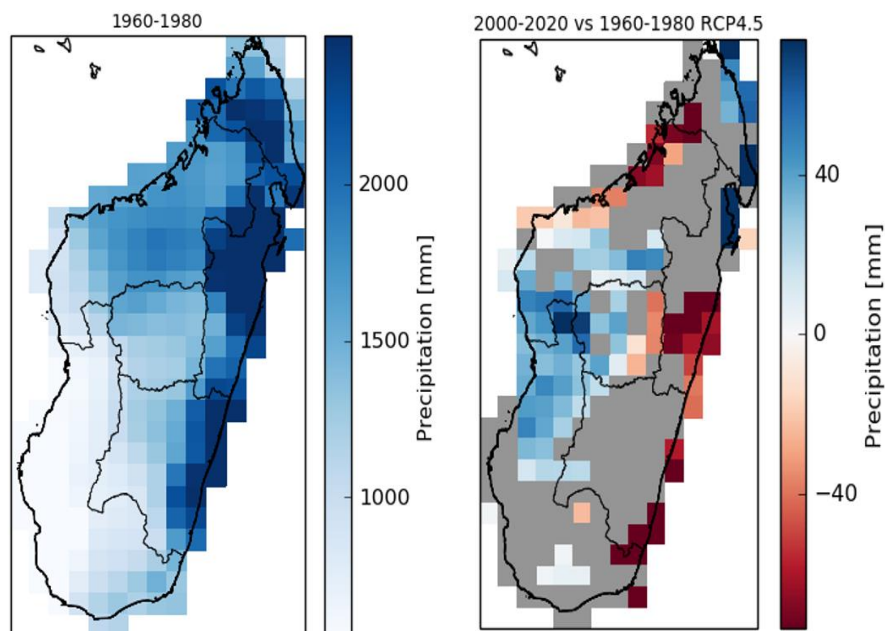
¹³ Adapted from World Bank 2020

Figure 2: Changes between 1960 and 1980; and between 2000 and 2020¹⁴



Precipitation trends. Even though data at the national level indicate no obvious trend in annual average precipitation, rainfall has changed significantly across different regions. It has increased in some parts and decreased in others over the last decades, in parallel with an increased length of dry spells (Figures 3 and 4B).¹⁵

Figure 3: Precipitation changes in different regions¹⁶



¹⁴ EWEMBI climate dataset, <https://climateknowledgeportal.worldbank.org/country/madagascar/climate-data-historical>

¹⁵ <https://climateknowledgeportal.worldbank.org/country/madagascar/climate-data-historical>

¹⁶ EWEMBI, <https://climateknowledgeportal.worldbank.org/country/madagascar/climate-data-historical>

Figure 4A: DEFIS+ project area (in green)

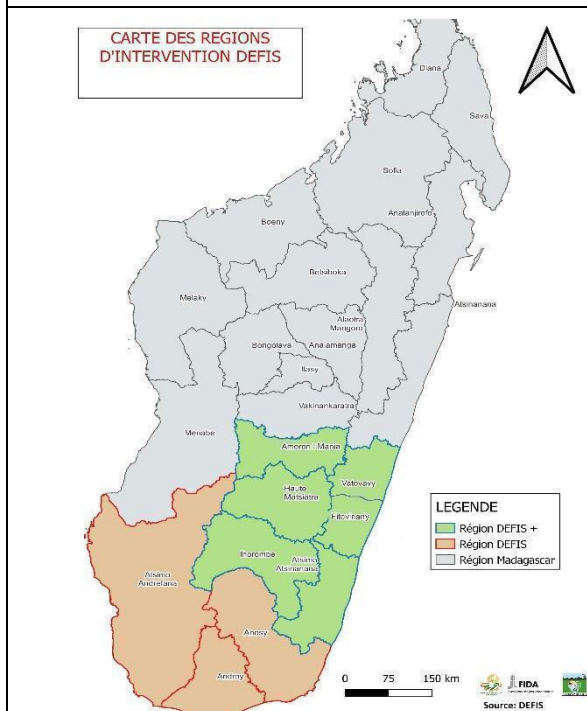
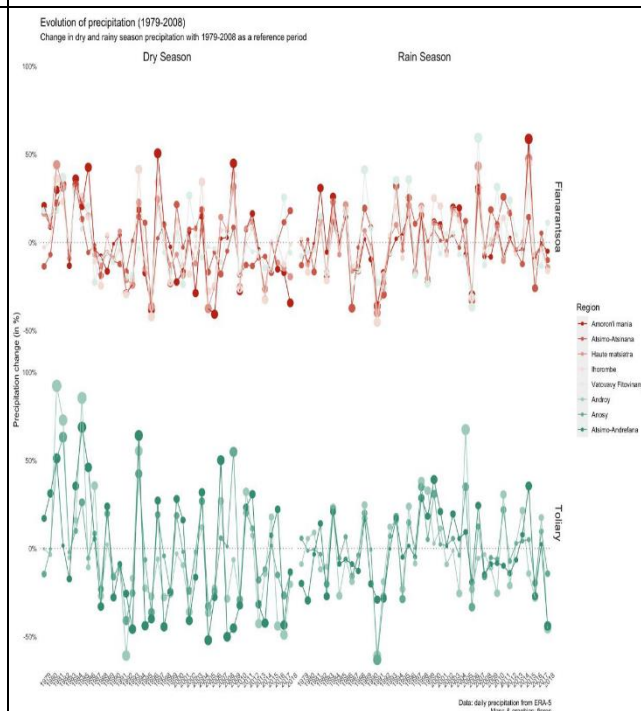


Figure 4B: Evolution of precipitations in Southern Madagascar¹⁷



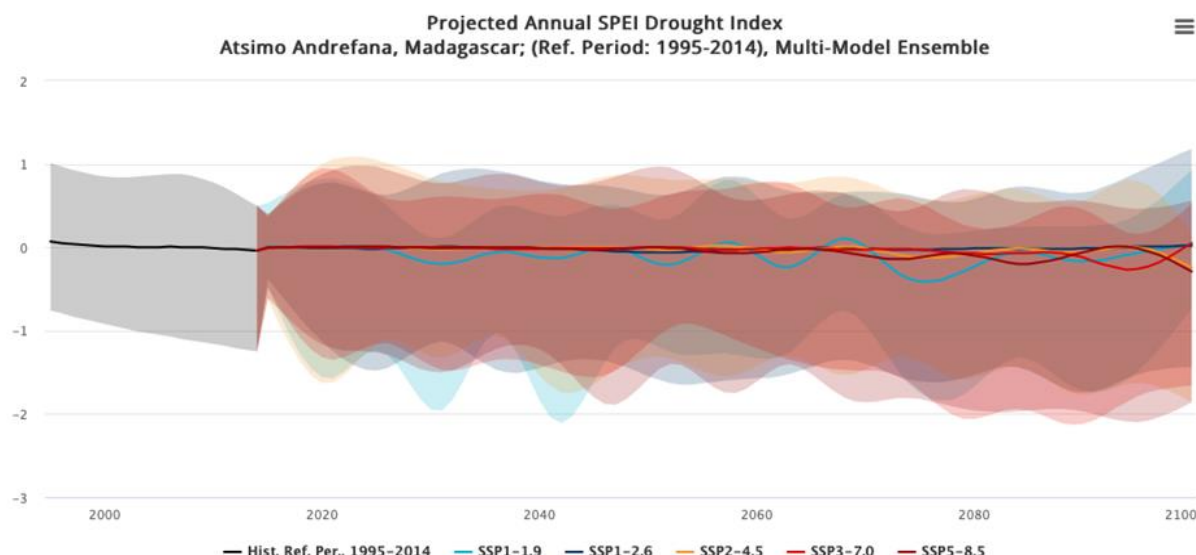
In contrast to the rainy season, the dry season is characterised by low rainfall during May to September, especially in the regions of Amoron'i Mania, Haute Matsiatra and Ihorombe. La Niña brings about the possibility of slightly increased precipitation during the wet season in the western part of Amoron'i Mania and Haute Matsiatra and a normal season in Vatovavy, Fitovinany and Atsimo Atsinanana. El Niño contributes to the risk of drought in Amoron'i Mania, Upper Matsiatra and Ihorombe and the southern parts of Atsimo Atsinanana, Vatovavy and Fitovinany as well as to a slightly wetter season on the coast of the Vatovavy and Fitovinany regions and the north coast of the Atsimo Atsinanana region. During the 2015/2016 El Niño event, there was a 28% reduction in rainfall (JFM), resulting in approximately 80% yield loss for 2016. Data provided by the National Meteorological Agency (NMA) highlights that up to 60% of the south regularly faces water shortages, which have experienced a marked increase, both in frequency and length, since the beginning of the century. Furthermore, rainfall patterns have become more erratic and extreme rain events more intense, thus exacerbating the risk of floods and negatively impacting the irrigated perimeters and lowlands in Madagascar. Thus, the increased intensity of rainfall variability, cyclones and El Niño Southern Oscillation (ENSO) events exacerbate the frequency and duration of droughts in the south of Madagascar.

Temperature projections. Projected temperature increases in southern Madagascar are estimated to be between +1.2 and +2.1°C by 2050. In the Mid-Warming Scenario (RCP4.5), in the near future, the multi-model mean temperature is projected to increase from 0.5°C (HC) to 0.7°C (HC) in the dry season. By mid-century, average temperature could increase from 1.1°C (HC) to 1.4°C (HC) in the dry season and from 1.2°C (HC) to 1.4°C (HC) in the rainy season. Under RCP4.5, in the near future, the multi-model mean number of days subject to extreme heat events is projected to increase from 9.2 days to 18.5 days.¹⁸ By mid-century, the average number of days subject to extreme heat events could increase from 32.4 days (HC) to 54.7 days (HC).

¹⁷ Finres, 2020 (Annex 2)

¹⁸ Finres (2020). Climate Risk Assessment for IFAD Funding Proposal to Green Climate Fund.

Figure 5: Projected annual SPEI drought index



Precipitation projections. The number of days with intense rainfall (above the 95th percentile) is slightly increasing, whilst the maximum daily rainfall amount shows a slight downward trend with drought events from 1.5°C on the rise (Figure 5).¹⁹ A change in average annual precipitation, ranging from -8 to +1% is projected by 2050 and an increase in the amplitude of inter-annual rainfall variability, i.e., greater extremes, is predicted (Annex 2a).

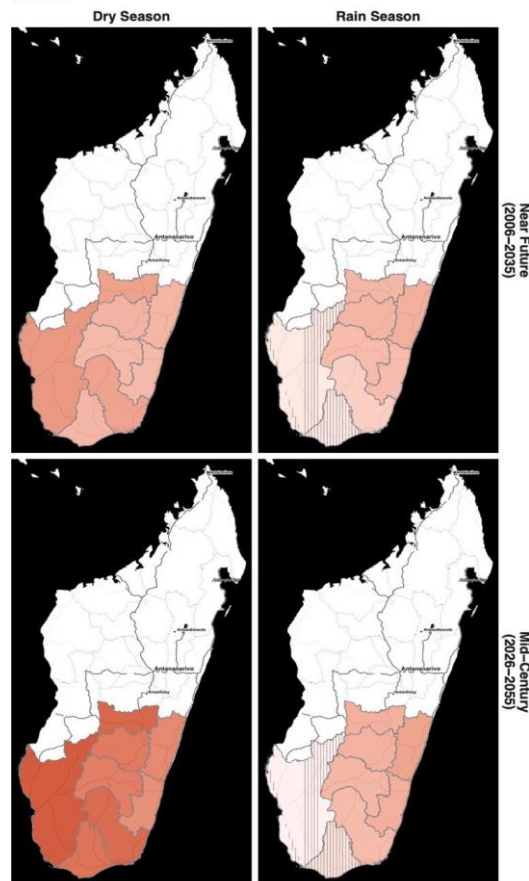
In the regions of Vatovavy, Fitovinany and Atsimo Atsinanana, a decrease in rainfall occurs mainly during the cool months (June, July, August and September). The inter-annual variability of precipitation manifests itself in two ways: in terms of annual amount and at the beginning of the season. Wet, normal and dry years can occur depending on weather conditions, but the probability of exceeding 1500 millimetres per year in the Ihorombe region does not exceed 20%. The projected changes in precipitation and mean temperature are depicted below (Figures 6, 7, 8, and 9).

¹⁹ USAID, Madagascar Climate change risk profile, 2016 and IPCC: https://www.ipcc.ch/report/ar6/wg1/downloads/factsheets/IPCC_AR6_WGI_Regional_Fact_Sheet_Africa.pdf

Figure 6: Projected changes in precipitation and mean temperatures²⁰

Precipitation in the Mid-Warming Scenario (RCP4.5)

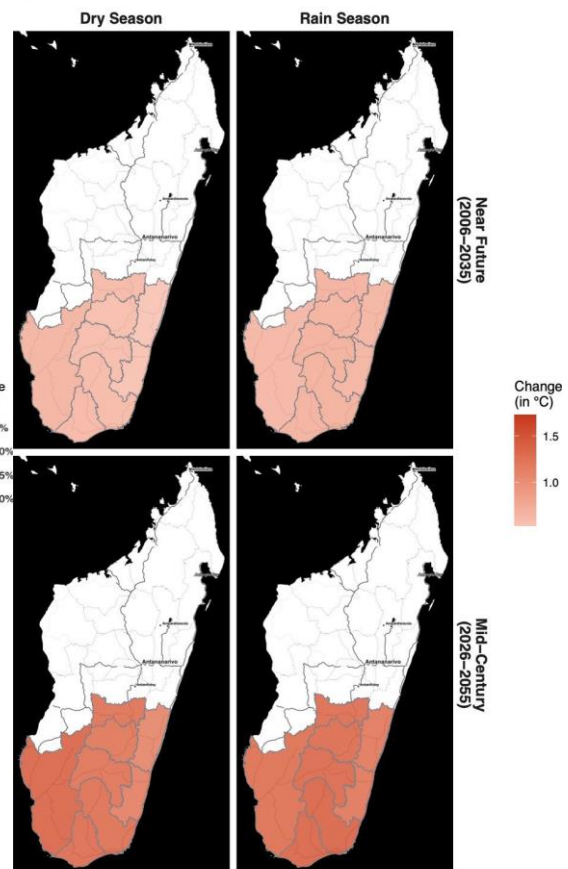
Change in precipitation for the rainy and dry seasons in the near-future (2006–2035) and for mid-century (2026–2055) compared to seasonal precipitation in the reference period (1979–2008). The hatching indicates model disagreement, here below 75%.



Data: CanESM2, CSIRO-Mk3, EC-EARTH and HadGEM2
Maps & graphics: finres

Temperature in the Mid-Warming Scenario (RCP4.5)

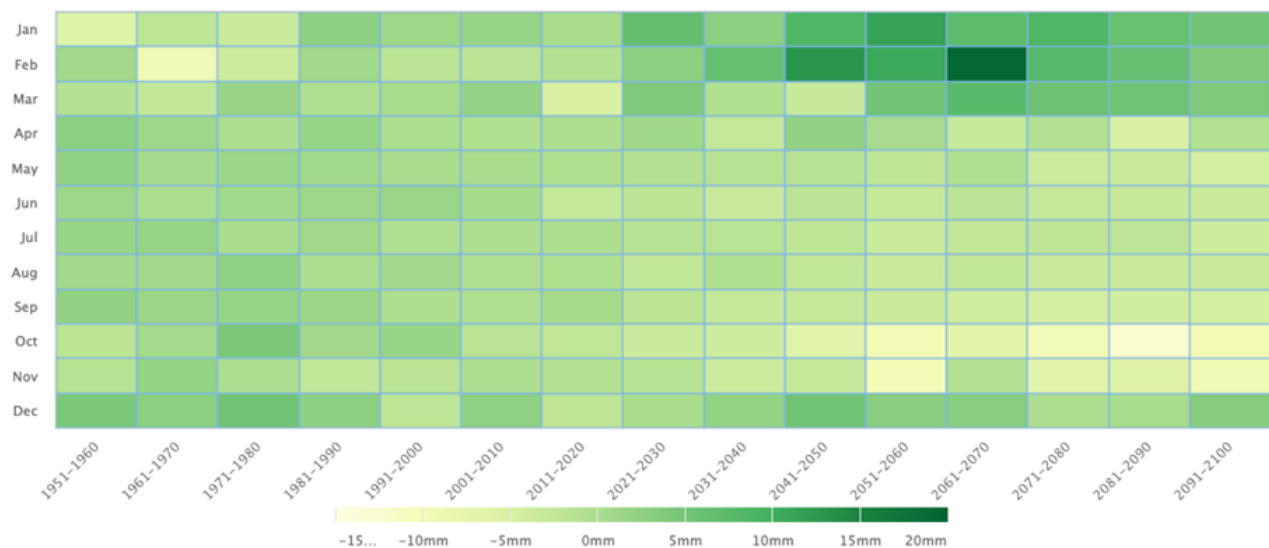
Change in mean temperature for the dry and rainy seasons in the near-future (2006–2035) and for mid-century (2026–2055) compared to temperature in the reference period (1979–2008). The hatching indicates model disagreement, here below 75%.



Data: CanESM2, CSIRO-Mk3, EC-EARTH and HadGEM2
Maps & graphics: finres

Figure 7: Projected precipitation anomaly

Projected Precipitation Anomaly
Atsimo Andrefana, Madagascar; (Ref. Period: 1995–2014), SSP2-4.5, Multi-Model Ensemble



²⁰ FINRES, 2020 (Annex 2).

Figure 8: Projected climatology of precipitation 2080-2099

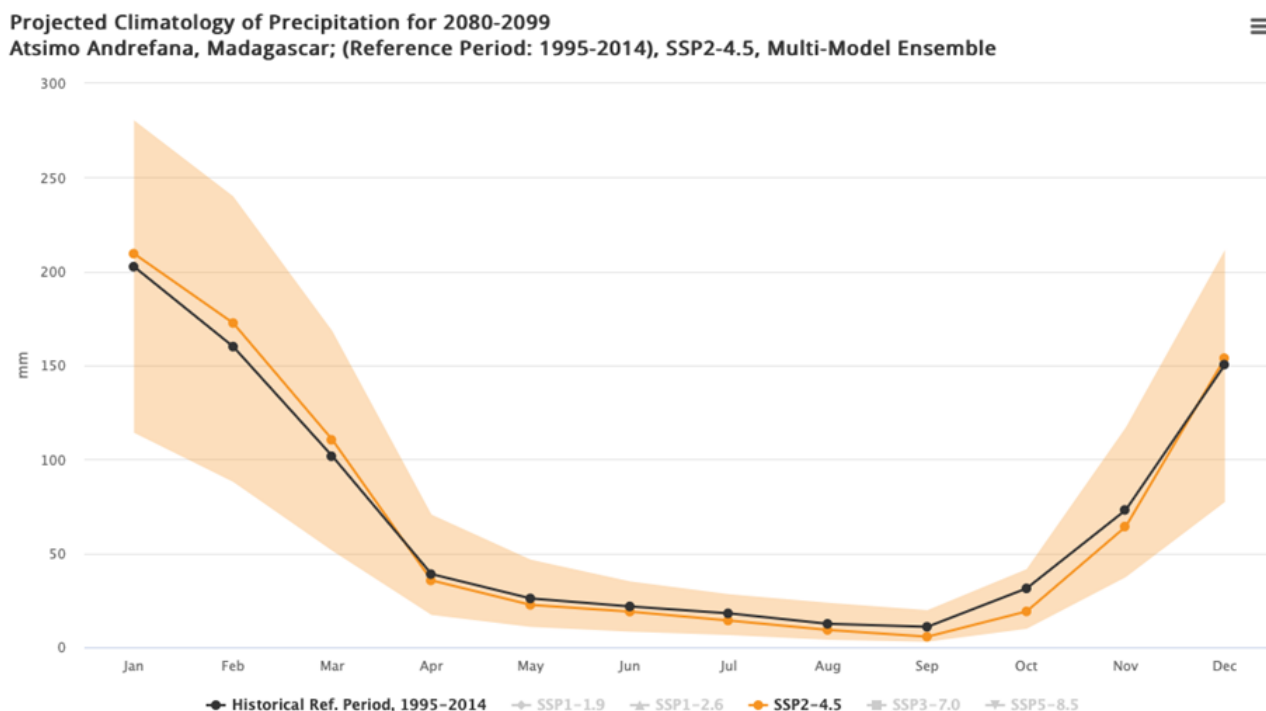
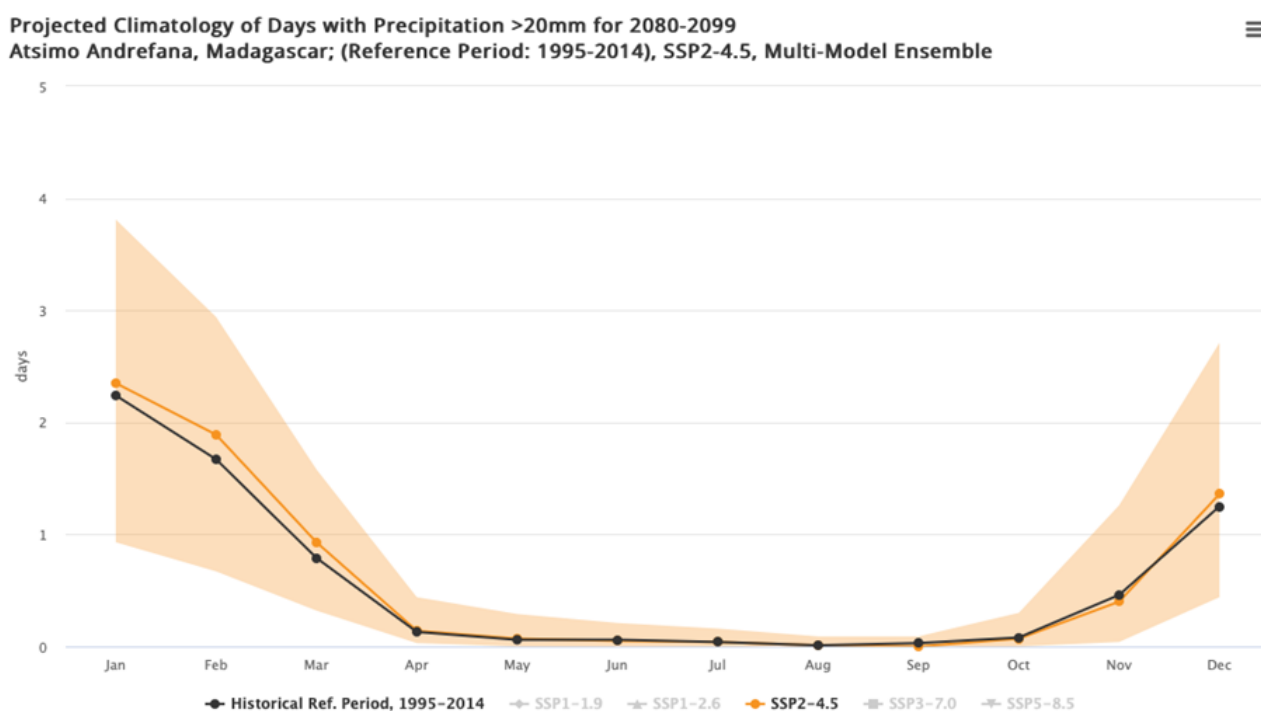


Figure 9: Projected climatology of days with precipitation > 20mm, 2080-2099



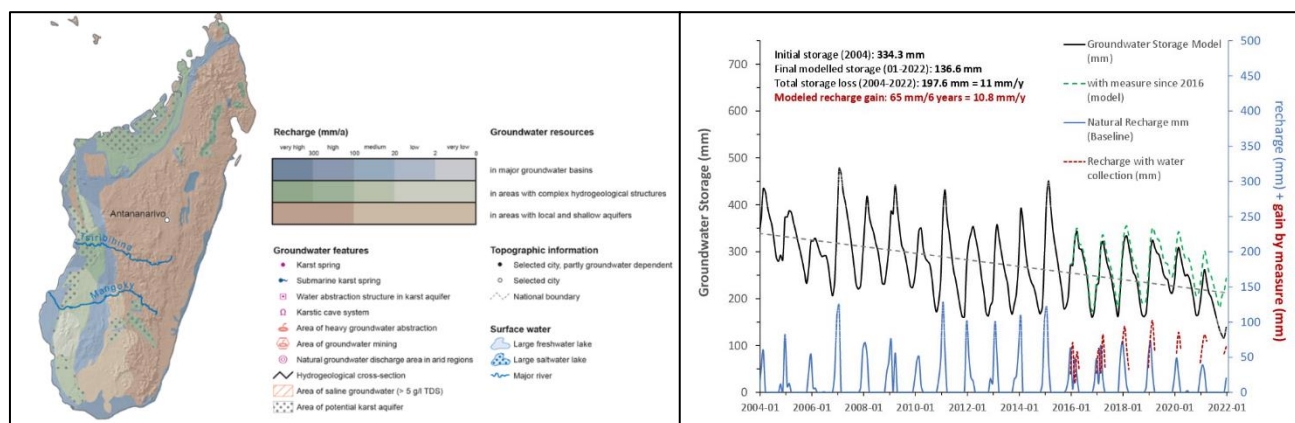
Climate hazards. From 1980 to 2010, fifty-three (53) natural hazards - including, droughts, earthquakes, epidemics, floods, cyclones and extreme temperatures - affected Madagascar and caused economic damages of over US\$1 billion. Of these, cyclone-related damage has been the most significant. High poverty rates and a lack of functional institutions increase vulnerability to natural and climatic hazards such as floods, droughts, cyclones, extreme temperatures and sea level rise.²¹ According to the Think Hazard Index, there is a high risk of

²¹ <https://climateknowledgeportal.worldbank.org/country/madagascar/vulnerability>

floods, landslides, cyclones, wildfires and extreme heat in the country in the coming years.²² The great variability of rainfall in the DEFIS+ project area (Figure 4A,B) leads to either droughts or situations where the water balance is very much in surplus (Annex 2a).²³ As the groundwater storage is limited, the surplus will create runoff and is not available as groundwater during the dryer season. Madagascar is subject to recurrent droughts and the drought phenomenon is cyclical for the southern semi-arid areas (in particular in the Vatovavy and Fitovinany Regions), with some areas receiving less than 400 mm of rainfall each year, and is exacerbated by the El-Niño phenomenon.²⁴ Historical records indicate that, particularly in south of Madagascar, droughts are common leading to famines in 1903, 1910, 1916, 1921 and 1943 among other years. Recently, Madagascar experienced an unprecedented climate induced famine in 2016²⁵ and 2021, following a four-year drought, the worst recorded in 40 years²⁶. This drought has depleted groundwater resources as shown by GRACE data by NASA and integrated modelling of the water balance using MSWEP data (Figure 10).

Climate change is the primary cause of recurrent droughts and cyclones, which are the leading causes of food insecurity and frequent needs for emergency assistance in the project area.²⁷ For example, the 2017-2018 drought reduced agricultural production by 60% and threatened 1.3 million people with food insecurity. According to UNICEF, the proportion of Malagasy children under 5 suffering from malnutrition, which has been declining from 1992 to 2018, is now increasing due to the recent famine. Madagascar's economy shrank by 4.2% in 2020, despite a fast-growing population. Furthermore, droughts also lead to a decline in the fertility of the natural environment. Degradation usually begins with a change in vegetation and with a decrease in biomass production. The soil, which is less protected, is subjected to the mechanical action of precipitation, which causes a modification of the surface conditions and the progressive destruction of the soil. A spiral of degradation is formed, and without any intervention, it leads to irreversible desertification. Desertification is an environmental problem, exacerbated by climate change, that is inseparable from the issue of sustainable development of drylands. Droughts affecting the project area are expected to increase, disrupting yields, crops, livestock production and the resilience of the agricultural sector. Monitoring the occurrence of droughts for agricultural production activities is very important for building human and climate resilience.²⁸

Figure 10: Development of groundwater storage in the projected area from GRACE Data (NASA, Earthobserver) and from the Global Land Assimilation System (NASA, Earthobserver).



The groundwater bodies in the study area are predominantly Precambrian basement rocks (Figure 10 left, map by BGR, WHYMAP, UNESCO, 2023, brown colour). Their key characteristic is that they have very limited storage,

²² <https://thinkhazard.org/en/report/150-madagascar>

²³ The droughts are meteorological i.e., being the result of a rainfall deficit compared to normal during a given period (season) and over a given region, agricultural i.e. If this situation affects crops or hydrological i.e., a decrease or scarcity of water resources (rivers, lakes and water bodies)

²⁴ UNICEF. (2021). Wash Technical Paper. Accessed at:

<https://www.unicef.org/madagascar/media/5661/file/Simplified%20Drought%20Monitoring.pdf>

²⁵ FAO (2021). The impact of disasters and crises on agriculture and food security. Accessed at <https://www.fao.org/3/cb3673en/cb3673en.pdf>

²⁶ Harding, A. (2021). Madagascar on the brink of climate change induced famine. Accessed at <https://www.bbc.com/news/world-africa-58303792>

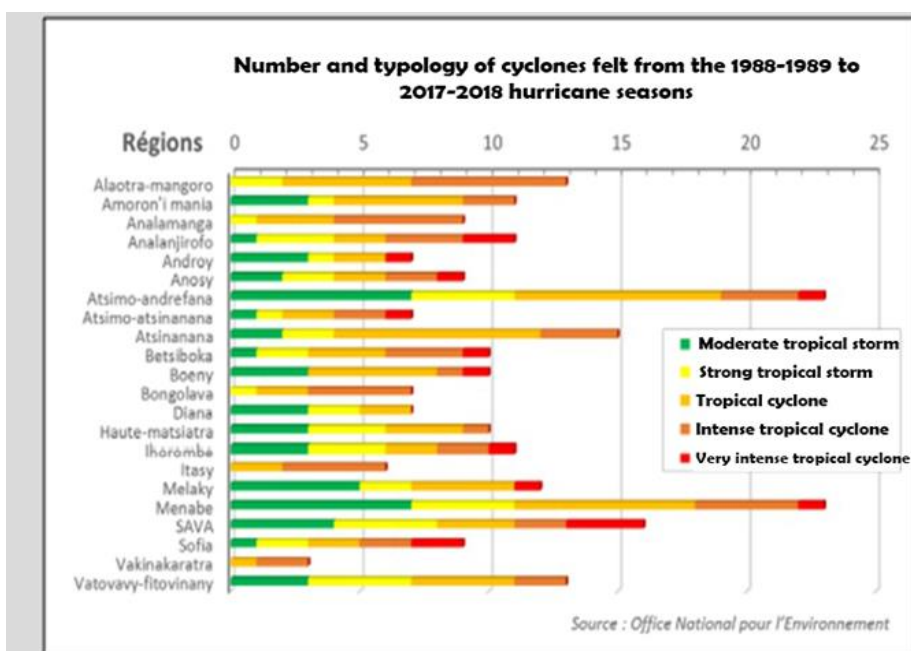
²⁷ The UN Migration Agency (2017). Evidencing the Impacts of the Humanitarian Crisis in Southern Madagascar on Migration, and the Multisectoral Linkages that Drought induced Migration has on other Sectors of Concern. Accessed at <https://www.iom.int/sites/g/files/tmzbd1486/files/country/docs/Madagascar/IOM-Madagascar-Southern-Madagascar-Assessment-Report-EN.pdf>

²⁸ Crop evapotranspiration is measured using FAO's Irrigation & Drainage Paper No.56 approach, evapotranspiration (et₀) is estimated following the Penman-Monteith equation and crop-specific K_c coefficients for the four main phenological phases. By combining evapotranspiration with crop coefficients at the different phenological stages of the crop, ET_c provides an indication of crop water use under normal conditions.

storing only between 0.1% and 5% of their volume. As a result, these basement rocks are poor aquifers for groundwater storage and deplete quickly during drought. The measurement of groundwater storage with satellites (Figure 10 right) from 2004 until 2022 shows a decline in groundwater storage from 334.3 mm to 136.6 mm for the south-eastern basins of Madagascar, corresponding to a loss of groundwater of 11 mm/year. The natural recharge has been extracted (shown in blue, Figure 10 right). The potential impact of the measure of collecting surface runoff at road intersections, upgraded in the study area as part of the project, and using the collected runoff as artificial recharge for a scenario calculation since January 2016 is shown (green dotted curve). The potential addition by the measure can compensate for the climate induced loss in storage (red curve) and yields 10.8 mm/year of recharge according to the scenario model which compensates for the storage loss.

Typology of cyclones. Due to its insular nature in the Indian Ocean, Madagascar, and the project area in particular, are frequently affected by tropical cyclones. Not only are tropical cyclones more frequent in this ocean basin, but they are also more intense than in other parts of the world. On average, 13 tropical systems with wind speeds exceeding 63 km/h are reported to form each year. Madagascar has one of the highest cyclone risks among African countries, with an average of three to four cyclones affecting the country every year. Climate change is associated with a higher frequency of cyclones, resulting in lower rainfall in the project area. However, the intensity of such cyclones is projected to increase by 46%, along with the intensity of associated floods. Due to the negative impact of climate change, the project areas and central regions (Amoron'i Mania, Haute Matsiatra, Ihorombe and Atsimo Atsinanana) are expected to experience more intense and frequent cyclones during the wet season and will be subject to more frequent flooding. On the other hand, during the dry season, these same regions are likely to experience periods of increased water deficits. The accentuation of such extreme climatic events could have a strong impact on agriculture, and particularly on rice growing, the main source of economic activity in the project area. Increases in average tropical cyclone wind speeds and associated heavy precipitation are projected to result in category 4-5 tropical cyclones (Figure 12). Intense rainfall events caused by strong storms and tropical cyclones, coupled with poor land use practices and increasing deforestation, can lead to significant and damaging floods across the country and a high number of human and animal deaths. Floods represent the second highest impact disaster, affecting more than 300,000 people during the 1961-2017 period. Over 30 floods or heavy rainfall events have affected Madagascar in the past 30 years, killing hundreds of people and affecting thousands (Figure 12). During the passage of Tropical Cyclone Dzumazil, the water levels of the Matitanana River in Vohipeno increased by 5 metres and the Manapatrana River in Farafangana increased by 4 metres causing significant flooding with extensive damage and high human casualties. During the rainy season of 2007-2012, the Vavovavy and Fitovinany regions appeared to be the most threatened by floods. The intensity of these hazards will be even greater due to an exacerbation of climatic extreme events such as an increase in the average intensity of tropical cyclones, the amplitude of storm surges and the associated rainfall, particularly if greenhouse gas emissions remain high.²⁹

Figure 11: Number and typology of cyclones from 1988-1989 to 2017-2018



²⁹ Atlas des risques naturels et climatiques de Madagascar, BRGM (Géosciences pour une Terre Durable), Cellule de Prévention et gestion des urgences, Pilot programme for climate resilience, World Bank, 2020.

In the project area, other climate hazards include soil erosion and landslides. With regard to soil erosion (Figure 13), the effects go beyond the loss of fertile land, with an increase in pollution and sediment deposits in streams and rivers and in the clogging of reservoirs and dams, leading to a threat to aquatic flora and fauna (rivers, lakes and lagoons). Thus, the erosion leads to a loss of fertility on arable land and therefore, a decrease in agricultural yields. *Lavaka* is a Malagasy word meaning “hole”, used in geomorphology to describe a spectacular form of erosion. It is characterised by a large, ten-meter-deep, roughly ovoid depression that narrows downstream to form an outlet. At this outlet, the materials spread out in the form of a cone. *Lavakas* are common on the slopes of the central highlands of Madagascar, at high altitudes between 900mt. and 1250mt.³⁰

Figure 12: Floods risk map

USAID, Madagascar, WRI Aqueduct 2019

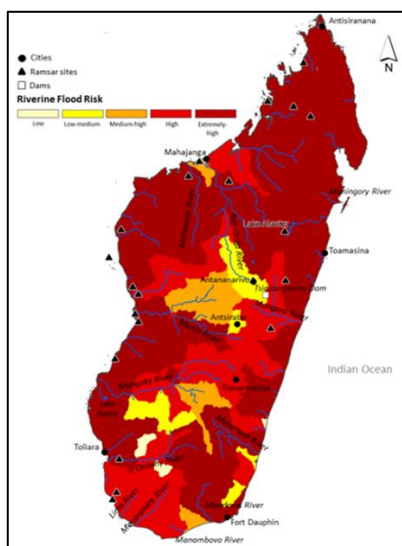


Figure 13: Soil erosion risk



Climate Impacts and vulnerabilities. Madagascar has been ranked the fifth most vulnerable country to climate change in the world³¹. ND-Gain measured Madagascar’s exposure and sensitivity to climate change as high and ability to adapt to the negative impacts of climate change as low and classified it in the 169th place out of 181 countries³². Being an island country by virtue of its geographical position, Madagascar is exposed to recurrent climate risks that greatly affect its economy, the living conditions of its population and its natural resource capital.³³ Sectors such as agriculture, water resources, livestock, human health, infrastructure, terrestrial and marine biodiversity and the environment are vulnerable to climate change to varying degrees. An example are the social infrastructures that have been destroyed (including schools, basic health centres, and dispensaries), agricultural infrastructures have been damaged and administrative buildings, urban roads and rural roads have been severely impacted by extreme weather events, particularly cyclones, all of which are directly linked with the negative impact of climate change.

Agriculture and Climate Change. Agriculture, the backbone of Madagascar’s economy, accounts for 20% of its GDP and employs 80% of the population, with women and youth representing an important proportion. The agricultural sector is predominantly rain-fed, characterised by subsistence farming and based on traditional practices and technologies, such as slash and burn, which contribute to deforestation and soil erosion, further aggravating the sector’s climate vulnerability. Indeed, rural populations’ dependence on low-yielding rain-fed staple crops and their over-reliance on single commodities (most often rice) make them extremely vulnerable to climate change.³⁴

Climate-related extreme events such as increased rainfall during the rainy season lead to increased sedimentation, soil erosion and siltation, which compromise the flat lowlands and lead to flooding and crop diseases. On the other hand, insufficient rainfall and increased drought conditions will result in (i) lower agricultural productivity, crop damage and yield reduction, (ii) higher rates of evapotranspiration, reduced humidity and increased soil degradation, (iii) more difficult access to drinking water; increased climate-sensitive diseases

³⁰ Atlas des risques naturels et climatiques de Madagascar”, BRGM (Géosciences pour une Terre Durable), Cellule de Prévention et gestion des urgences, Pilot programme for climate resilience, World Bank, 2020

³¹ Maplecroft, 2012

³² <https://gain.nd.edu/our-work/country-index/rankings/>

³³ Climate Trends and Future Climate Change in Madagascar 2019

³⁴ COI, 2011

(dengue, malaria, typhoid fever, etc.) and (iv) loss of biodiversity. For example, maize production is projected to decrease in large areas of Madagascar due to reduced rainfall.³⁵ Extreme climatic seasonal events such as cyclones can destroy crops and the flooding they generate, especially in heavily deforested areas, can leave behind a layer of sand that makes land uncultivable.^{36,37}

In irrigated systems, rice yields are expected to decline by as much as 10.98% over the next 30 years, whereas in non-irrigated systems, wheat will decrease by as much as 37.45% and maize by 30.68%³⁸, representing a significant threat to the livelihoods and wellbeing of the rural population. Extreme climate events will exacerbate these projections even further. Variation in rainfall patterns and temperatures, along with the growing intensity of extreme climate events, is modifying farming calendars and consequently, increasing crop failure.³⁹

Faced with the effects of climate change, farmers cannot adopt adaptation measures in the absence of reliable agro-met information. Meteo Madagascar is the natural provider of climate and agro-met services, but demand generally outpaces their capability to respond in terms of spatial and temporal accuracy. The lack of reliable agro-met information compromises farmers' adaptive capacity to climate change. The availability of agro-met information would help farmers better adapt to climate change; therefore, improved access to agro-met information is essential to reduce the climate vulnerability of rural and marginal communities in the project area.

Food insecurity, malnutrition, and climate change. All six regions in the project area have experienced extreme dry episodes and the situation has caused precarious conditions (malnutrition, undernourishment) in recent years, following the drought related to El-Niño and several cyclones.⁴⁰ In 2021, four consecutive years of drought reportedly resulted in one of the world's first famines caused by climate change registered in modern times, with people forced to eat locusts, insects and wild animals to survive.⁴¹ Famine is the highest internationally recognised level of food insecurity. Climate related drought effects have also led to agricultural losses of up to 60% in the most populated provinces⁴². Thus, it should be noted that famine and food insecurity are also driven by non-climatic direct and indirect drivers, such as sub-optimal agricultural practices that result in low productivity for the farmers as well as land degradation. These factors are addressed through the promotion of good agricultural practices, which include CSA. These practices will improve agricultural productivity at the household and community levels, thus improving food and nutrition security in communities.

Roads and the impact of climate change. Roads play a vital role in strengthening the socio-economic development and climate change resilience of isolated communities with critical connections to health, education and markets. The important interaction between the road network and the river network for the study area with basin boundaries is shown in Figure 14. Road networks are crossed by and aligned with rivers. Road construction so far has not explored and used the potential of surface runoff for harvesting. While one focus of the project is to improve rural road resilience towards erosion and floods, a second innovative aspect is to use road infrastructure for water harvesting. The water from upstream of the road is collected by check dams along the road.

Madagascar's poorly developed railway network and limited inland waterway transportation increase the country's reliance on road transportation. Most of the roads nowadays are in very poor condition, with the majority being unpaved and difficult to access, especially during the rainy season. With an estimated road network of 31 640 km, Madagascar has one of the lowest road densities in the world. In the year 2000, 1.6 % of major roads were exposed to river floods at least once a year. By 2080, this value is projected to increase to 2.0 % under RCP2.6.⁴³

³⁵ Shi and Tao 2014, Ngwakwe, 2019

³⁶ Llopis, 2018

³⁷ Gay-des-Combes et al. 2017; Niang et al. 2015

³⁸ Climate Adaptation in Rural Development Assessment Tool (CARD): <https://www.ifad.org/en/web/knowledge/publication/asset/41085709>

³⁹ http://www.commissionoceanindien.org/fileadmin/resources/Aclimate%20Etudes%20Asconit-Pareto/20110818_vulnerabilite-madagascar.pdf

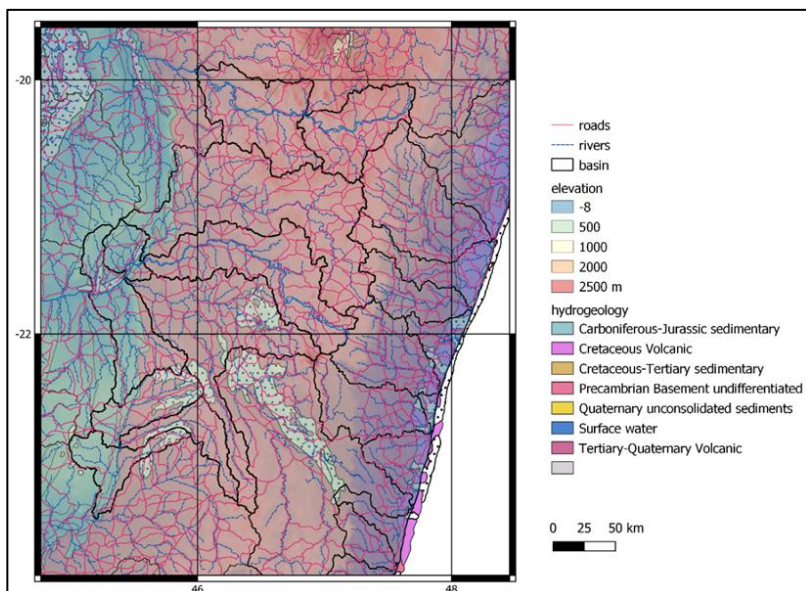
⁴⁰ https://reliefweb.int/sites/reliefweb.int/files/resources/IPC_AcuteFoodSec_Madagascar_2019July2020June_English%20summary.pdf

⁴¹ <https://www.aljazeera.com/news/2021/9/3/madagascar-is-on-brink-of-first-climate-induced-famine-un-warns>

⁴² Ibid

⁴³ GIZ, Climate Risk Profile: Madagascar, accessible at: https://agrica.de/wp-content/uploads/2021/01/GIZ_Climate-Risk-Profile-Madagascar_EN_final.pdf

Figure 14. Network of roads and river



Hence, investments will have to be made to build climate-resilient road networks and improve the resilience of local people through improved accessibility. In particular, floodwater is the single largest factor in road damage and is considered the prime enemy of road infrastructure. According to Chinowsky & Arndt (2012), water is responsible for 80% of damage to unpaved roads and 30% of damage to paved roads. Road damage makes them unusable and increases maintenance and rehabilitation costs.

Climate change will exacerbate the frequency and intensity of meteorological phenomena posing more risks for water-related damage to roads and the surrounding landscape. Therefore, to turn the problems above into opportunities for

beneficial water management and climate resilience, the internationally acknowledged [Green Roads for Water](#) (GR4W) approach will be adopted in DEFIS+. The aim of GR4W is to transform the way roads are built and maintained, by incorporating water management and climate change adaptation into the design and construction of roads. The GR4W is a simple approach that (i) reduces water-related damage to the roads, thanks to well-designed cross-drainage systems, maintenance works and the promotion of seasonal roads; (ii) minimises adverse impacts of roads on the surrounding landscape, such as flooding, waterlogging and land degradation; and (iii) manages water beneficially, either for the benefit of roadside water users (agriculture), or by improving the sustainability of water resources, by reducing water scarcity and climate disaster risks. Annex 2a and 2b contain an extensive analysis of climate change impacts, models and vulnerabilities of Madagascar.

Market, market access and climate adaptation strategies. The Global Competitiveness Index ranks the quality of the country's roads at 2.2 on a scale of 1 to 7 (World Bank 2017). Farm-to-market routes are commonly unpaved and highly susceptible to degradation from increasing flooding and erosion.⁴⁴ More than 30 floods or heavy rainfall events have affected Madagascar over the last 30 years. The most damaging floods were those caused by cyclones Eline and Gloria, which killed 20 people, and tropical cyclone Elita, which killed 363 people and affected one million, while also causing economic damages of over US\$250 million.⁴⁵ Improving transport mobility will lower disruptions in the movement of goods and people and contribute to reducing the GHG emissions of exhaust gases into the atmosphere as a result of reduced travel time, reduced number of operations while driving vehicles, stabilising speed at a level that allows for running of vehicles at optimum consumption levels, increasing considerable energy efficiency and reducing the consumption of fossil fuel. Accessing basic services, such as schools and hospitals, is problematic in rural areas of Madagascar.⁴⁶ Lack of access to health services and care increases vulnerability to climate change, diseases and malnutrition. Damage to infrastructure caused by extreme climatic events such as heavy rain, cyclones and sea level rise, is a critical issue, especially since the population is expected to increase by 40% by 2030. For example, in early 2000, a series of three particularly devastating cyclones affected more than a million people and caused nearly USD 85 million (equivalent to 2% of the GDP in 2000) in damage to agricultural infrastructure.⁴⁷ In terms of jobs, investments in emergency road works have amounted, for the last 10 years, to approximately USD 94 million.⁴⁸ Damages due to floods and cyclones from 2010 to 2015 were estimated to be 470-490 million USD a year. Other structural economic and institutional issues further undermine human resilience (e.g., limited access to credit, high interest rates, obstacles to market access, etc.).

Resilience to climate hazards. Addressing the aforementioned hazards in an integrated manner, as summarised in Tables 1 and 2A below, is crucial to increasing the resilience of smallholder farmers and their

⁴⁴USAID (2018) Climate Risk Profile. Madagascar. [online] Available at: https://www.climate-links.org/sites/default/files/asset/document/180404_USAID-ATLAS_Climate-Risks-in-Urban-and-Urbanizing-Geographies.pdf

⁴⁵GFDRL 2011 Climate Risk and Adaptation Country Profile, Madagascar.

⁴⁶<http://www.fao.org/3/ca3933en/ca3933en.pdf>

⁴⁷East African Agriculture and Climate Change: comprehensive analysis Madagascar, International food policy research institute, Dec 2012

⁴⁸Strategic program for climate resilience: Madagascar pilot program for climate resilience, 2017

agricultural production systems to climate change in the project area. This entails the adoption of improved farming practices and technologies (drought and stress tolerant crop varieties, soil moisture conservation, mixed cropping and crop diversification, irrigation efficiency, agroforestry and afforestation, etc.), as well as climate-proofing basic rural infrastructure such as soil protection from erosion and climate resilient roads. Annex 2 provides various options for climate change adaptation for the value chains included in the DEFIS+ project, those that are seen as the most technically feasible and cost effective are selected and prioritised. By enhancing resilience at the household level, the risk of further natural resource exploitation during “lean” periods is reduced. Watershed restoration and infrastructure rehabilitation will also strengthen ecosystem services, improve land productivity, mitigate the negative impact of climate change, and further reduce dependence on natural resources for economic gain.

The prioritisation of the climate change adaptation options is a process that starts with the use of the Adaptation Framework, Annex 23. The Adaptation Framework is a tool that uses a multi-criterion analysis system incorporating factors such as cost-benefit, climate-risk relevance, farmer capacities, mitigation co-benefits and biodiversity support when selecting adaptation options during project design. Its selection of the best adaptation measures is informed by analysis of climate change risks and impacts in the project locations. The available options are scored by technical teams on the ground with IFAD experts on the design teams. The prioritised options are the suitable ones with the higher scores in the completed tool for the project. These options are then further assessed with the climate modelling results and scenarios and costed for inclusion in the project.

Table 1: Overall adaptation measures proposed by DEFIS+ to improve resilience to climate hazards

Adaptation measures	Primary justification
Irrigation system	Lower precipitation throughout the rainy season (Figures 7 and 8). Irrigation is required to compensate for the decrease in rainfall. Combined with increasing temperatures, evapotranspiration could increase – measures limiting it are required to make irrigation more cost efficient. Irrigation could become especially relevant at the onset of the rainy season, as the number of dry days is expected to increase the most in October and November (Figure 5).
Road infrastructure	The Southern regions face flooding risks even though the number of extreme wet days is not projected to increase (Figure 9). Road infrastructure could be used for water catchment purposes as a response to decreased precipitation and a shorter rainy season. The Green Roads for Water (GR4W) approach will be adopted in DEFIS+. This modern, sustainable, and eco-friendly approach can guarantee a small footprint impact. More information is available in Annex 2a and 2c.
Evaporation limiting measures	The number of hot days in the rainy season is projected to increase over time, especially at the onset of the rainy season (Figure 6). Measures protecting crops, particularly seedlings against heat stress (agroforestry good practices and natural shading) are essential. Such measures would also reduce crop stress induced by increased evapotranspiration.
Crop shifting & improved seed selection	Lower precipitation throughout the rainy season could cause crop production to fall below optimal levels (Figures 7 and 8). The increase in precipitation in January–March does not offset the reduction in October–November. Need for crops with shorter growing cycles as a response to the overall decrease in length of growing season. An extensive analysis with additional information is available in Annex 2a and 2b.
Climate information	Potential progressive delay in the onset of the rainy season (Figures 7 and 8) – lower precipitation in October and November. An overall decrease in the length of the growing season is analysed and discussed in Annex 2a. Climate information is essential to improving the resilience of the local population to climate shocks.

Table 2A: From climate change impacts to identified responses at the farm level

Key climate change impacts identified by the project	Impact	Major challenges associated with key climate c. impacts	Short- and long-term response measures to alleviate climate negative impacts
Temperature increase will enhance the intensity of rainfall and delay their onset	Crop yields reduced	Farmers (and their farming systems) must contend with:	Support farmers, through agricultural extension/advisory services on practices that build resilience to climate extremes:
Extreme weather events, such as droughts, floods, El Niño	Crop yields reduced Damage to irrigation and road infrastructure	• Short-/medium-term changes in the timing, distribution and availability of water, with associated heat stress, resulting in negative impacts on crops	• Promote landscape restoration and reforestation • Adopt agroforestry good practices
Temperature increases will result in associated increased water deficit stress, reduced river flow and groundwater reserves associated with increased rainfall intensity and surface runoff	Increased drought and evapotranspiration	• Reduced soil moisture levels, which adversely impact crop productivity • Long-term water scarcity, aggravated by inefficient irrigation infrastructure • Increased risk of extreme weather events and related infrastructure loss	• Adopt locally adapted cropping systems, practices and varieties that are more resilient to extreme climate events • Adopt renewable energy-efficient technologies • Develop/support water harvesting techniques • Promote water use efficient irrigation schemes
Extreme weather events, such as cyclones, sea level rise, droughts and floods	Losses of rural infrastructure, land degradation (erosion, sedimentation) and local flooding and waterlogging which in turns causes losses in agricultural activities	More frequent and extreme weather events pose new risks to roads, landscapes and water resources	Support the short- and long-term rehabilitation and climate proofing of rural infrastructure, especially irrigation schemes and feeder roads linking farmers to markets and feeder roads close to agricultural activities

Target regions and the beneficiaries of the project. The DEFIS+ project will complement and catalyse climate adaptation and mitigation investments in six regions (Amoron'i Mania, Haute Matsiatra, Vatovavy, Fitovinany, Ihorombe and Atsimo Atsinanana, Figure 4A) and target 447,200 direct beneficiaries. These regions were selected based on their vulnerability to climate change, high incidence of rural poverty, food and nutrition insecurity and high potential for agricultural development and for facilitating access to markets.

The six regions of intervention of DEFIS+ can be grouped into two zones:

- (i) The south-east zone (Atsimo Atsinanana, Vatovavy, and Fitovinany) is characterised by a humid climate, high exposure to cyclones, erratic and intense rainfall; and
- (ii) The south-central highlands zone (Ihorombe, Matsiatra Ambony and Amoron'i Mania) is characterised by a humid climate and increasing droughts and floods.

The design of DEFIS+ also addresses the urgent adaptation and mitigation needs of Madagascar as outlined in its NDC, which identifies southern Madagascar as the most vulnerable region to climate change. At least 50% and 30% of DEFIS+ direct beneficiaries will be women and youth, respectively. The disaggregation of the beneficiaries is presented in Table 2B.

Table 2B: Beneficiaries of the DEFIS+ project.

Beneficiaries

	Direct	Indirect	Totals
Women	223,600	1,771,860	1,995,460
Men	223,601	1,771,859	1,995,460
Totals	447,201	3,543,719	3,990,920

Complementarity. DEFIS+ will complement DEFIS's efforts to transform the climate vulnerable and low-performing agricultural sector from its subsistence nature to a climate resilient market-oriented and profitable sector of the economy, which also contributes to improved food and nutrition security. DEFIS focuses on economic development through improving agricultural productivity while DEFIS+ will focus on climate resilience building of the communities specifically their livelihoods and the ecosystems they depend on. Both initiatives (DEFIS and DEFIS+) will work hand in hand, making a decisive contribution to the attainment of the Sustainable Development Goals (SDG) in Madagascar. DEFIS+ will also complement the GCF-funded project of Conservation International and the European Investment Bank (EIB) entitled "Sustainable Landscapes in Eastern Madagascar" which is implemented in the watershed not far from DEFIS regions by rehabilitating and climate-proofing key rural infrastructures and establishing synergies to further develop the value chains targeted by the respective projects. In the southeast area, DEFIS+ will complement UN Environment's project entitled "Adapting Coastal Zone Management to Climate Change in Madagascar Considering Ecosystem and Livelihoods" (PAZC) in the regions of Vatovavy and Fitovinany by providing communities with improved farming techniques and promoting crop diversification and climate-smart agriculture. DEFIS+ will also scale up the results of the African Development Bank/GEF "Enabling Climate Resilience in the Agriculture Sector in the Southwest Region of Madagascar" project in the six DEFIS+ target regions. Finally, DEFIS+ will complement the GIZ-funded "Adaptation of Agricultural Value Chains to Climate Change" (PrAda) project by supporting the development of value chains and the AfDB's PAPAM project by replicating and scaling-up the agro-ecological production techniques of its areas of intervention.

Additionality. DEFIS+ is strategically designed to increase the climate resilience of the socio-ecological systems in six regions. The GCF resources will be very important for making DEFIS+ investments in the six target regions more resilient to climate change and for promoting activities that will help reduce GHG emissions and provide other benefits that will be discussed in this and the next section. As a result, DEFIS+ resources from the GCF will provide benefits for adaptation that are not possible with just national economic development projects. Without the GCF's resources, interventions to enhance climate resilience in the six regions would not be possible.

With GCF resources, DEFIS+ investments in the project area will also contribute to climate change adaptation by improving the aquifers that are critical to supporting agricultural systems. Thus, GCF resources for efficient water management systems and GR4W will be critical in improving the water retention in the catchment area, thereby contributing to recharging aquifers. The construction of GR4W involves environmentally friendly, simple, and low-cost techniques to systematically manage the run-off across the entire catchment.^{49;50} The support for fuel-efficient cooking stoves to reduce fuelwood use will also contribute to maintaining or improving groundwater sources and combating deforestation. Deforestation has negative impacts on the water table and surface water systems as it loosens soil, favours runoff, decreases the absorption of water in the soil and increases GHG emissions.

Enhance national policies and strategies on climate change. DEFIS+ is consistent with and complementary to the government's policies, plans and strategies to address climate change issues, linking them with climate resilience, poverty reduction, food security and nutrition. Table 3 provides a summary of DEFIS+ contributions to the relevant policies and strategies.

⁴⁹ Madhu, S.A (2013). Green Road: an ecological road construction method for the preservation of mountain environment and landscape in Nepal

⁵⁰ Van Steenberg, F., Arroyo-Arroyo, F., Rao, K., Hulluka, T. A., Woldearegay, K., & Deligianni, A. (2021). Green Roads for Water. Washington, DC: World Bank)

Table 3: DEFIS+'s contribution to national strategies and policies in the climate change sector

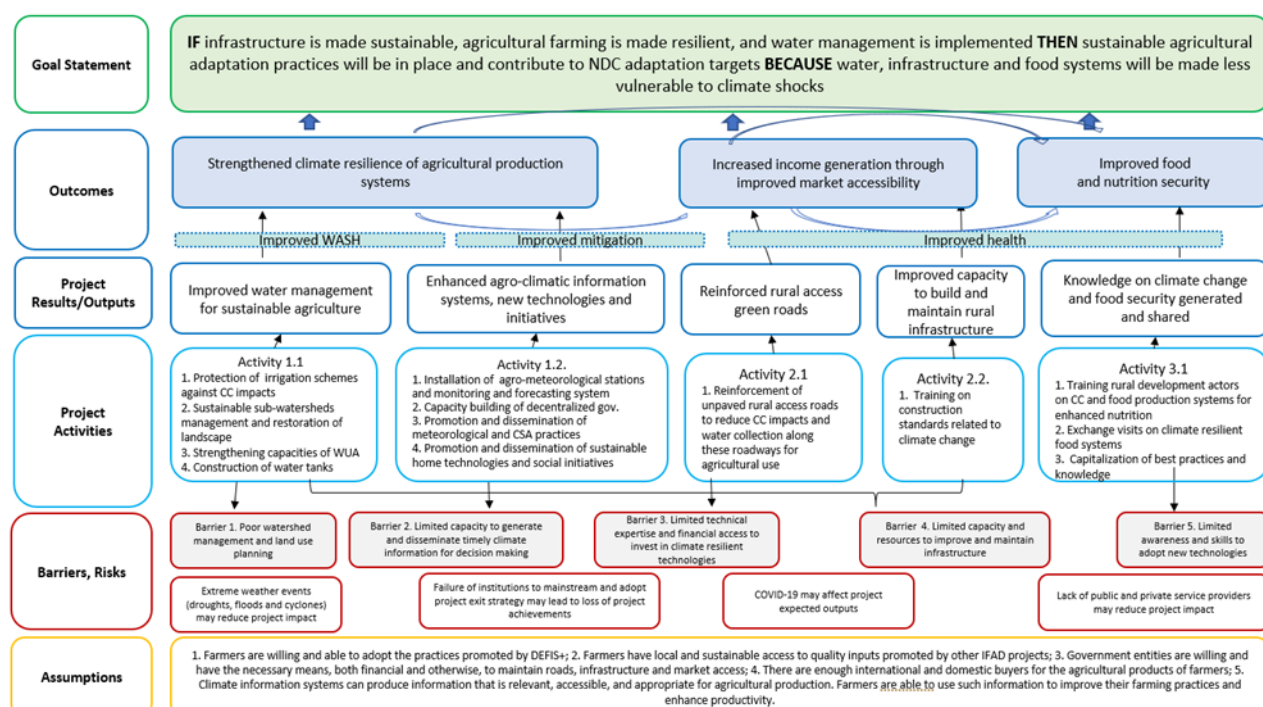
Policy or strategy	Goals	Contribution of DEFIS+ to climate change policy or strategy
National Adaptation Plan of Madagascar (NAP)	Madagascar's NAP revolves around six strategic priorities: Axis 1 (Regreening Madagascar, through reforestation and rehabilitation works); Axis 2 (Green and Blue Economy, through promoting sustainable value chains and job creation); Axis 3 (Climate Change and Energy Transition, through enhancement of climate change resilience and energy sovereignty); Axes 4, 5 and 6 on education, information, green diplomacy and governance.	DEFIS+ activities will contribute to the implementation of these NAP strategic axes, in particular Axes 1, 2 and 3.
National Risk and Disaster Management Policy	It is a tool of reference and guidance for decision-making in the context of Risk and Disaster Management. It is a transversal, inter-sectoral and multi-stakeholder framework for risk management, both in terms of response to climate shocks, but also prevention and mitigation.	DEFIS+ contributes to the implementation of the policy by taking into account the climate prevention and mitigation dimension in project activities.
National Climate Change Strategy for the Agriculture, Livestock and Fisheries Sectors (SNCC / AEP)	This document outlines the strategy adopted by the Agriculture-Livestock-Fisheries Sector in the context of climate change. The strategy aims to establish technical, social, economic and financial bases in order to reduce the climate vulnerability of the agriculture, livestock, and fisheries sectors to the impacts of climate change.	DEFIS+ integrates consideration of climate change (reducing vulnerability, increasing resilience and reducing greenhouse gas emissions) in all project actions.
National Climate Change Strategy and action plan of the dissemination of Climate-Smart Agriculture in Madagascar	Strategy and action plan for strengthening livelihood resilience to climate change. The capacities of all actors can be strengthened so as to stimulate the scaling up of the dissemination of CSA and that actions should be carried out at the level of the 22 regions to have a national scope.	The DEFIS+ project largely contributes to the implementation of the national strategy through the main strategic axes: foster the adoption of AIC practices through optimization, the adoption of CSA practices through the facilitation of access to means and factors of production as well as capacity building actions to combat the negative effects of climate change and improve productivity.
National Policy to face Climate Change	Comprehensive and multisectoral policy framework adopted by Madagascar with a view to achieving sustainable development to face climate change.	The DEFIS+ project largely contributes to this policy by the nature of its activities to the fight against climate change in several vital sectors.
National REDD + Strategy (Reduction of Emissions from Deforestation and Forest Degradation)	The REDD+ mechanism in Madagascar contributes to the reduction of 14% of GHG emissions from the forestry sector, through the increase of forest cover and the control of forest fire emissions from the forestry sector, also through the increase of forest cover and the control of deforestation and forest degradation in its investment areas.	DEFIS+ contributes to the implementation of 2 major strategic axes of the national strategy: (i) promotion of the development and efficient use of land and water in rural areas; (ii) improvement of the standard of living of local populations through good agricultural practices and the reduction of unsustainable fuelwood for cooking.

Nationally Determined Contributions (NDCs)	Madagascar NDCs includes both a mitigation component, aiming to reduce 14% of the national emissions and an adaptation component which aims to strengthen agriculture's resilience to climate change.	DEFIS+ will promote activities to improve climate resilience, reduce the vulnerability of smallholder farmers to climate shocks by investing in agriculture and livestock systems, maintain ecosystem services and access appropriate climate data by improving information systems.
Third National Plan of Action on Nutrition (NPANIII)	The Plan, which is aligned with SDG2, seeks to improve the nutritional status of the most vulnerable populations by implementing complementary multi-sectoral interventions, including food security and social protection, resilience to climate change, and strengthening the development-emergency nexus.	DEFIS+ will promote more tolerant seed varieties with a higher nutritional value. The project activities will also contribute to food and nutritional security, enhance the livelihoods of the most vulnerable, and improve the climate resilience of those in need.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Rationale: The Theory of Change diagram (Figure 15) demonstrates how the project shifts the BAU from poorly managed climate risks in the DEFIS+ regions to a climate resilient and sustainable approach for smallholder farmers and their communities. The project's outcomes and outputs are in-line with GCF's adaptation goals and objectives. The paradigm shifting aspects of the project will include the data driven climate-proofing of modern and sustainable infrastructure, water security, CSA and a focus on the last-mile engagement of vulnerable rural communities. The theory of change shows that **IF** infrastructure is made sustainable, agricultural production is made resilient, and improved water management is implemented **THEN** sustainable agricultural adaptation practices will be in place and contribute to NDC adaptation targets **BECAUSE** water, infrastructure and food systems will be made less vulnerable to climate shocks.

Figure 15: Theory of Change Diagram



Madagascar is a low-income country that ranks 164th out of 189 countries on the Human Development Index and frequently experiences climate hazards.⁵¹ The vast majority of the Malagasy population, estimated at 75% in 2019, lives in extreme poverty. Food insecurity and malnutrition are also prevalent in the country. This reality is particularly prevalent in rural areas, where the population's main activity is low-productivity subsistence agriculture. Moreover, the climate change vulnerability of Madagascar and its population is exacerbated by their high exposure to the effects of climate change that adversely affect living conditions, the environment and basic infrastructure.⁵²

The climate change vulnerability in the DEFIS+ project area is linked, inter alia, to: (i) the decrease in the availability and reliability of water for food and agriculture; (ii) the limited access to and degradation of irrigation systems; (iii) the high level of soil erosion and the degradation of natural resources, as a consequence of deforestation and unsustainable slash and burn farming practices in rainfed areas; (iv) the limited availability of climate resilient agricultural inputs and technologies and relevant knowledge and adoption of farm management practices; (v) and the limited access to rural basic infrastructures. Thus, the vulnerability of rural people in the project area is further exacerbated by the underlying high incidence of extreme poverty, reaching 87.6% of the population living below the absolute poverty line.⁵³ Women and girls experience the greatest impacts of climate change, which amplifies existing gender inequalities and poses unique threats to their livelihoods, health and safety. To be able to overcome these barriers, the financial resources of the GCF will be used to reduce the vulnerability to climate change of the rural communities in the DEFIS+ six regions (Section B5).

In view of the climate rationale detailed in the ToC, the planned activities under DEFIS+ will aim at: addressing the impacts of climate change risks on agricultural production, supply chain infrastructure and road infrastructure; increasing the availability of water for irrigation and agricultural productivity; and enhancing the resilience of communities living around road infrastructure to the impacts of climate change. This can be done through:

- The establishment of early warning systems that provide information on the start and cessation dates, duration and amount of rainfall to help farmers determine their planting schedule and crop selection. This will also include the integration of phytosanitary surveillance, agricultural warnings, drought alerts and food and nutrition surveillance, in particular.
- Strengthening awareness raising on climate change among men, women and youth, including policymakers (IEC and behaviour change). Climate resilience will require a national program of information, awareness, education and capacity building at all levels: population, local authorities and decision makers.
- Establishing a large-scale restoration program for the most threatened ecosystems. Madagascar has already committed to restoring four million hectares of degraded forest landscapes by 2030, with appropriate planning and methods.⁵⁴ However, to significantly increase this commitment, international climate finance and this project can be secured to help support climate change adaptation activities across the country.
- The implementation of reforestation programs for watersheds, which could fulfil a quadruple objective, namely: (i) the preservation of water resources while limiting erosion phenomena; (ii) the creation of climate refuge for endangered species; (iii) large-scale reforestation; and (iv) sequestration of GHGs.
- Improving water infrastructure and promoting small- to medium-scale irrigation to help farmers manage variable and unpredictable heavy rainfall.⁵⁵ This will reduce walking distance to fetching water for domestic use for women reducing their workloads.
- Improving soil and water conservation strategies, upgrading production infrastructure, intensifying production and providing training and technical assistance to promote the adoption of new climate-friendly techniques to help farmers adapt to the changing climate and reduce drudgery associated with farm workloads.^{56,57}
- Scaling up the dissemination of conservation and CSA through the replication of innovative practices currently conducted on a small scale. This measure has co-benefits in terms of GHG emission mitigation and water resource conservation.

⁵¹ The human development index value of Madagascar is 0.528 in 2019, UNDP

⁵² Eckstein D, Hutfils ML and Winges M. Who Suffers Most From Extreme Weather Events? Weather-related Loss Events in 2017 and 1998 to 2017

⁵³ <http://documents.worldbank.org/curated/en/538821468271809604/pdf/781310PRIORITY0English0Apr900May012.pdf>

⁵⁴ SNRPF, 2017

⁵⁵ Asafu Adjaye, 2014

⁵⁶ Dawson 2016, Desbureaux and Damania 2018, Zougmore et al. 2018; Asafu-Adjaye 2014, Rakotondravony et al., 2018

⁵⁷ Delille 2011, Gay-des-Combes et al. 2017 ; Zougmore et al. 2018

- The current unsustainable rain fed agricultural practices need to be transformed into a climate-resilient, market-oriented and profitable economic sector supported by adequate climate-resilient infrastructure and timely and reliable climate information. This will benefit the end-users and guarantee higher incomes, better food and nutrition security, and stable access to water and markets.
- Developing appropriate program(s) including training technicians and workers on the GR4W approach. This modern, sustainable, and eco-friendly approach will be used as the basis for the rehabilitation and maintenance of national and rural roads, adapted to resist extreme climatic events.

Building blocks. The three key building blocks for the **paradigm shift** are as follows:

- i. Strengthen the climate resilience of production systems, through:
 - a. The development of efficient water management systems, while ensuring the integrated management of the watersheds;
 - b. Regular access to reliable-climate data to inform risks assessments and planning;
 - c. Diffusion and adoption of climate resilient inputs and of new technologies and management practices that reduce GHG emissions.
- ii. Improve the climate resilience of basic rural infrastructure, particularly unpaved rural access roads with cross drainage systems to improve access to food markets and protect the landscape around roads with runoff water collection systems set up to harness water resources collected along the roadways.
 - a. Improved roads will allow for increased access to markets. This is crucial for the generation of income, the sharing of knowledge, and the access to basic services and acquisition of basic necessities. Thus, a modern approach to road construction and rehabilitation has a minimal carbon footprint, while BAU road management has a large carbon footprint.
 - b. Improved roads will allow for sustainable water management, increase water availability for farming, and reduce the isolation of communities, which directly contributes to lowering their climate change vulnerability.
 - c. Improved roads will lower the risk of accidents and also allow remote communities to access facilities (such as health care centres) especially for women and children, thus contributing to overall health (co-benefit).
- iii. Scale up technological innovations on climate change adaptation and knowledge management among family farmers and decentralised actors, for enhanced nutrition, through:
 - a. Training on climate change and capacity building services;
 - b. Knowledge management on the adaptation of food production systems to climate change.

This building block on knowledge generation and management will reinforce the previous three and ensure a paradigm shift in the daily lives of the smallholder farmers supported by the project.

The proposed approach has high transformative potential as it will combine key adaptation interventions that will strengthen the capacities of family farms and their production systems, create an additional positive feedback loop by improving the overall climate resilience of beneficiaries and contribute to reducing emissions.

Barriers. However, some barriers and risks may hinder the achievement of the outcomes and the overall goal of the project. Mitigation measures for such risks are already integrated into the project design.

Barrier 1: Poor watershed management and land use planning, lead to soil erosion, siltation of irrigated perimeters and lowlands and disruption of road infrastructure, which results in low agricultural productivity and undermines the functionality of the ecosystems on the landscapes. This will be overcome through activities under Outcome 1 through the focus on sustainable agriculture and watershed management as well as Outcome 2 through the GR4W approach and climate proofing of the rural infrastructure.

Barrier 2: Limited capacity to generate and disseminate timely climate information for decision making, for farmers, and the rural population to be able to predict the changes in climate patterns in order to better plan ahead and adjust, shift crop calendars, adapt cultivation practices and techniques. This will be overcome through the activities under Outcome 1 which consists of strengthening the climate resilience of agricultural and animal

production systems through the improvement and maintenance of ecosystem services, and Outcome 3 which includes increasing access to and adoption of technological innovations for climate change adaptation.

Barrier 3: Limited technical expertise and financial access to invest in climate resilient technologies, which hinders the dissemination and access to climate smart technologies and therefore the resilience of communities and their livelihoods. In particular, to tackle Barriers 1, 2 and 3 the following activities are key:

- i. Exchange visits on the adaptation of food systems to climate change and the application of sustainable farming practices;
- ii. Capitalization of best practices and knowledge on climate change adaptation measures;
- iii. Installation of agro-meteorological stations and development of water, flood and drought monitoring and forecasting system;
- iv. Capacity building on access to finance and governance (SRM & DRAEP).

Barrier 4: Limited capacity and resources to improve and maintain infrastructure, which hinders rural populations from accessing basic services and increases food insecurity at the regional level.

Barrier 5: Limited awareness and skills to adopt new technologies, which result in inefficient use of resources such as water and use of traditional methods and therefore low adaptive capacity of communities and their livelihoods.

In particular, to tackle Barriers 4 and 5, the following activities are key:

- i. Barrier 4 will be overcome through outcome 1, and 2 respectively, focusing on the promotion of water efficient technologies, sustainable GR4W initiative, access to markets and income generation activities.
- ii. Barrier 5 will be overcome through outcome 3, by implementing modern and green sustainable technology that will enhance the climate change resilience of rural communities against climate shocks.

In addition minor barriers to those listed above require risk mitigation tools and contingency plans, which are extensively discussed in Annex 2a (Feasibility Study).

The assumptions that apply to the results of the proposed project and the respective activities and that would allow the project to succeed in its objectives are listed below:

- Farmers are willing and able to adopt the practices promoted by DEFIS+, also because they have local and sustainable access to quality inputs promoted by other IFAD projects such as DEFIS;
- Government entities are willing and have the necessary means, both financial and otherwise, to maintain roads, infrastructure, and market access;
- There are enough buyers (local and regional) for the agricultural products of farmers;
- Climate information systems can produce information that is relevant, accessible and appropriate for agricultural production. Farmers are able to use such information to improve their farming practices and enhance productivity.

Policy and gender. The DEFIS+ activities have been defined according to the climate change adaptation perspectives in relation to the priorities set out in the Madagascar National Climate Change Adaptation Plan (NCA) and the NDCs that the country has adopted following the Paris Agreement and the National Policy for the Fight against Climate Change, which focuses on the sectors of Agriculture-Livestock-Fisheries, Water Resources, Public Health, Biodiversity and Forestry, Coastal Zones, Infrastructures, Management/Reduction of Climate Risks and Disasters. This strategic priority refers to the NDC which foresees, between 2020 and 2030, the strengthening and updating of multi-hazard Early Warning Systems by integrating climate hazards warnings, phytosanitary surveillance, crop warnings, flood and drought alerts, and food and nutrition surveillance. In addition, Table 3 includes a list of DEFIS+'s contribution to national adaptation strategies and policies in the climate change sector.

Gender empowerment and social inclusion is an integral part of the overall DEFIS+ and will therefore be mainstreamed into project activities, complemented by targeted interventions to overcome any specific barriers as necessary. This approach will enable project opportunities to be expanded to women and youth, while also addressing the main gaps and barriers they face to inclusion. This approach recognises the need to identify and address social norms that perpetuate gender empowerment and social inclusion inequalities and exacerbate

barriers to access productive agricultural assets, project services and attaining climate resilient and improved livelihoods. An extensive analysis of the gender dimension is included in Section G.2 and a gender action plan is included in Annex 8. Youth outreach will be improved by working with Technical Vocation centres that are supported through FORMAPROD/ PROGRES, which are IFAD financed projects in Madagascar.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1:	☐	☐	☐	☐	☒	☒	☐	☒
Outcome 2:	☐	☐	☐	☐	☒	☒	☒	☒
Outcome 3:	☐	☐	☐	☐	☒	☒	☐	☒

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Improved health	☐	X	☐	☐	☐	☐
Improved WASH	X	X	X	X	☐	☐
Improved mitigation	☐	☐	☐	☐	☐	X
Biodiversity conservation	X	☐	☐	☐	X	X

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Background of the project. The DEFIS+ project will be part of a larger ongoing programme named DEFIS, which benefits from 40 years of IFAD presence and experience in family farming in Madagascar. DEFIS is a multi-actor funded programme (IFAD, OFID, the government and beneficiaries) that will scale up tested results and best practices focusing on the value chains identified in Annex 2 and prioritised based on the value chain studies (covering climate change, production potential and challenges in value chain development). DEFIS covers two funding cycles for IFAD for a total of 10 years. DEFIS was designed under IFAD's tenth replenishment cycle (IFAD10, 2016-2018) which, based on country driven demands, focused on agricultural value chain development as the pillar of an integrated approach to rural economic development. The first batch of financing was approved by the IFAD's Executive Board (EB) in December 2017, entered into force in March 2018 and is expected to complete activities in 2027. In December 2021, IFAD's EB approved the second batch of financing. As part of IFAD's increased focus on climate resilience, in IFAD12 (2022-2024) there is a substantial increase in the availability of funding for climate activities (40% of total IFAD financing) and thus the GoM requested the DEFIS+

project to increase climate resilience of the food production and socioecological systems in six out of nine regions of DEFIS.

Complementarity. DEFIS and DEFIS+ complement each other because DEFIS aims to improve productivity in select agricultural value chains, while DEFIS+ aims to increase the resilience to climate change of smallholders benefiting from the services of the improved agricultural value chains. The decision that DEFIS covered nine regions and DEFIS+ only six out of nine (Figure 16) was taken by the National Designated Authorities and the GoM because the other three regions were included in other projects. Thus, to avoid overlapping with these new projects, it was decided to exclude them from DEFIS+. DEFIS+ design took advantage of and capitalised on experiences and emerging lessons identified during the ongoing implementation and supervision of the DEFIS project. In addition, DEFIS+ benefitted from exchanges with other actors in rural development, particularly projects financed by IFAD (AD2M, AROPA, FORMAPROD) and those of other development partners. All these initiatives will be carried out with a view to continuously learning and improving performance. The inclusion of the poorest and most climate change affected people in local development dynamics requires specific targeting mechanisms. These mechanisms for targeting vulnerable people (AROPA) and boosting micro-projects (AD2M and FORMAPROD) with climate change and gender components have been tested and approved.

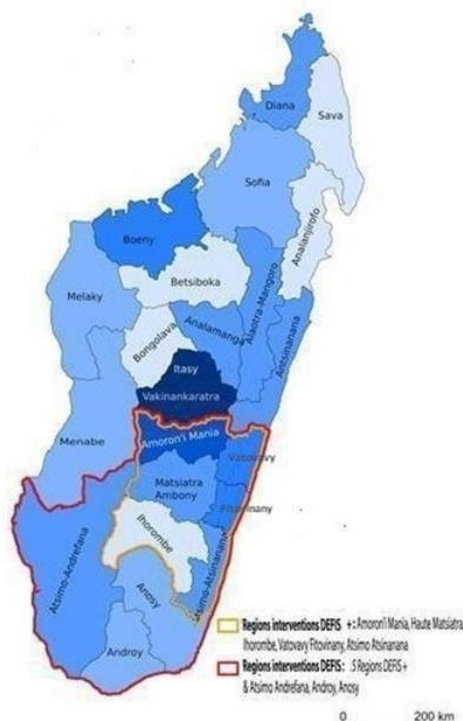
In addition to complementing on-going projects and programmes, DEFIS+ is perfectly aligned with the GCF Sectoral Guides, as summarised in Table 4.

Table 4: DEFIS+ alignment with GCF Sectoral Guides

Sector	Pathway	Interventions	Description
Climate, agriculture and food security	Pathway 1: Promoting climate resilient agroecology	Improved climate-resilient crop varieties, crop diversification (including promotion of foods with high nutritional value such as Non-Timber Forest Products and Neglected and Underutilised climate-tolerant food species), improving land and water management, and appropriate financial mechanisms.	DEFIS+ investments respond to the main climate hazards of droughts and floods and the specific risks they pose to agricultural production, while building more resilient communities through improved farming systems and practices. DEFIS+ activities focus on climate change adaptation while also ensuring productivity and diversification gains for the farms and production value chain actors, and food and nutrition security for the populations. DEFIS+ production technologies and practices are therefore climate-resilient, and focus on low-emission agriculture, through agroforestry practices and the use of renewable energy.
	Pathway 2: Facilitating climate informed advisory and risk management services	Information about daily weather, future climate risks, how to face them, and what risk management services are available.	DEFIS+ includes an expansion of the agro-meteorological network to improve the collection of meteorological data and climate information. The regional officers will also receive training to interpret the data to inform the seasonal planning in a timely manner to help smallholder farmers plan and manage natural resources.
Climate hazards and water security	Pathway 2: Strengthening integrated water resource management and predict climate hazards	Preserving existing water resources, identifying new sources of water supply, and protecting communities from water related climate hazards.	DEFIS+ investments include the protection of irrigation schemes against climate change impacts, sustainable sub-watersheds management and landscape restoration, strengthening the capacities of Water Users Associations and the construction of REEP tanks for improved water availability for both crops and small livestock.
Ecosystems	Pathway 1: Carbon sequestration: ecosystem-based management of	Landscape-level protection, and restoration to maintain	DEFIS+ investments in landscape restoration, CSA, and agroforestry will contribute to maintaining resilient and healthy ecosystems that will deliver a

	terrestrial and freshwater ecosystems	the ability to store and sequester carbon.	range of services that will contribute to adaptation, carbon sequestration, sustainable livelihoods for the smallholders, and other benefits for biodiversity, local communities, and wider society in the targeted areas.
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Figure 16: Regional graphical interconnection of DEFIS+ with the IFAD programme (DEFIS)



Objectives of DEFIS+. DEFIS+ will be implemented over six years in six regions (Ihorombe, Haute Matsiatra, Amoron'i Mania, Vatovavy, Fitovinany and Atsimo Atsinanana— Figure 16, Figure 4A). In targeting the beneficiaries, DEFIS+ will also adopt a participatory approach that will make it possible to involve local authorities and representatives of Farmer Organisation (FO) beneficiaries in the identification and validation of direct and indirect beneficiaries. The average income of farmers (households per year) by region is presented in the below table, and in general, it is low compared with other categories in the same regions.

Region	Amoron'i Mania	Atsimo Atsinanana	Upper Matsiatra	Ihorombe	Vatovavy and Fitovinany
Average income of farmers, household per year (Ariary, Ar)	3,017,924 Ar	2,424,177 Ar	3,508,226 Ar	3,138,589 Ar	4,295,280 Ar
	666.70 US\$	535.54 US\$	775.02 US\$	693.36 US\$	948.89 US\$
(United States Dollar, US\$)	exch. 1US\$=	4 526,65 Ar			

Currently, farmers sell very little produce due to i) extreme climate hazards that destroy large areas of crops every year; ii) low production levels; iii) low access to markets due to poor roads; iv) crop diseases (i.e., insects, viruses and infections); v) poor weather forecast information and, vi) inaccurate climate change predictions. The low production levels also limit household food consumption levels, resulting in food and nutrition insecurity. Self-targeting will make it possible to define and validate with the local populations the eligibility criteria based on the level of poverty, climate vulnerability, access to resources and markets, food insecurity and malnutrition status. Thus, it will facilitate the identification of poor and vulnerable beneficiaries within the communities/villages and/or the area with their participation and that of the FOs.

Geographically, the DEFIS+ project will intervene in areas with a relatively high concentration of poor and extremely poor people, where climate hazards are not predicted on time, food insecurity, infection diseases, and malnutrition prevalence are high, and the negative effects of climate change affect the productive capital of the populations. Thus, specific actions proposed for women and young people in vulnerable situations will facilitate their access to knowledge, productive capital, income-generating activities and green jobs.

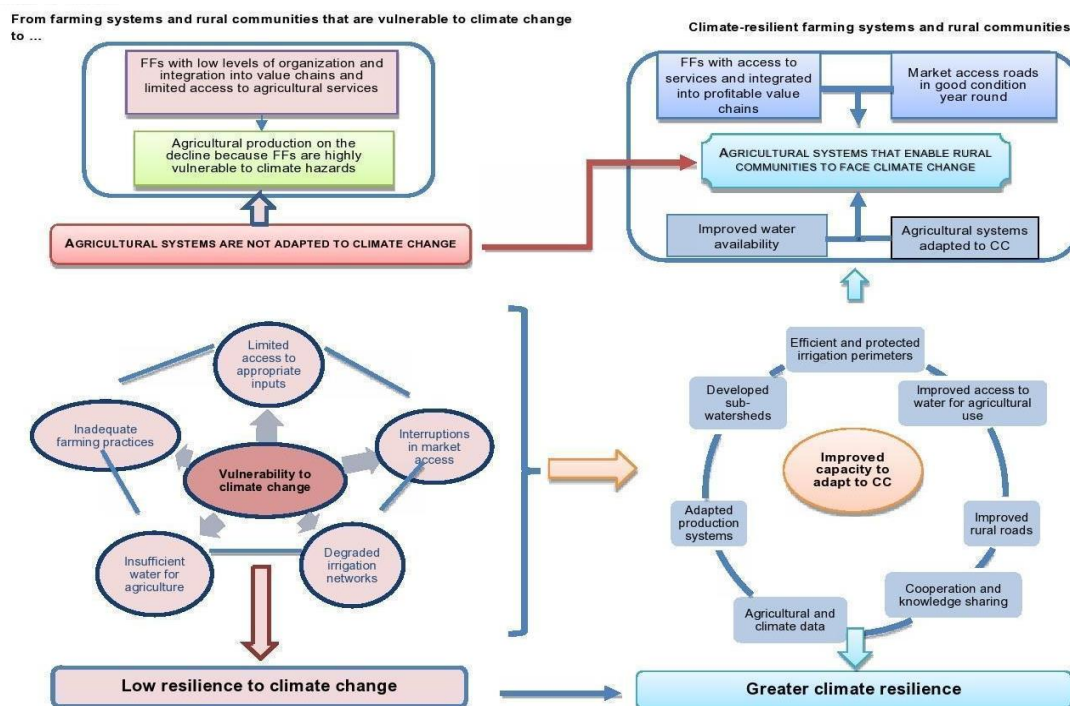
Expected impacts and outcomes of DEFIS+. The expected impacts of DEFIS+ are: (i) increased resilience to climate-related shocks and food and nutrition insecurity and improved livelihoods of family farmers; (ii) improved resilience of ecosystems and ecosystem services; (iii) increased resilience of basic rural infrastructure to the effects of climate change; and (iv) reduced GHG emissions from forestry and land use.

The expected outcomes are:

- Strengthened climate resilience of agricultural production systems and increased carbon sequestration through: (i) the adoption of energy and water efficient technological innovations and climate smart practices that reduce GHG emissions, (ii) improved and secure water availability, including the protection of sub-watersheds from climate shocks and the restoration of landscapes; and (ii) enhanced agro-climatic information system.
- Increased income generation through improved market accessibility through (i) enhanced robustness of roads; (ii) training of trainers on new construction standards for small-scale, climate-resilient rural infrastructure for local communities; and (iii) runoff water collection systems set up to harness water resources collected along the roadways.
- Improved food and nutrition security, through scaling up innovations on climate change adaptation, knowledge management and the exchange of best agricultural practices.

The outcomes, outputs and activities of DEFIS+ are extensively discussed below, and a summary of the project approach is depicted in Figure 17.

Figure 17: Project's approach



Outcome 1: Strengthened climate resilience of agricultural production systems

DEFIS+ will place emphasis on the dissemination and adoption of climate smart agronomic practices, stress tolerant varieties, species and fodders, as well as additional investments needed to improve water use efficiency, and the resilience of basic infrastructure to climate shocks. The dissemination of improved climate-friendly agricultural and livestock management practices as well as energy efficiency tools will be supported through Farmer Field Schools (FFS), which allows for a long-term paradigm shift towards sustainable agriculture and green energy solutions. In addition, through the FFS, the project promotes the transfer of skills in improved cropping practices, integrated pest management, and the use of sustainable agricultural inputs.

Output 1.1: Improved water management for sustainable agriculture

In order to sustain agricultural production, certain measures are a priority. Investment in and development of water resource management systems (irrigation, drainage and retention) is a major issue. However, it is also necessary to overcome the pressures linked to competing uses, through an integrated and sustainable management of these resources. The objectives are to reduce climate change vulnerability in the targeted watersheds by strengthening the agricultural and livestock sectors on the basis of an Integrated Water Resources Management (IWRM) approach and improving governance in order to protect and enhance the natural resources within the limits of each watershed or sub-watershed. To achieve the IWRM foundation, water cycle management analyses will be carried out at the watershed level and will make it possible to indicate or specify the main issues linked to water resources, both in terms of quality and quantity. Thus, the analysis will identify the problems and their causes within the sub-watersheds, inform the IWRM system to be implemented and determine the scope of activities in priority areas to enhance water preservation mechanisms.

The activities planned under Output 1.1, which will be undertaken using the IWRM, are the following:

Activity 1.1.1. Protection of irrigation schemes against climate change impacts

A large majority of existing irrigation schemes in different regions of Madagascar and in the DEFIS+ regions were built by the rural engineering department or by other projects in the 1960s. The infrastructure is old and needs to be rehabilitated and reinforced because of the evolution of water and soil problems linked to climate change (floods, erosion, cyclones, decrease in water resources during the dry season, etc.). GCF funds will therefore allow the financing of the rehabilitation of irrigated perimeters and the construction of new ones with improved drainage systems. Thus, the regeneration of the irrigation schemes and any other activity that enables further water extraction will be carefully designed once the water balance is analysed without further harm to communities and exacerbation of climate issues related to water scarcity. This will happen in parallel with IWRM system implementation. Furthermore, the rehabilitation/construction of irrigation schemes will contribute to increased production and diversification (including the production of nutrient-dense foods such as cereals and legumes) to improve the food and nutrition security at a household and community level with particular benefit to women and young children. This activity will be complemented by Activity 1.1.3 (Strengthening capacities of Water Users Associations (WUAs)) to ensure the sustainability of the investments. Special attention will be given to ensuring and increasing women's access to irrigated plots through engagement with traditional authorities and community engagement to provide allocation quotas for women and youth and avoid potential backlash. In addition, a comprehensive water cycle management analysis will be conducted to identify the main issues linked to water resources, both in terms of quality and quantity and in relation to implications for women and men, in the DEFIS+ regions. WUAs will be provided with small start-up and spare equipment to maintain these rehabilitated irrigated perimeters. However, complex construction will be conducted by contractors or procured by construction parties. Thus, procurement is carried out by the person in charge of procurement at the Ministry of Agriculture and Livestock, with the support of the DEFIS+ Procurement Manager.

Activity 1.1.2. Sustainable sub-watersheds (adjacent to irrigated schemes) management and restoration of landscape

This activity entails watershed development plans by the communities. The climate adaptation plans will include priority landscape restoration, ecosystems restoration, natural water purification, rainwater harvesting aimed at reducing workloads for women and capacity building to promote non-regret solutions. This will be complemented with the adoption of biological (planting herbaceous or shrub species, along curves or perpendicular to prevailing winds) and structural (benches, terraces, techniques to prevent gullies and lavakas) measures to protect sub-watersheds near irrigated perimeters from wind and water erosion, improve vegetation cover, improve water infiltration and reduce erosion. The plans will apply IWRM principles, bringing together the different actors for joint planning. As a result, the technical departments of MINAE, the Ministry of the Environment and Sustainable Development, the project's implementing partners, and representatives of the beneficiaries, farmer" organizations, the chamber of agriculture, and the technical departments concerned will hold technical workshops to validate the adaptation plans.

Activity 1.1.3. Strengthening the capacities of Water Users Associations to manage water and apply sustainable water management practices.

WUA members and local technicians (including building inspectors, engineers, architects and designers) will be trained on the Hydro-agricultural Infrastructures Against Floods and Inundations (NIHYCRI), climate change risk management and climate change adaptation practices. The NIHYCRI-related training builds capacity on the installation, operation and maintenance of climate change adaptation technologies, the implementation of early warning action protocols and operational procedures for long-term monitoring. DEFIS+ will ensure the application of these standards to small- and large-scale water infrastructure through capacity building to be financed by the private sector and the Government and provided by specialised technical institutes in Madagascar. Special efforts will be made to strengthen women's participation in WUAs and their representation in leadership positions (with a quota of at least 30%) and the development of water resource preservation and management plans.

Activity 1.1.4. Construction of climate smart water tanks for drip-irrigation and for small livestock.

This climate adaptation technique will consist of building a surface smart climate water storage system to irrigate the terraces and reduce runoff, along with the strengthening of design offices and construction companies in terms of NIHYCRI, the protection of the irrigated perimeters and the development of anti-erosion devices on the watersheds.

Climate smart water tanks can either be conventional water storage tanks (gravity water tanks) or buried sand-filled water tanks (REEPs)⁵⁸. The storage capacity of the tanks is estimated between 50m³ and 250m³ or even more, depending on the availability of water resources (rainfall). Other than REEPs, there is also the ISAA or Giant Impluvium for gravity collection and storage of large amounts of rainwater for agriculture, which consists of a rainwater harvesting system via road gutters to be connected via runoff channels dug in hillside and transport them to large reservoirs (hill reservoirs) sealed and equipped with a drainage system to distribute water to crops, animals, and/or villages. The two rainwater harvesting techniques represent a modern climate change adaptation tool, saving time and labour often associated with water collection especially by women. It is labour intensive and necessitates tools and techniques that are mastered by Malagasy farmers and materials that are easily available in Madagascar, without the need for importing material from abroad, drilling, pumping, electricity generators or sophisticated engineering equipment. This technique has a very limited climate and environmental footprint. This activity will include WASH trainings on hand washing and hygiene targeting households.

The **expected results of Output 1.1** are: (i) conducting comprehensive water cycle management analyses; (ii) improved climate resilience of 15 000 hectares of irrigated schemes; (iii) anti-erosion measures are implemented on 8 100 hectares of sub-watersheds that drain into irrigated perimeters, of which 5 500 hectares of landscape are restored by women and youth, and 900 technicians are trained; (iv) 1 400 WUA members and technicians are trained on climate change adaptation, water resource preservation and management plans; (v) 150 units of climate smart water tanks are installed; and (vi) 268 321 farmers, including women and young people, are engaged in the effective management of sub-watersheds and irrigated perimeters.

Output 1.2: Enhanced agro-climatic information systems, new technologies and initiatives

The activities planned under this Output are the following:

Activity 1.2.1. Installation of automated agro-meteorological stations and development of flood and drought monitoring and forecasting system

DEFIS+ will support the upgrading of agrometeorological information systems by strengthening the network for collecting and processing agro-meteorological data in target areas, which is part of the national network. The project will support the collection of accurate climate information (temperature, precipitation, wind, humidity, etc.) and promote the expansion of the existing national network through the acquisition and installation of automated agrometeorological stations. The stations will be located in the DEFIS+ project area to fill the gap in the existing network of stations. The national hydromet agency will be the implementing partner and will be responsible for the operation and maintenance of the stations. Comprehensive collection of meteorological data and climate information and its correct interpretation will ensure that detailed weather, climate, and agricultural hydrological forecasts will support precise short- and long-term climate projections, predict extreme climate events long in advance, manage water resources, crop modelling and seasonal planning in a timely manner to help smallholder farmers plan. Satellite data will be assessed together with hydromet variables, drought indices and climate risk indices for a drought early warning system including the use of 1- to 10-day flood forecasting for better accuracy.

The Directorate General of Meteorology already provides some climate services and makes use of different platforms to disseminate information, but many of them are out-of-date. These services can be further upgraded and enhanced with the help of the project. A state-of-the-art flood and drought monitoring system will be developed for focused and tailored climate services for hydrological hazards. The system will be driven using precipitation from the scientifically validated Multi-Source Weighted-Ensemble Precipitation (MSWEP) dataset, which merges stations, statistical models, and satellite information using a sophisticated algorithm. Based on the system, the following products will be promoted:

- Customised rainfall and drought information on a daily/seasonal/longer term basis linked to early warning systems for management actions and decision making based on the climate update and monthly climate summary is sent to all agriculture and livestock personnel by email;
- An agricultural newsletter is published quarterly and includes information on seasonal weather outlooks (accessible online);

⁵⁸ REEPS is a technology combining the collection, storage and mobilisation of water that the ENERGIS FD Company developed and patented in 2011 to solve the problems of drying up and evaporation of water collected in the water table and on the surface. It is based on a biological retro filtration in the surface and is a system for storing and reusing rainwater at a lower cost while rigorously respecting the ecological aspect.

- Special brochures (visual illustrations) are published (e.g., on mulching techniques, agroforestry, etc.) that are specifically designed to help farmers in times of climatic stress (e.g., drought) with translation to local languages;
- An ICT system will be established and will operate for the development of new advisories and warnings for agriculture and water management and their dissemination through the media, including SMS and radio.
- DEFIS+ will offer a diverse range of options for accessing the information to accommodate the specific needs of women, and for this reason, a gender needs assessment will be carried out with the aim of providing gender-specific information.

Activity 1.2.2. Capacity building of decentralised government (SRM & DRAEP)

The project will also provide support at the regional level to the staff of the Regional Meteorological Services (SRM) and Regional Directorates for Agriculture, Livestock and Fisheries (DRAEP) through capacity development for the analysis, communication and dissemination of agrometeorological and climate information. National institutions and relevant civil society organisations will be trained and mobilised to better inform farmers and their rural organisations and build their capacity for the integration of climate risk into their decision-making processes and good farming practices. In these trainings, specific attention will be given to the gendered dimension of access and preferences regarding climate information.

Training will be provided to SRM staff for data collection and information generation to support the use of new technologies such as the flood and drought monitoring system, GIS, and the development of new products, including reduced and additional timescales for weather forecasting to produce information adapted to end users.

Popularization of climate information will be through local agents, agents of the Regional Directorate of Agriculture, technicians, FFS, agents of the SRM and the use of mass media, local radio and fairs. Farmers will have access to localised and timely weather services and long-term weather forecasts and will receive training on how to manage weather risks and better plan their farming and livelihood activities.

To overcome obstacles to local livelihood planning, the project will also establish new partnerships to strengthen the producer-user interface, including engaging with NGOs to establish sustainable and cost-effective service transfer modes. This will also build user-level capacity for the interpretation of climate data and advice.

Activity 1.2.3. Promotion and dissemination of Meteorological and Climate Smart Agriculture practices

DEFIS+ will promote the dissemination of meteorological and CSA information to strengthen the climate resilience of dryland farming systems and rainfed upland farming systems (with respect to irrigated areas) through the promotion of climate resilient agricultural production, including afforestation, mixed cropping systems and improved technologies such as adapted varieties, planting materials and water harvesting.

DEFIS+ will rely on the local network of seed multiplier farmers and seed producing sites. The multiplication and dissemination of locally climate adapted seeds and planting materials will be carried out using community-based participatory approaches (e.g., FFS), integrated with meteorological information and disseminated among the farmers and stakeholders. Learners at the FFS level will not only be supported in resilient agricultural techniques but will also be provided with seeds for the first production season. The seed production centres exist in the DEFIS+ regions, but their skills will be improved as will the materials and equipment for the production of basic seeds. Thus, DEFIS+ will support the Agnarafaly Seed Production Center (CPSA) and Kianjavato Center in the multiplication and promotion of certified or quality declared seeds.

In addition, agroforestry and landscape management systems will be promoted for the improvement of micro-soil bunds, trenches and pits to prevent erosion, as well as several agroforestry and landscape operations at the local level. Water pumping will be done using solar pumps to ensure reduced diesel and gasoline use (full scale up and nationwide adoption). The summary of the CSA activities and locations is available in Table 5.2.

Activity 1.2.4. Promotion and dissemination of sustainable home technologies and social initiatives

DEFIS+ will promote low-cost on- and off-farm modern technologies, such as improved cooking stoves and handmade briquettes, by using local materials, thus reducing deforestation. Other technologies include solar pumps for irrigation, biogas production along with livestock systems, solar milling and dryers for product processing. These technologies will contribute to improving efficiency and production while also reducing

women's time burden and workload (thereby impacting their health and nutrition status and indirectly those of their young children). DEFIS+ will promote these technologies and invest in the training of trainers for the use of the technologies and youth associations and groups for the maintenance of the technologies. The scale of the technologies will be assessed through a feasibility study and an assessment of the adoption of the technologies as well as their supply chains. The technologies will be both household/farm level and multi-household/farm level and will be procured through the DEFIS+ procurement team and local municipalities.

Some of these tools and technologies (e.g., cooking stoves) will be provided to vulnerable households for demonstration and catalytic purposes i.e., generating demand. In IFAD's experience working with the most vulnerable and poor populations in rural areas, transfer of goods is necessary in situations where beneficiaries do not have sufficient wealth and liquidity to purchase productive assets. The free distribution targeted at the most vulnerable households is important to enable them to improve their production systems and also demonstrate the effectiveness of the technologies to promote adoption by other community members who can then invest in the technologies. Doing so, they will be able to emerge from poverty and perhaps support other businesses and move other people out of poverty. Without external support, the change in their economic conditions is less likely. Well targeted support to the most vulnerable households will not distort markets because these individuals, at the bottom of the wealth pyramid, are not able to access markets with their own resources today. The project is expected to contribute to move them out of poverty so that they eventually become buyers themselves and consequently stimulate the growth of the market. DEFIS+ will also engage youth groups and associations to participate as entrepreneurs in the supply chain of the technologies once the demand is generated working through other IFAD financed projects such as FORMAPROD/PROGRES.

The **expected results of Output 1.2** are: (i) six automated agro-meteorological stations are installed and operational; (ii) a flood and drought monitoring and forecasting system is developed and operationalized; (iii) 24 DRAEP and SRM teams are trained and involved in processing and disseminating agro-meteorological data; (iv) 178 881 farmers will use agro-meteorological and CSA information during the agricultural production seasons; (v) 268 321 farmers use low-cost and modern technologies; (vi) support is provided to CPSA for the production of basic seeds of stress-tolerant varieties; (vii) at least three new methods of integrated pest management techniques adapted to the context of smallholder farms in the project area are promoted and resilient fodder production is integrated on 1 500 ha; and (viii) CSA, afforestation and agroforestry approaches are implemented on 10 000 hectares of land.

Outcome 2: Increased income generation through improved market accessibility

In Madagascar, floods and road submersion by runoff are frequent, especially in the DEFIS+ area. Besides, most of the communal roads are unpaved, so they are practically impassable in the rainy season. There are also many crossings given the important hydrographic network of the area, but their condition often handicaps the accessibility of the villages and the outlets for production. This makes it difficult to get products to market. Extreme climatic events, such as floods and cyclones, in combination with poor quality and conventionally designed roads, often cause erosion and flooding in the surrounding environment or the road embankments, which result in losses of agricultural land and products. DEFIS+ will, therefore, invest in: (i) harvesting water for agricultural purposes using rural roads while improving their resilience; and (ii) implementation of a wide variety of GR4W techniques to secure the transport function of roads year-round, protect the environment and agricultural activities around rural roads.

According to Madagascar's NDC and the National Adaptation Plan, the adoption of climate resilient infrastructure is among the sectoral priorities for adapting to climate change. In fact, to link small producers to the market, DEFIS+ will strengthen the climate resilience of rural roads, in particular with regard to rainwater, which is the main factor of damage and the main enemy of rural roads infrastructure. Other support and technology transfer already implemented in other countries under the GR4W approach will be useful for the implementation of the climate resilience-building activities for rural road infrastructure under DEFIS+. Also in this case, there might be some unexpected negative externalities resulting from road upgrading, such as GHG emissions although in the case of Madagascar given the low levels of road infrastructure development the overall impact will be low.

The GR4W techniques that will be used in the DEFIS+ project will be drawn from the recently published World Bank Guideline on GR4W^{59,60}, which documents a wide variety of strategies and techniques on how to use roads as instruments for beneficial water management and climate resilience in different geographies. According to the GR4W Guidelines, the opportunities are abundant, and an extensive analysis is available in the infrastructure feasibility study (Annex 2c).

The rehabilitation of rural roads will improve access to markets, a priority identified in Madagascar's national development plan. It will lead to an increase in traffic and therefore an increase in GHG emissions. However, this increase in emissions, which is linked to the implementation of the National Development Plan and efforts to increase the resilience of the most vulnerable population, is considered marginal. It will not contribute significantly to national GHG emissions and does not contradict the country's NDC or the Paris Agreement, which recognises that the fight against climate change is always linked to national development and food security. Madagascar accounts for just over 0.10% of global GHG emissions.

Output 2.1: Reinforced rural access roads

To minimise damage, water will be systematically directed to the land rather than running along the road surface. To direct water to the land, different design parameters will be considered, such as regular slope reversal, avoidance of sunken roads and inclusion of basic drainage. In addition, a wide variety of techniques will be used to preserve unpaved roads, enable water harvesting and reduce erosion including water bars, rolling dips, infiltration bunds and vegetative measures among others. More information is available in Annex 2c.

The activities planned under this Output are the following:

Activity 2.1.1. Reinforcement of unpaved rural access roads to reduce climate change impacts and water collection along these roadways for agricultural use

In the context of DEFIS+, the following GR4W techniques will be explored while rehabilitation works are performed through structured labour-intensive work:

- Using road-surface cross drainage systems: a series of rolling dips (small depressions with a small bump) or water bars (small, slanted humps) to divert water from the road surface to the land for productive use. These structures have two purposes: (i) prevent water from flowing beside the road surface by diverting it from the road, (ii) through the diversions water is conveyed to an area where it can be beneficially deployed, e.g., recharge of groundwater, feeding a spring or directly flowing into an on-farm pond to be used for irrigation.
- Using infiltration bunds to slow down the side erosion from the roads and promote recharge of aquifers. This is a typical issue on many unpaved roads where road runoff builds up, causing erosion and sedimentation. It is best combined with vegetative protection to reduce the erosive effect of runoff from the road template, in particular if no side drain is provided and the road template is out sloped or crowned.
- Monitoring groundwater recharge percentages will employ borehole dippers, also known as Water Level Dip Meters. DEFIS+ will use Water Level Dip Meters to determine the water level within a borehole, piezometer pipe or sump.⁶¹ It should be noted that the potential concomitant impacts, such as extension of flood duration and reduction of peak discharge with the proposed structures for underground water systems, are well captured in the DEFIS+ ESMP's section related to the construction of water tanks – including measures of mitigation and indicators of achievement/sources of verification.
- Complementing a range of bioengineering methods to revitalise vegetation in the road adjacency. Bioengineering is the use of plants for slope stabilisation and runoff control. It involves using plant parts such as roots, cuttings and stems as a cost-effective and locally adaptable means of erosion control. It ranges from planting deep-rooted species to a combination of vegetation with civil engineering structures. It helps not only to preserve the slopes, in addition, it buffers the water through higher infiltration capacity.
- Routine maintenance, periodic maintenance and rehabilitation/reconstruction works will need to be systematised and applied along the construction and rehabilitation of unpaved rural roads. DEFIS+ will give special attention to ensuring that women are involved in maintenance and rehabilitation/reconstruction works.

⁵⁹ Van Steenberghe, F., Arroyo-Arroyo, F., Rao, K., Hulluka, T. A., Woldearegay, K., & Deligianni, A. (2021). Green Roads for Water: Guidelines for Road Infrastructure in Support of Water Management and Climate Resilience.

⁶⁰ This tool can be found on p. 86 of MetaMeta's publication "Guided Learning on Roads Water Harvesting".

⁶¹ Geotechnical Centre. (2019). Water Level (dip) Meter <https://25489-813672-raikfcquaxqncofqfm.stackpathdns.com/wp-content/uploads/2018/08/Water-Level-Dip-Meter-V1.1.pdf>

The **expected results of Output 2.1** are: 540 km of unpaved rural access roads are rehabilitated or improved with climate-resilient planning, including cross-drainage systems, and the project contributes to recharging aquifers. This also includes maintenance services for the entire duration of the project.

For Activity 2.1.1, IFAD as AE will ensure the EE conducts, prior to undertaking any construction works, comprehensive technical assessments and feasibility studies for each of the sub-projects identified and this shall be done prior to launching of the design and construction works. The designs to be implemented for each of the sub-projects shall be informed by the output and results from these feasibility studies.

The feasibility studies to be conducted encompass analysing the baseline situation, identifying location-specific climate risks and devising appropriate measures for climate proofing the infrastructure under consideration against the identified climate hazards and integrating the climate resilience measure into the designs. The methodology involves a systematic assessment utilizing data, local knowledge, stakeholder engagement, and the development of the appropriate adaptation approach. The expected deliverables include a detailed technical and feasibility report, climate-proofing design approaches, and the recommendations for integrating the resilience measures into the construction to be submitted to the GCF'. This also applies to Activities 1.1.1, 1.1.2 and 1.1.4.

IFAD, in collaboration with the EE, has clearly defined criteria that were used in the selection/prioritization of the roads and irrigation schemes to be rehabilitated/retrofitted under this project. Hence, if a particular sub-project that has been earmarked and listed in the FP could no longer be implemented due to changes in the site condition during implementation, an eligible replacement would be identified by applying the same criteria. The selection of sub-projects is based on the identification and prioritization of projects using specific criteria aimed at maximizing their impact on adaptation and resilience building in the regions in Madagascar considered.

Output 2.2: Improved capacity to build and maintain rural infrastructure

DEFIS+ will equip technical staff with the basic design, construction, operation and maintenance of Green Roads, well-tuned to the local conditions, so as to make rural roads and the surrounding environments more resilient to climate change.

The activities planned under this Output are the following:

Activity 2.2.1. Training on construction standards related to climate change

The Malagasy Standards for the Construction of Road Infrastructure Resistant to Floods and Geological Phenomena, developed in 2020, are based on the Guide for Road Protection Against Floods in Madagascar (GPRCIM). The Standards will be part of the training that will be provided to key municipal staff as well as several other construction areas. Thus, they will be trained on the operational procedures for the long-term maintenance and the scaling of the technologies of climate-resilient roads and of water collection according to the operation and maintenance plans and road construction standards. The training will include the design and implementation of a long-term monitoring framework for road-specific climate proofing measures.

In accordance with the categories of personnel working in the municipalities, one recommendation would consist of designating the Mayor as the first person in charge in matters of Authority capable of enforcing these standards, s/he must be assisted directly by the technicians of the Town hall who would be trained in these standards and will have to ensure the relay between the municipality and the road contractor.

The **expected result of Output 2.2** is: 178 881 regional and local technicians are trained on road construction standards related to climate change, such as the Flood Protection Standards. Half of them are women and vulnerable people.

Outcome 3: Improved food and nutrition security

To promote greater uptake and ensure that food and nutrition security will be long-lasting, knowledge products on best practices will be elaborated and disseminated under DEFIS+.

Output 3.1: Knowledge on climate change and food security generated and shared

The activities planned under this Output are the following:

Activity 3.1.1. Training rural development actors on climate change and food production systems for enhanced nutrition

Training smallholder farmers will build their capacities, especially those of women, to improve agricultural resilience and productivity by effectively identifying their needs, managing rainwater and soil moisture and reducing or potentially eliminating water stress during periods of short-term drought. The innovative aspects of DEFIS+ are linked to the adopted methodology. The FFS methodologies will create a tiered training system that will draw on farmer experience and institutional agronomic science to ensure the adoption of climate smart farming practices. Training of trainers and farmer-to-farmer training will ensure peer support, exchange of experiences based on on-farm practices (for example, conservation agriculture and drip-irrigation) and increased institutional experience and capacities for agricultural extension. Rain-fed farmers will be able to manage soil, water and biomass in order to maximise water availability through on-site or off-site water collection and improve water productivity through the use of water-saving technologies. To complement the training on climate change, training on cross-cutting topics, such as nutrition and gender empowerment in the context of climate change will be organised and integrated into the FFS.

DEFIS+ will support each FFS (approximately 25 people per training course) through: (i) the installation of a plot of 0.5 ha, made available by a producer/member of a CBO or community within the development pole (with priority given to women); and (ii) the application of technical itineraries and new practices at the level of individual farmer's plots. The main beneficiaries of the FFS will primarily be young people (male and female) who will use their plots for training, facilitating demonstrations, exchanges and reporting results. The FFS organisation must therefore be flexible to allow women to participate actively. DEFIS+ will use the FFS structures already established under DEFIS and integrate the climate change resilience building practices and technologies.

The FFS approach is intended to support the strengthening of farmer knowledge through field observations. The FFS approach is an effective peer to peer learning system that enables farmers to identify challenges they face in their agricultural production activities and identify solutions in a collaborative facilitated manner. Among other things, the project will prepare learning materials for the FFS, train farmers in climate change adaptation practices for sustainable production, provide tools and training to enable them to adopt more efficient soil, water and input management practices, and train farmers in agricultural techniques such as the Sustainable Intensive/Improved Rice Cultivation (SRI/SRA) systems.

Activity 3.1.2. Exchange visits on climate resilient food systems

To promote the exchange of knowledge and best practices related to climate resilient farming practices, DEFIS+ will support exchange visits and study tours at the community and local levels. Specific measures will be taken to ensure the involvement of women in these exchange visits (e.g., using a participant quota of at least 50%; organising women-only exchange visits and planning visits to successful female producers).

The participants will be identified in a participatory, transparent, and consensual process involving the beneficiary communities, representatives of local and regional farmer organizations, strategic partners (Government and State representatives) and DEFIS+'s team. Eligibility criteria based, on age, gender, degree of vulnerability, level of schooling, socio-economic status, etc., will be defined. Thus, the approach will ensure that target groups, especially the most vulnerable, are the beneficiaries of DEFIS+ support.

Activity 3.1.3. Capitalization of best practices and knowledge on food and nutrition security measures

Local skills and knowledge will be leveraged, for instance, through the development of construction operations manuals, in order to scale up innovations. DEFIS+ will also support knowledge capturing studies on the climate adaptation and mitigation options and strategies developed by smallholder farmers, by developing and proposing the design and composition of a package of materials, and communication formats, e.g., Documentary Clip Abécédaire game, educational games, radio broadcasts, technical data sheet and voice messages. These knowledge products will pay special attention to gender dimensions for access and use.

As part of this capitalization process, DEFIS+ will consolidate the efforts made by programs financed by the World Bank to enrich and improve the database on LAUREL (Land use planning for enhanced resilience of landscape) for assessing, simulating, evaluating and re-orienting land use and land use change processes and to develop a land degradation baseline and a Prototype Land Use Change Simulation Platform (LANDSIM-P). The platform is essential for evaluating changes in land use and forms of degradation or restoration in order to make decisions on the orientations to be taken on the watershed within the framework of territorial planning.

It should be noted that DEFIS+ has already arranged and discussed the specific requirements and procedures for obtaining licenses, permits, and authorizations for access to private/public land with all stakeholders and will provide, on a case-by-case basis, the necessary licenses, authorizations and permits. For example, in the case of private property, access to land rights requires either a land title issued by the Ministry of Land Management and Land Services (MATSF), or a land certificate issued by decentralized authorities (communes) in areas where there are Land Offices. This issue and all its details are extensively discussed in Annex 9.

The **expected results of Output 3.1** are: (i) 447 201 farmers are trained on farming practices adapted to climate change, mitigation, use of stress-tolerant seed varieties and materials through FFS, as well as nutrition (including healthy diets) and gender; (ii) 30 exchange visits are held to disseminate good practices on climate change adaptation; (iii) land degradation baseline is enriched (updated with assessment of current levels of degradation and regional specificity); and (iv) at least 20 thematic knowledge products are published on best practices for climate change adaptation.

The activities to be financed by each co-financier are summarised in Table 5. Also, the table includes the implementing partners for the GCF financed activities.

Table 5: Summary of the activities split between GCF and IFAD financing

GCF Financed Activities	IFAD Financed Activities
Outcome 1. Strengthened climate resilience of agricultural production systems	
1. Protection of irrigation schemes against climate change impacts (PCU/Service providers/ Contractors) 2. Sustainable sub-watersheds management and restoration of landscapes (PCU/ Service Providers/WUAs/ FOs/ Communities) 3. Strengthening consultants and construction companies' capacities to apply improved standards (PCU/ILO/Technical Institutes/ Government Agency) 4. Strengthening local communities' capacities in sustainable integrated watershed management (PCU/Service Providers/ FOs) 5. Training/exchange visits/tours for Water Users Associations members in water use efficiency (PCU/WUA) 6. Installation of automated agro-meteorological stations (Meteorological Department/CPGU (Emergency management and prevention unit)/PCU) 7. Capacity building of decentralised government agencies (SRM&DRAEP/ PCU/ Technical Institutions) 8. Promoting new (CSA and renewable energy) technologies: (PCU/ Service Providers/ FOs/ Communities) 9. Construction of water tank for drip-irrigation and development of sustainable mitigation technology (PCU/Service Providers/ Contractors/ FOs)	1. Rehabilitation of irrigation systems 2. Expansion of existing irrigation systems 3. Development new irrigated perimeters 4. Formation of Water User Association for new schemes 5. Organisational behaviour training for existing Water User Associations. 6 Support youth enterprises in agriculture and livestock management 7. Leadership and functional literacy training for women and youth 8. Improvement of agricultural and climate change statistics 9. Improving access to quality seeds and vegetal material and genetic climate resilient materials in rural communities 10. Establishment of seed multiplication systems for uptake of improved varieties 11. Support youth enterprises in agriculture developing innovative mitigation and carbon sequestration tools

10. Support for the production of basic climate resilient seeds and crop varieties (PCU/ FOFIFA (National Center for Applied Research in Rural Development)/Seed Production Center/ extension Services) 11. Support to the Seed Production Center for the development of crops and plants (PCU/FOFIFA (National Center for Applied Research in Rural Development)/ Seed production Center)	
Outcome 2. Increased income generation through improved market accessibility	
1. Reinforce unpaved rural access roads to reduce the impacts of climate hazards: drainage systems, embankments (PCU/ILO/Service Providers/ Contractors/ Government Agency) 2. Training on construction standards related to climate change (PCU/ILO/Technical Institutes) 3. Training and capacity building on GR4W techniques	1. Development of rural access infrastructures by labour-intensive approach 2. Market studies 3. Capacity building of Farmer Organizations 4. Establishing partnerships among Farmer Organizations and Market Operators 5. Establishment of commodity platforms 6. Improved access to finance
Outcome 3. Improved food and nutrition security	
1. Training rural development actors on climate change (PCU/ CPGU (Emergency management and prevention unit) and other government Agencies/ Service providers) 2. Exchange visits on adaptation of food system to climate change and carbon sequestration (PCU/FOs/ Communities) 3. Capitalization of best practices and knowledge on climate change adaptation measures (PCU/ Service Providers/ FOs) 4. Diffusion of resilient agricultural practices (PCU/ Extension Services/ Service Providers/ FOs), and promotion of NTFP-based enterprises 5. Communication and awareness-raising of beneficiaries on the effects of climate change (PCU/Service Providers (NGOs)/ FOs/ Communities)	1. Local support and nutrition education, with gender and climate change 2. Institutional support to national Farmer Organizations 3. Institutional support to regional Farmer Organizations 4. Organization of Farmer Fields Schools (FFS) 5. Extension of systems for food security and climate change vulnerability

Table 5.1 Summary of crops and prioritisation

Type of crop	Location*	Level of priority**
Cassava	Center	High priority for DEFIS+
Groundnut	Center	High priority for DEFIS+
Rice	All DEFIS+ areas	High priority for DEFIS+
Maize	Center	High priority for DEFIS+
Coffee spp	Centre	High priority for DEFIS+
Robusta coffee	South-East	High priority for DEFIS+
Onion	Center	High priority for DEFIS+

*South-East (Vatovavy, Fitovinany, and Atsimo-Atsinana) Center (Haute Matsiatra, Amoron'i Mania, and Ihorombe).** High, medium, low

Table 5.2. Summary the Climate Smart Agriculture activities and locations

#	Type of activity	Location	Benefits
1	Resilient seeds research and production	Convention Research (Vatovavy, Matsiatra) with centers Haute	- Drought tolerant seeds to stabilise production during times of water stress from droughts thus reducing negative climate impact in farm systems - Increased productivity - Preservation of local varieties - Conservation of genes that can be used in the selection of new stress-resistant traits
	Specialized equipment for seed production	All DEFIS+ regions	- Modern tools to be used for the multiplication and dissemination of seeds of varieties resilient to climate change - Reduction of use of pesticides
2	Agroforestry (including afforestation, restoration, mixed cropping systems)	All DEFIS+ regions	- Enhanced biological diversity - Creation of microclimates in the agricultural production systems - Reduction of soil erosion due to windbreaks and limitation of evapotranspiration - Improvement of carbon sink capacity - Improvement of water storage capacity, soil fertility, biological activity and limiting pest attack - Added economic value due to economic services (timber, wood energy, fruits, etc.)
3	Land management (soil improvement)	Terraces only in Amoron'i Mania, Fianarantsoa, Vatovavy, but all landscape restoration and reforestation activities in all DEFIS+ zones.	- Rehabilitation of the terraces - Protection of soil from erosion, fertility, and water conservation - Rehabilitation of irrigation and drainage systems - Improved yields and crop diversification
4	Solar pumps for agriculture irrigation	All DEFIS+ regions	- Enhance the use of renewable energy in agriculture - Improved sustainability of farm systems - Reduction in GHG emissions - Enhanced productivity
5	Water efficient technology	All DEFIS+ regions	- Reduction and conservation of scarce resources, such as water

			- Improved water management and supply of water in dry seasons and periods of drought - Reduction of soil erosion and mitigation of the risk of floods
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Table 5.1 summarizes the adaptation measures for important food crops by area and priority. Some of the interventions such as the agro-forestry and improved soil and water conservation measures are no-regret climate change adaptation measures that are accessible to and feasible for the farmers to adopt. Furthermore, an extensive discussion is available in Annex 2b on the various climate change adaptation options. Table 5.2 synthesises the CSA benefits for the DEFIS+ regions.

DEFIS+ capitalizes on the lessons learned from past IFAD projects. A summary of such lessons is included below.

- Drip irrigation technology is emerging as a suitable solution in Madagascar dry regions.
- The innovative underground water collection technology (REEP) has proven to be very relevant in the current context of drought, helping to provide access to water. It will be further developed within the framework of DEFIS+ for the collection of surface water at the level of road networks.
- The use of improved rice cropping techniques (SRI/SRA) has proven its success with a remarkable increase in yield and rice productivity in its intervention zones and will be further developed by DEFIS+.
- The implementation of agroecological blocks in the DEFIS regions (not the DEFIS+ regions) has confirmed the effectiveness of this technique in terms of climate change resilience. The first perceived changes concern a better control of the agricultural calendar, the use of quality seeds, a better knowledge of the optimal doses of fertilisers (particularly organic fertilisers), better water management, and the adoption of better pest control techniques. The blocks will be further developed in DEFIS+ regions, taking into account the specific agro-ecological characteristics of each area.
- IFAD has trained champion farmers and livestock breeders from the FFSs who are able to disseminate their skills to other producers. The training sessions for trainers in the FFS and the association of producers into POs have helped to strengthen social cohesion.
- IFAD projects have included a module on climate change in the Gender Action Learning System (GALS) (a community-led empowerment methodology that uses principles of inclusion to improve income, and the food and nutrition security of vulnerable people in a gender-equitable way), which besides improving gender relations is also allowing for the adoption of good practices in adapting to climate change and preserving the environment. The methodology works with households (both females and males) to influence shifts towards positive gender relations and social norms and behaviours.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

The Executing Entity is the Republic of Madagascar (RoM) through its Ministry of Economy and Finance (MEF) and Ministry of Agriculture and Livestock (MINAE). The GCF funds will be received by MEF and the Funded activities will be implemented by MINAE. A legally binding Financing Agreement (Subsidiary Agreement), which will be signed by MEF shall detail the specific roles and responsibilities of the relevant governmental ministries, MEF and MINAE, involved in the implementation of the Project. The Subsidiary Agreement shall establish the terms for the financing and implementation of Outcomes, Outputs and Activities and include the relevant obligations of the Executing Entity in respect of the implementation of the overall Funded Activity, including compliance with the eligibility criteria (as set out in the "Eligibility Criteria") and the Project Implementation Manual. The Executing Entity through MINAE will select the beneficiaries of all Activities of the Project (the "Final Beneficiaries") and procure service providers for specific Project Activities in accordance with the Eligibility Criteria and the PIM. The Executing Entity through the MINAE will enter into contracts for procured services with the selected service providers.

From a legal point of view, MEF will sign the Financing Agreement as the government's representative, but there are cases where it is MINAE that signs the FA as the governor of IFAD. In the case of DEFIS+, MEF will represent the Government of Madagascar in accordance with Decree 2019-093 setting the attributions of the Ministry and its general organization. MEF ensures the planning, coordination and monitoring-evaluation of the country's medium- and long-term development programs and projects at all levels and the coordination of cooperation at the national level. MEF also manages, controls, and evaluates external funds and contributes to the harmonization

of cooperation with donors. Both MEF and MINAE are fully committed to DEFIS+ and ensuring its successful implementation.

The DEFIS+ activities to be funded by the GCF (DEFIS+) will be fully integrated⁶² into the DEFIS program, which is a large agricultural program of the Malagasy Government that has been placed under the technical supervision of MINAE. MINAE has been working with IFAD for 40 years and houses a permanent unit that oversees program and project implementation (CAPFIDA). Owing to an implementation arrangement that rests on country systems at both national and decentralised levels, DEFIS+ interventions will be in synergy with the on-going and planned investments of all technical and financial partners in the areas of climate change, nutrition and gender.

Management, coordination and supervision of DEFIS+ activities

National Steering Committee

A National Steering Committee (NSC) will oversee the activities of DEFIS+ at the national level. The Secretary-General of MINAE will chair the NSC, which will comprise the following members: (i) the General Project Coordinator of MINAE; (ii) a Director in the Ministry in charge of Finance; (iii) a Director in the Ministry in charge of regional planning; (iv) a Director in the Ministry in charge of the Environment; (v) the National Climate Change Officer; (vi) a Director in the Ministry of Public Works; (vii) a representative Director in the Ministry in charge Social Protection and Gender; (viii) The President and three senior representatives of the Chamber of Agriculture; and (ix) the President and one senior representative of the Federation of Chambers of Commerce and Industry. The secretariat will be provided by the Coordinator of the PCU. The NSC will meet twice a year and its main role is to: (i) validate DEFIS+ annual work plans and budget (AWP/Bs); (ii) evaluate progress made on project activities, their results and their alignment with strategies for rural areas and (iii) manage project-related grievances that cannot be resolved at the regional level.

Regional Steering and Monitoring Committee

Regional oversight will be provided by the Regional Steering and Monitoring Committees (RSMCs). The members of the RSMCs are: the Head of the Region; the Directorate of Rural Development; Directors of the Regional Directorate of Agriculture and Livestock and Fisheries and the Inter regional Directorate for the Environment and Sustainable Development (DIREDD); representatives of the Regional Steering and Allocation Committee; the Assembly of Regional Agriculture (RACs); and Regional Farmer Organisations. The Coordinator at the regional level will assume the secretariat of the RSMC. The RSMC will meet twice a year and its main role is to: (i) review DEFIS+ annual work plans and budget (AWP/Bs); (ii) assess progress made on project activities and their results and (iii) manage project-related grievances.

Programme Coordination Unit

The Programme Coordination Unit (PCU) will coordinate and manage DEFIS+ resources and activities in accordance with the financing agreement, project manuals (operations, financial management and administration procedures, monitoring and evaluation) and AWP/Bs, under the supervision of the RSMC, the NSC and IFAD.

The PCU will be responsible for: (i) the management of the project's human, financial and physical resources; (ii) the preparation of AWP/Bs and project procurement plans; (iii) the coordination of the implementation of the activities; (iv) the monitoring and evaluation (M&E) of the project's activities and results; (v) the coordination and monitoring of the work/activities of partners and service providers; (vi) reporting and submitting activity and financial reports to MEF, MINAE and IFAD on a regular basis (minimum twice a year in the case of IFAD, so as to fulfil IFAD's semi-annual reporting obligation vis-à-vis GCF); (vii) liaising regularly with regional government authorities and decentralised services and other actors in the DEFIS+ intervention zones; (ix) ensuring the compliance of IFAD's safeguards during project implementation activities (i.e. ESMP, Gender Action Plan).

Planning and coordination of activities at the regional level will be provided by two inter-regional coordination units located in: (i) Fianarantsoa for the Ihorombe, Haute Matsiatra and Amoron'i Mania regions; (ii) Manakara for the Vatovavy, Fitovinany and Atsimo Atsinanana regions.

A participatory approach will be used to identify and prioritise rural roads. Once identified, the feasibility studies and technical designs will be undertaken during the first and second year of project implementation. The feasibility studies will include environment and social impact assessments and the development of site-specific risk management plans. The works on the roads will be undertaken during years two, three and four of project implementation to allow for the evaluation of use and benefits in the last two years of project implementation. The sequencing of studies and works will also apply to the irrigation infrastructure to be rehabilitated. The execution will include beneficiaries' contributions through materials and labour, control and supervision by the enterprise

⁶² The full integration refers to activities being incorporated (timing, sequencing, approach) in the implementation to ensure the results and overall objectives outlined in DEFIS are achieved in a cost effective and efficient manner.

and by the rural engineering department. Further details on the sequencing of the studies and works in the different regions are included in Annex 2c.

For the activities related to the low emission agricultural production system, the implementation approach will start with the identification of sites based on a needs assessment, beneficiary awareness raising and technical feasibility. Structuring and mobilisation of in-kind contributions from beneficiaries (5%) will be developed to enhance ownership and sustainability of the adopted measures. Lessons learned from previous IFAD projects: use of drip technology, underground water collection technology, surface water collection at road level for agricultural use, agro-ecological blocks, will be capitalised.

To strengthen knowledge generation on climate change, the implementation approach will consider the profiling of the target group (location, roles, and importance in economic life). An analysis of the socio-economic profile of the target group (women, youth, illiterate) and an inventory of training needs and capacity building in the sites will be conducted to ensure an alignment between needs and knowledge. Training streams/themes will be validated and prioritised, followed by the development of training plans by theme and by target group category. Forms of post-training support will be identified.

For the implementation of the activities, the PCU will adhere to the Project Implementation Manual that will be elaborated detailing the approach. On the ground implementation will entail establishment of operational partnerships with state institutions and service contracts with private operators and NGOs awarded through competitive processes. For both private and public sector operators, the condition for the continuation of their collaboration on the project is the delivery of the annual results defined in the conventions/contracts/protocols. Identified partners included government institutions such as the National Office of Nutrition and Regional Offices of Nutrition, and NGOs such as FAFAFI, CTAS, AFAFI, CFAMMA, FOFIFA, FIFAMANOR and private sector players such as STOI, Société 3S, SSM, OM Kompiva, agrival LFL, Tozzi Green, MAC SIOU KING Alixia (ENTREPRISE PLUS), Sté 3S Union MAMIBE and La Myriade.

DEFIS+ climate change adaptation activities will be implemented to complement the interventions of other partners, namely the GIZ-supported PRADA project; AfDB's PAPAM project; the CI/EIB Sustainable Landscapes in Eastern Madagascar Project, among others. The project will also take into account initiatives for which other agencies plan to submit funding proposals to the GCF.

To support the PCU team (Figure 18) with the implementation of climate change adaptation activities, six personnel will be recruited through a competitive process carried out in accordance with the modalities in effect for IFAD-supported projects. The positions are: one climate change adaptation specialist; one finance officer; one M&E officer for the climate change adaptation activities; one M&E assistant for knowledge, GIS and database management; one social and environmental safeguards expert who will also be responsible for the environmental and social management plans (ESMPs) and one gender expert, both also involved in M&E procedures; the latter four will be integrated in the DEFIS M&E unit; all personnel will be hired on one-year contracts that may be renewed after a performance assessment has been done. The CAPFIDA will also support the PCU by providing advice and expertise on specific aspects, including monitoring and evaluation, knowledge management and financial management.

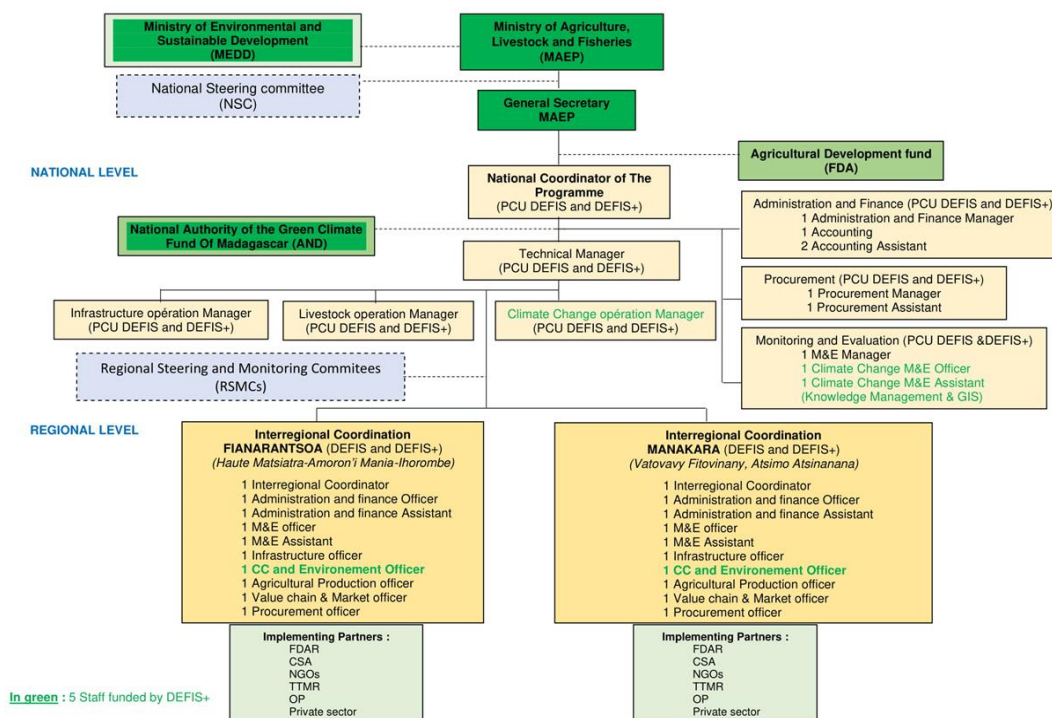
National Designated Authority (NDA)

Through its responsibilities, the National Designated Authority (NDA) of the Green Climate Fund of Madagascar will contribute to the monitoring and evaluation of the project throughout its cycle. Thus, it will monitor the implementation of project activities in the field and assess project indicators including the reduction of greenhouse gas (GHG) emissions and the assessment of the degree of communities' vulnerability. Thus, the NDA's Adaptation Unit (NDA-AU) will have an active role in the national adaptation planning processes and its contribution will be incorporated into relevant DEFIS+ activities. The monitoring will be based on secondary data produced by recurrent statistical surveys and framework surveys carried out at the national level and on the results of studies and surveys allowing for measuring the changes brought about by the project at the target level. Results including progress on the Gender Action plan will be assessed during supervision missions. The impact of the activities on GHG emissions will be analysed using the EX-ACT tool. Adaptation activities will be analysed and monitored in consultation with the NDA-AU. Further details are available in Annex 11 on Monitoring and Evaluation.

Joint supervision of the program will be provided by the Government of Madagascar (MINAE, MEF, MEDD DNA; NDA, and ONE) and IFAD. Supervision missions will be carried out twice a year and their main goals will be: (i) to assess progress on the implementation of each of DEFIS+ components; (ii) evaluate progress on the expected outcomes and outputs for each of the project's components in comparison to what had been established in the concept note or other project documents; (iii) assess the quality of targeting and gender equality; (iv) analyse the different fiduciary aspects of the project (funding allocations, payment and disbursement commitments, conformity

of expenditures with AWP/Bs and concept note, accounting records) and review outcomes of the external audits conducted in accordance with IFAD handbook for Financial Reporting and Auditing. A midterm review will be held to evaluate the actions taken in relation to the project's objectives, analyse the effectiveness and appropriateness of the methodologies used, verify the level of disbursement of resources and determine the need to reallocate available resources.

Figure 18: Organisational chart of the DEFIS+ program



Fiduciary aspects

The financial management and procurement procedures to be used for the GCF funds are the ones specified in IFAD's Manual of Administrative, Financial and Accounting Procedures, which has been validated by the Government of Madagascar and IFAD. The project's accounting will be done on approved TOM2PRO software and will follow the rules and practices of other IFAD-funded projects in Madagascar, expanding the scope to include all reporting dimensions required by GCF to track expenditures. The project will disburse funds according to the plan of activities defined in the AWP/B, which includes the disbursement and procurement plans.

Flow of funds

The GCF financing will be made available to Madagascar in the form of both grant and loan funds. This will be operationalized through:

- a Funded Activity Agreement (FAA), between GCF and IFAD, ruling the contribution of GCF to IFAD for which funds will flow into a dedicated bank account held by IFAD;
- a subsequent Subsidiary Agreement (SA), between IFAD and Madagascar (i.e. "Financing Agreement" as per IFAD standard terminology) on which basis the proceeds of the GCF contribution (i) will be provided (on-lent/on-granted) to Madagascar in the agreed denomination currency (i.e. USD) and (ii) will flow from IFAD to the Borrower (Madagascar-MEF) into a Project Designated Account (DA) in USD opened at the Central Bank of Madagascar (CBM).

In the FAA, the "GCF's Financial Terms and Conditions of Grants and Concessional Loans" will apply to the GCF contribution (grant, loan) to IFAD. The same terms and conditions will be maintained in the SA with regards to the GCF funds on-lent/granted by IFAD to Madagascar. It is also relevant to mention that, in the SA, "IFAD Lending Terms and Conditions" will apply to any IFAD co-financing (with own IFAD resources) to Madagascar.

Disbursement and payment procedure

The administration of the external funds of a public institution, in Madagascar, is regulated by decrees N°2015-1457 and N°2016-1160, which stipulate that all funds for projects managed by public institutions must be deposited in a designated account (DA) opened at CBM. MEF authorises the opening of the Project DA, in foreign currency (e.g., USD) at CBM. MEF (the borrower) approves the requests for disbursement of funds into the Project DA, which are prepared by the Project and Line Ministry (Lead Project Agency-LPA) on the basis of the supporting documentation defined in the Subsidiary Agreement (i.e., "Financing Agreement" as per IFAD standard terminology). Afterwards, funds will be converted from the foreign currency into the national currency upon transfers from Project DA to operational accounts opened at commercial banks in Ariary (MGA), the national currency. MEF controls the flow of funds to the operational accounts. The opening of operational accounts is authorised by MEF on request of the Project/LPA. The replenishment of operational accounts will be done after justification of the utilisation of a part of the previously allocated amount. The operational accounts will be used to make payments, for activity-related expenses, to third parties (Implementing Partners, Contractors, Service Providers) on the basis of signed agreements (MoU/Contracts).

Counterpart financing

The Government of Madagascar will take measures to exempt the project's expenses from taxes and duties, such as including annual budgetary provisions for exemptions on taxes and duties charged on products imported by the DEFIS+ project. The value-added tax and import duties and taxes are subject to the rules on the execution of public expenditures. The office of the Treasury of Madagascar will assume the taxes and duties vis-à-vis suppliers. The Government will also contribute its own internal resources to the account opened with the Treasury. Beneficiaries will also contribute in cash or in kind. The beneficiaries' contribution, whether in cash or in kind will be fully tracked during implementation.

Audit of accounts

The internal audit will be carried out by MINAE units established for this purpose, which are: the CAPFIDA internal controller who reports to project coordinators; and this department's Management and Internal Audit Control Unit, which reports to MINAE General Secretariat. Furthermore, the project's accounts will be verified annually through external audits conducted by independent and qualified auditing firms in accordance with IFAD Handbook for Financial Reporting and Auditing, and in compliance with the applicable GCF requirements (i.e., expenditure eligibility, categorisation vis-à-vis GCF budget categories, etc.). A certified copy of the auditor's report will be submitted to IFAD a few weeks before IFAD's deadline for sending such a report to GCF.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Climate change and the effects of the El Niño/La Niña phenomena are increasing the frequency and severity of extreme weather events such as droughts and floods and will continue to be major impediments to sustainable development. Indeed, climate change and its effects aggravate the vulnerability of populations that are already affected by poverty, food and nutrition insecurity. The DEFIS+ activities are targeted at climate change adaptation with mitigation co-benefits. In the context of common but differentiated responsibilities and respective capabilities and its fair and ambitious NDC, the country has estimated the amount of the adaptation cost to be USD 28.713 billion. Reiterating the very low responsibility in terms of GHG emissions the NDC highlights, Madagascar spends a substantial part of its annual budget on social services and infrastructure in response to the adverse effects of climate change (impacts of floods, droughts and cyclones). The precarious economic situation of Madagascar entails that the implementation of the NDC is conditional on the availability of external financial support, especially through the financial mechanisms established under the UNFCCC, hence, this request for GCF resources is mainly in the form of a grant (74% while 26% will be a loan)⁶³. The GCF resources will supplement the limited national resources, while also ensuring debt sustainability for Madagascar as a country that is highly vulnerable to the effects of climate change.

Due to its history of recurrent political crises and economic stagnation, Madagascar is one of the poorest countries in the world with a highly narrow fiscal space for investments in key sectors, that will stimulate inclusive and broad-based economic growth such as agriculture, social services and basic infrastructure. The degradation of ecosystems (land, water and biodiversity) and the narrow base of basic infrastructure have been accentuated by the recurrence of extreme events resulting from the effects of climate change. Madagascar's Country Strategic Opportunities Program (COSOP)⁶⁴ jointly developed with IFAD highlights the limited performance of the agricultural sector as one of the main causes of rural poverty. This poor performance is mainly due to exposure to climate change, low levels of public and private investment, low productivity of production systems and the weak institutional capacities of sector actors. The average yields of most food crops are well below their potential. For example, the average annual yield for irrigated rice is estimated at 3.0 tons/ha against a potential of up to 5.0

⁶³ The Economic and Financial Analysis provides further details on the justification for grant and loan resources.

⁶⁴ The COSOP outlines IFAD's engagement in a specified country and is jointly developed with the Government for a time period of approximately 5-6 years.

or even 6.0 tons/ha; for maize, 0.9 tons/ha against a potential of 4.0 tons/ha; for beans, 0.8 ton/ha against 2.0 tons/ha; and for cassava, 7 tons/ha against 30 tons/ha. The agriculture sector is deemed a risky business to invest in, due to the uncertainties inherent to weather, yields, prices, government policies, global markets and other factors that can result in fluctuations in farming activities and related income generation. Such market failure is exacerbated by climate change and disincentives the private sector to invest in agriculture, therefore justifying the request for GCF financing. There are several barriers that contribute to the suboptimal allocation of goods and services in the areas targeted by DEFIS+. Farmers have limited access to productivity enhancing technologies due to their remoteness. Without Farmers' Organisations, farms located in the southern part of Madagascar generate a limited marketable surplus. With climate change and the associated extreme events, some production areas can remain inaccessible for several months following a flood.

The positive results of IFAD's long-standing investments in Madagascar make it clear that the development of inclusive agricultural value chains (from inputs, production, post-harvest and marketing) is a powerful vehicle for rural transformation and improved food and nutrition security. With the normalisation of the political and social situation in the country, DEFIS+ investments will contribute to restoring, reinforcing and securing the productive bases of rural households, by connecting them to production and market support services and building their resilience to climate change and food and nutrition insecurity. This will contribute significantly to Madagascar's efforts to transform its economy sustainably and adapt to climate change. The objectives and priorities of DEFIS+ fit perfectly within the framework of national policies aimed at transforming the agricultural sector from one that is predominantly subsistence-based with low productivity levels and unsustainable practices such as slash and burn farming to one that is more resilient and sustainable, and which contributes significantly to improving food and nutrition security.

The GCF resources are requested mostly in the form of non-reimbursable grant given the country's situation and needs particularly the increasing debt situation and the target beneficiaries being vulnerable populations. The outstanding debt contracted or guaranteed by the State (CGE) stood at 33.8% of GDP at the end of 2017 compared with 26.0% at the end of 2013. In Madagascar, a recent analysis conducted by the IMF has highlighted that the public debt could reach about 49.0% of GDP in 2025, higher than the 46.0% of GDP in 2020.⁶⁵ It is thus recommended to maximise the use of non-reimbursable grants (Reports on the analysis of debt sustainability of Madagascar, Ministry of Finance and Budget Ministry of Economy and Planning June 2018). The Economic and Financial Analysis further emphasises that without such a non-reimbursable grant from the GCF, project activities would not be financially viable. The vulnerability of the target populations is exacerbated by climate change and the extremely low adaptive capacity at various levels makes it particularly difficult to use non-grant financial instruments. The limited public funding available for the proposed DEFIS+ activities will not allow timely action and will result in further degradation of natural resources and increasing food and water insecurity among the rural populations of southern Madagascar. Attracting private capital for innovative financing is very challenging and would require a significant expectation of profit to mitigate the perceived risk, such high profitability would be incompatible with the target beneficiaries. In 2019, the World Bank Doing Business ranked Madagascar the 161st country out of 190, and specifically 132nd in terms of access to credit.⁶⁶ Therefore, without the participation of the GCF as a key investor to build the foundation, it would be almost impossible to raise further private funds for climate change-related investment in Madagascar. Furthermore, the downside risk for smallholder farmers resulting from cost and sales pressures indicates the need for grant-based support as well as mechanisms to provide stable and affordable inputs.

As elaborated in Table 4 above, DEFIS+ aligns with the GCF Sector Guide on Agriculture and Food Security especially in regard to the investment pathway on Climate Resilient Agriculture, which focuses on adaptation in agriculture and takes into account the mitigation co-benefits of reforestation and renewable energy technologies and climate-smart practices used for agriculture and value chains development. It is also aligned with the GCF Sector Guide on Ecosystems and Ecosystems services through the pathway on landscape-level protection, and restoration to maintain the ability to store and sequester carbon. DEFIS+ aligns with the Sector Guide on Water Security through the pathway on strengthening integrated water resources management and water management. DEFIS+ targets the most vulnerable areas and livelihoods to the impacts of increasing temperatures and increased rainfall variability. The adaptation interventions focus on the production of staple foods, cash crops and foods that have high nutritional value, by making them more resilient to climate change impacts, especially by implementing activities with small scale farmers related to improved water management, irrigation and resilient agricultural practices and seeds coupled with landscape restoration, and agro-climatic information systems.

⁶⁵ International Monetary Fund. African Dept. 2021 Republic of Madagascar: Request for a 40-month Arrangement Under the Extended Credit Facility—Press Release; Staff Report; and Statement by the Executive Director for Republic of Madagascar.

⁶⁶ <https://www.doingbusiness.org/en/rankings#>

B.6. Exit strategy (max. 500 words, approximately 1 page)

Government institutions, local communities and agricultural development institutions will be involved in DEFIS+ implementation, which will ensure the sustainability of the investments from an institutional standpoint. In addition, DEFIS+ will feed into national policy dialogues and strategies, particularly those pertaining to climate change, to ensure that the project's components are integrated into and can inform the country's legal and policy framework. This will further reinforce the synergies between DEFIS+ and the national programs implemented by MINAE and MEDD, as well as with the activities led by research organisations, such as FOFIFA.

DEFIS+ will follow an integrated approach that uses a set of interconnected interventions to generate new opportunities for rural communities. This approach will create the enabling conditions for sustainable development to ensure that local producers benefit from increased and long-lasting socio-economic returns as they improve their resilience to climate change, reduce vulnerability to climate shocks, food and nutrition insecurity. Partnerships between producers and market organisations, as well as the inclusion of smallholder farmers in the formal economy will generate social and economic benefits that last well beyond the project's duration.

In particular, the DEFIS+ exit strategy will consider:

- The establishment of simple and efficient water harvesting systems that require minimal technical skills and that can be easily managed and maintained by beneficiaries during and beyond the project's lifetime. Training and awareness-building activities for the beneficiaries on climate-resilient agricultural practices, including efficient water use and management, maintenance of irrigation infrastructure, and value addition to agricultural products. The rehabilitation and maintenance of the non-transferable infrastructure are partly financed by users of the hydro-agricultural network, through their contributions (user fees). In addition, the Regions, annually include expenditure projections in their budgets as does the State, through the Fund for the Restoration and Maintenance of the Irrigations Infrastructure Networks (FRERHA).
- Proposed DEFIS+ interventions are consistent with the development priorities of the Malagasy government. This is important, because it is this consistency that will ensure the institutionalisation of the project activities and their sustainability. At the level of the GR4W development for example, DEFIS+ interventions will be linked to the World Bank-funded Madagascar Road Connectivity and the Madagascar Road Sector Sustainability projects in Malagasy regions, including Atsimo Atsinanana, which is in the DEFIS+ target area. Thus, these two projects constitute an important baseline investment that DEFIS+ will leverage at various levels, including drawing lessons from and building on existing institutional arrangements for road improvements in rural Madagascar. DEFIS+ will catalyse but also synergize with these projects in regions where they will overlap for enhanced socio-economic and environmental benefits.
- A maintenance programme for roads, storage and packaging infrastructures, will be established by the communities and financed through user contributions/ service payments and Regions budgets. The climate-resilient nature of the road design and construction will reduce maintenance costs and reinforce the investments' long-term sustainability.
- By supporting the MFIs, DEFIS+ will develop the partnership of proximity, leveraging the advantages of MFIs as local institutions with easier access, simplified procedures and project design and lower interest rates compared to other sources of local finance such as loan sharks. DEFIS+'s collaboration with MFIs will contribute to the development of the MFI sector and reinforce access to finance for agriculture.
- Women's access to land will be secured and guaranteed long after the project ends. This will be done based on IFAD past experience and the context, namely depending on the existence or not of land offices and/or municipal development plans, the solutions could be, among others: the ex-ante land agreement between the WUA and the Municipality. DEFIS+ will facilitate access to land for women and young people, especially female heads of household, in the new irrigated areas to be constructed under DEFIS. For existing irrigated areas, DEFIS+ will make it a condition of access to funding that women and young people are included among the beneficiaries of the land. Awareness-raising and the mobilization of traditional authorities are necessary to overcome certain traditions that prevent access to land, especially for women heads of household.
- An Operations and Maintenance Plan will be included in the Project Implementation Manual (PIM), with detailed information on the financial management of key infrastructure, including solar-powered facilities, water-harvesting systems, and check dams that can guarantee the sustainability of the project. The project will also target youth groups and associations to be engaged as entrepreneurs in the supply chain of the different technologies including the maintenance of equipment at the local level.
- One of the critical elements for ensuring sustainability to underpin the exit strategy is capacity development. DEFIS+ has been designed with capacity-building interventions partly for local technicians and farmers. In this regard, the project will train 2 000 trainers, actors and technicians on construction standards that enhance road resilience to climate change; 2 500 farmers on the utilisation of new technologies (renewable energy and energy-efficient small-scale technologies, water efficient technology); and 45 000 farmers and their

households on nutrition. DEFIS+ will also build capacities of the decentralised government. Developed capacities of different stakeholders are part of the exit strategy that will contribute to the project's sustainability.

Madagascar's Law 90-016 places the maintenance, management and control of hydro-agricultural networks in the hands of the beneficiaries constituted in WUAs. The Union maintains and protects the primary infrastructure common to the WUAs, including the main canal, the sand trap, and above all the dams. WUAs, maintain the secondary infrastructure (secondary, tertiary canals, etc.) specific to one or more WUAs. The "dina" or social conventions, which apply to all members of the WUAs, are rules agreed upon by all members to ensure the proper functioning of the association. For each violation of the "dina", the damage must be noted, followed by a fine which is drawn up and then applied after signature by both parties (the commission and the perpetrator). The exit strategy and the responsibility for the management of DEFIS+ infrastructures are summarized in Table 6.

Furthermore, in terms of exit strategy, DEFIS+ will support the integration of producers into markets and create strong long-lasting relationships of trust between producers and market operators. Two scenarios exist: an available product offer or building on an existing market. Depending on whether the products are available or build on an existing market, the result will be private actors engaged through aggregation such as cooperatives of private producers or the support of Farmer Organisations (FOs) through local microfinance.

Table 6: Exit strategy for DEFIS+ infrastructure

Exit strategy		
Infrastructure type	Operations and Maintenance	Long term financing for O+M
Irrigation systems	<u>Autonomous schemes:</u> Structure operation: Water User Associations /Farmer Organization	<u>Autonomous schemes:</u> User fees (100%)
	<u>Partnership schemes (with strategic infrastructure):</u> Water User Association/Private or Public Organization	<u>Partnership schemes:</u> User fees (Water User Association 50%)/Regional contribution (30%)/State contribution (20%)
Storage facilities	Agricultural cooperative	Service charges (100%)
Primary processing facilities (cleaning, drying, sorting)	Agricultural cooperative	Service charges (100%)
Roads	Municipal Roads: Municipality	Municipality expenditure (100%)
	Regional and agricultural/economic roads: Regions	Regions' expenditure (100%)

C. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	53 861			million USD (\$)		
GCF financial instrument	Amount	Tenor		Grace period	Pricing	
(i) Senior loans	12 816				Max 0.00% interest rate; 0.25% service fee; 0.50% commitment fee	
(ii) Subordinated loans						
(iii) Equity						
(iv) Guarantees						
(v) Reimbursable grants						
(vi) Grants ⁶⁷	41 045					
(vii) Results-based payments						
(b) Co-financing information	Total amount			Currency		
	96 986			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
IFAD Grant ⁶⁸	<u>Grant</u>	<u>37 434</u>	<u>million USD (\$)</u>			
IFAD Loan ⁶⁹	<u>Senior Loans</u>	<u>59 553</u>	<u>million USD (\$)</u>	10 years moratorium 40 years repayment period	Fee commission of 0.75% per annum, with a 40-year repayment period and 10 years moratorium %	
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	<u>150 847</u>			<u>million USD (\$)</u>		
(d) Other financing arrangements	The DEFIS programme financing includes USD 33.7 million from the Government of Madagascar, USD 20 million from the OPEC Fund for International (OFID) and USD 14.3 million from the programme beneficiaries. The Government contribution is mainly through					

⁶⁷ Grants are non-reimbursable.

⁶⁸ The amount includes funds from the ongoing IFAD grant amount approved in 2017 as 50% of the approved USD 26.5 million will be available to disburse during the implementation of DEFIS+.

⁶⁹ The amount includes funds from the ongoing IFAD loan amount approved in 2017 as 50% of the approved USD 26.5 million will be available to disburse during the implementation of DEFIS+.

and contributions (max. 250 words, approximately 0.5 page)	tax exemptions and resources from its public investment programme. The beneficiaries will contribute in-kind through their labour and share of costs towards assets such as post-harvest and value-added equipment. The OFID funds will be used primarily for infrastructure development. GCF lending terms and conditions will apply to IFAD. In turn, IFAD is expected to apply the same terms to Madagascar, in the Subsidiary Agreement, with regard to GCF funds. More precisely, the Loan Terms and Conditions are: max 0.00% interest rate; 0.25% service fee; 0.50% commitment fee. The service fee and commitment fee will cover the Fund's administration and resource mobilization costs. IFAD has provided Annex 13 "Co-financing Commitment Letter" with detailed information on co-financiers in relation to the co-financing agreements. It is important to note that the beneficiary contributions are not included in the detailed budget table as they are considered as parallel co-financing. The same applies to the in-kind contributions. The detailed budget table only includes cash financing and co-financing. The tax exemption is under the DEFIS programme. As per IFAD requirements, the cash counterpart contribution will be monitored and reported.
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C.2. Financing by component

Part of the IFAD co-financing (approx. USD 30.513 million) has already been disbursed for preparatory activities such as the provision of advisory services to farmers, the purchase of improved inputs for the agricultural production seasons, and the conduct of feasibility studies, including environmental and social impact assessments, for some of the infrastructure to be developed. The funds already disbursed under the DEFIS programme have been deducted from the overall budget.

Component	Output	Indicative cost thousand USD (\$)	GCF financing		Co-financing		
			Amount thousand USD (\$)	Financial Instrument	Amount thousand USD (\$)	Financial Instrument	Name of Institutions
1. Strengthened climate resilience of agricultural production systems	1.1 Improved water management for sustainable agriculture	31 070,594	24 798,014	12 747,705 (Loan) 12 050,309 (Grant)	6 272,580	Loan	IFAD
	1.2 Enhanced agro-climatic information systems, new technologies and initiatives	75 570,774	19 452,709	68,000 (Loan) 19 384,709 (Grant)	56 118,065	Grant 12 240,149 (Grant) 43 877,916 (Loan)	IFAD
2. Increased income generation through improved market accessibility	2.1 Reinforced rural access roads	11 316,682	3 393,604	Grant	7 923,079	Loan	IFAD
	2.2 Improved capacity to build and maintain rural infrastructure	948,976	948,976	Grant	0	N/A	N/A

3. Improved food and nutrition security	3.1 Knowledge on climate change and food security generated and shared	5 919,521	2 279,409	Grant	3 640,112	1 479,323 (Loan) 2 160,789 (Grant)	IFAD
Coordination and program management	Program coordination and management	26 020, 896	2 988,291	Grant	2 032,605	Grant	IFAD
Indicative total cost (USD)		150 847, 444	53 861,003			96 986,441	

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☒ No ☐

Adapting smallholders and farmers to climate change requires not only building the resilience of production structures (irrigated perimeters and adjacent sub-watersheds, rural roads) to climate risks, but also strengthening the capacities of farmers and rural development actors to: (i) better analyse production systems, constraints and the changes they face; (ii) acquire new knowledge and learn new farming practices to adapt their production systems to climate risks; and (iii) adopt social skills, such as good nutrition and gender practices, that can also help them strengthen their resilience to climate change and food and nutrition insecurity.

DEFIS+ capacity building activities are summarised in Table 7a and will focus on: (i) collective management of watersheds and irrigated perimeters and their protection against erosion; (ii) efficient water management for irrigated perimeters and mechanical and biological protection measures for watersheds; (iii) introduction of agroforestry techniques and climate-resilient species and varieties in dryland upland farming systems through the organisation of farmer field schools and exchange visits; (iv) new construction norms linked to climate change; (v) training of rural development actors in the use of agrometeorological tools and climate change adaptation measures; (vi) household training on nutrition and gender; and (vii) training in DEFIS+ management. The cost of investment in capacity building is USD 14,565 million, which is approximately 27% of GCF financing (see details in Table 7a below).

Table 7A: Estimated costs of capacity-building

Component	GCF Total amount '000USD	Capacity-building	
		'000 USD	% of project total
C.1. Strengthened climate resilience of agricultural production systems	44,251	11,305	21%
C.2. Increased income generation through improved market accessibility	4,343	949	2%
C.3. Improved food and nutrition security	2,279	2,239	4%
Coordination and program management	2,988	71	0%
Total	53,861	14,565	27%

There are also plans to organise technology transfer activities, including: (i) introduction of new water conservation technologies; (ii) production of seeds of stress-resilient species and varieties; (iii) introduction of

new integrated pest management practices; (iv) green roads; and (v) renewable energy technologies. The total cost of the technology transfer activities is estimated at USD 9.1 million, or 17% of the GCF funding.

For the operation and maintenance of hydro-agricultural infrastructure: at the regional level, a public institution, the Regional Hydro-agricultural Network Rehabilitation and Maintenance Fund (RHNRMF), is endowed with legal personality and administrative and financial autonomy. The headquarters of the fund is located in the capital of the region concerned. The role of each regional RHNRMF is to receive and manage funds intended for rehabilitation, technology transfer, maintenance, and management of all strategic works in the regional territory that are not transferable to other areas. The RHNRMF contributes to the repairs necessary in the event of damage caused by floods, cyclones or natural disasters to all the hydro-agricultural networks in each region.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

DEFIS+ proposes a comprehensive approach to sustainably improving resilience to climate shocks and enhancing nutrition and food security for vulnerable households in the target areas. DEFIS+ will enhance the adaptive capacity of agricultural systems and vulnerable rural communities to the adverse impacts of climate change so that farmers are better able to confront the increasing frequency and intensity of drought and changes in rainfall patterns that often result in floods. This requires modernizing the present agricultural production systems, which are reliant on rainfall and whose infrastructures and practises are no longer suitable in cognisance of climate change. DEFIS+ proposes to reduce smallholder farmers' exposure to climate and economic shocks by climate-proofing irrigation infrastructures and rural roads to increase agricultural water availability, reduce the risk of market access disruptions caused by extreme climate events and promote and disseminate climate smart agricultural practices. DEFIS+ will support 447,200 people, of whom 50% are women.

At the field level, the diffusion of water-saving technology such as the implementation of drip-irrigation systems and the SRI system allows for efficiency in the availability of water, reduces vulnerability to water shortages and increases productivity per hectare. This will increase the recoverable crop area that contributes to high levels of production. The support of DEFIS+ through the dissemination of drought and stress tolerant varieties and the adoption of integrated pest management methods and other improved agronomic practices will contribute to further reducing production risk and sustainably increasing productivity. Through DEFIS+, the resilience of production systems will be strengthened by investing in ecosystem restoration, including the promotion of an integrated landscape management approach (under the watershed). This reduces the risk of erosion and will improve the sustained productivity of farms through the retention of fertile topsoils. Some activities such as the dissemination of drip-irrigation systems overwhelmingly benefit women who are the main producers of vegetable crops and will contribute to the nutritional improvement of households. Investments in infrastructure reinforcement are aimed at the rapid evacuation of flood waters during extreme weather events and the reduction of risks associated with infrastructure degradation, thus ensuring sustainability and reducing maintenance costs in the future. Finally, DEFIS+ will promote low-cost on- and off-farm renewable energy and energy-efficient technologies, such as improved cooking stoves and handmade briquettes, by using local materials, thus reducing deforestation and contributing to carbon sequestration. Other technologies will also include solar pumps for irrigation, biogas production along with livestock systems, solar milling and dryers for product processing. These technologies will contribute to improving efficiency and production, while also reducing women's time burden and workload (thereby impacting their health, education, and nutrition status and indirectly those of their young children).

By combining training on climate-smart agriculture with nutrition education and gender equality, DEFIS+ will contribute to improving the knowledge and practices of farming households to not only improve their agricultural performance and incomes, but also to link these positive outcomes with improved resilience to climate change, nutrition, gender equality and empowerment. By supporting the installation of automatic agrometeorological stations, the project will enhance the generation of climate data, which will feed into on-going initiatives that provide decision makers with updated climate information. Through awareness-raising sessions and formal training on the impacts of climate change and adaptation techniques, DEFIS+ will also directly contribute to greater awareness of climate threats and risk reduction processes. Furthermore, collaboration with MFIs may develop complementarities between financial and non-financial services to agriculture, by defining the respect of technical aspects as a condition to access MFI funding. By promoting integrated financial services, MFIs will contribute to a positive potential impact on agriculture productivity and the sustainability of climate change adaptation gains.

The adoption of the Green Roads for Water approach through capacity building of the technicians in various Government institutions will provide a pathway for scaling up the results of climate proofed road infrastructure into additional regions in Madagascar. The GR4W approach for DEFIS+ will also contribute to enhancing aquifer productivity, which will have positive impacts on agriculture in the part of the country that faces climate change impacts on water availability and socio-economic activities.

Through these interventions, DEFIS+ will directly improve the climate resilience of 447,201 individual beneficiaries and indirectly impact the lives of 3,543,719 beneficiaries in the six regions of the project. Improving their ability to sustainably combat the negative effects of climate change, produce greater yields and gain access to markets. This

results in greater income generation for rural communities, resulting in reduced dependence on the exploitation of natural resources in lean periods.

DEFIS+ contributes effectively and efficiently to the Updated Strategic Plan (USP-2) for the GCF 2024–2027 (Table 7B). The USP-2 reflects a commitment to smart, transformative action in the face of the climate crisis. By addressing the complexities of climate change through enhanced adaptation, including mitigation, social equity, private sector engagement, and rigorous evaluation, the GCF is poised to play a crucial role in catalysing global efforts to build a sustainable and resilient future. The USP-2 builds upon the successes and lessons learned from its predecessor (USP-1), emphasizing a comprehensive and integrated approach to climate finance. At its core, the strategic plan sets some important key objectives aimed at accelerating climate action, fostering resilience and promoting sustainable development. In this context, DEFIS+ plays an important and crucial role in contributing to the USP-2 strategic vision and collective determination in the pursuit of a climate-resilient world.

Table 7b. Contribution of DEFIS+ to USP-2

Strategic Priority	Description	DEFIS+ Contribution
Country Programming	More than 100 developing countries directly supported to advance NDCs, NAPs or LTS	It supports the NDC, NAP, NRDMP, SNCC/AEP, NPC, and NPAILI of Madagascar.
Direct Access	Doubling the number of DAEs with approved GCF funding proposals through strengthened climate programming capacity and increasing the allocation of GCF resources through DAEs.	DEFIS+ does not directly work with DAEs as there is no DAE in the country so far. However, during its implementation should any DAEs be approved for Madagascar, relevant partnerships will be established.
CIEWS	50 to 60 developing countries protected by new or improved early warning systems.	The project develops an early warning system that provides information to help farmers determine their planting schedule, integration of phytosanitary surveillance, provide agricultural warnings, drought alerts, food, and nutrition surveillance. Six automated agro-meteorological stations will be installed and become operational.
Food	190 to 280 million beneficiaries adopting low-emission climate-resilient agricultural and fisheries	447 201 will directly benefit from climate-resilient agricultural strategies
Ecosystems	120 to 190 million hectares of terrestrial and marine areas conserved, restored or brought under sustainable management	DEFIS+ strengthened climate resilience by improving access to efficient water management systems, enhancing irrigated schemes, restoring sub-watersheds, landscapes, and reforestation.
	Total (no double-counting)	41 200 ha
	1.Sustainable management (sub -total) 70	Sustainable management of restored 15 000 ha of irrigated schemes.
	2.Conserved/protected (subtotal)	8 100 ha of land improved on water infiltration with anti-erosion systems.
	3.Restored/natural regen/agroforestry etc (subtotal)	18 100 ha of which terrestrial forest: 6 330 ha and terrestrial non-forest: 11 770 ha.

⁷⁰ Including the number of conservation and restored areas?

Infrastructure	45 to 60 developing countries develop climate resilient infrastructure; including ecosystem-based approaches	DEFIS+ contributes to develop climate resilient infrastructures such as the Green Roads for Water (GR4W) in Madagascar
Clean Energy	20 to 30 developing countries supported	DEFIS+ promote the use of renewable energy and low-cost on- and off-farm modern technologies, such as improved cooking stoves, as well as solar pumps for irrigation, solar milling and dryers for food processing.
Transport, buildings, industry	18 to 25 developing countries supported	DEFIS+ supports road rehabilitation (not construction) in remote rural areas of Madagascar.
Locally led Adaptation	40-70 approved proposals for adaptation projects, including for locally led adaptation action.	DEFIS+ supports climate resilient agriculture development, energy saving technology and watershed management that will contribute to local adaptation plans in the target regions.
Innovation and Market Creation	900 to 1500 local private sector early-stage ventures and MSMEs provided with broad-based seed and early-stage capital for innovative climate solutions, business models and technologies	N/A
Greening Finance	90 to 180 national and regional financial institutions supported to access GCF resources, and other green finance, particularly for MSMEs	N/A

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

Several projects in the south of Madagascar will benefit from the DEFIS+ project because this project will contribute to transforming the climate vulnerable, and unsustainable agricultural sector into a climate-resilient, market-oriented with modern infrastructure, and profitable economic sector, which also significantly contributes to mitigating climate shocks and improving food and nutrition, security, gender equality and empowerment. The structural and capacity transformation will be achieved by addressing vulnerabilities to climate change and supporting all stakeholders to improve their resilience, efficiency and links to markets. The paradigm shift for a green and sustainable climate resilience society that DEFIS + supports entails:

- Climate proofing basic production and market related infrastructure to improve their resilience. The integrated watershed and catchment management approach will scale up national experiences in bottom-up participatory watershed and catchment management with the aim of both protecting natural resources and increasing income. Community- driven sustainable land and water management plans and activities will be supported. The planning process, which includes social engineering for behavioural change and implementation of the plans will also enable knowledge sharing and joint learning among peers in the WUAs and the FFS.
- The partnerships mobilised for DEFIS+, the scale of productive investments planned for sustainable agricultural development and the focus on building local capacities constitute a major paradigm shift in regions in southern Madagascar renowned for humanitarian and emergency assistance. The focus on scaling up well-tested approaches and technologies building on innovative approaches and lessons learned from previous IFAD investments (including DEFIS and other IFAD projects in the country) is an important feature of the proposed interventions.
- Access to stress tolerant (drought, heat, flooding, salinity, etc.) and nutritionally enhanced varieties and planting materials developed by research centres. The production, multiplication and dissemination at community level of these climate resilient technologies will be promoted as will the adoption of integrated pest management and other agroecological approaches. Supporting national research centres and the FFS will ensure the enabling environment for improved seeds, promotion and adoption by smallholders.

- Greening road infrastructure will have an overall positive impact on agriculture, water availability and soil erosion associated with running surface water. Thus, maintenance and restoration of green roads will have a minimal environmental impact and large benefits for civil society.
- Development of agrometeorological information. Due to the dynamic nature of exposure and vulnerability, impact scenarios for crops, land use and agri-infrastructure will vary widely. The project will use the DSS platform to understand risk (using hazard exposure and vulnerability), including gender-related considerations and applications such as Met Apps and others provided by the Regional Meteorological Directorate.

At the national level, DEFIS+ as well as other projects supported by IFAD, will contribute to the implementation of the strategy laid out in the National Policy for Action on Climate Change and the NDC (2015) that aim to support the development of sustainable, integrated and resilient agriculture models and water resources management. More specifically, the adaptation practices promoted by DEFIS+ respond to the UNFCCC Technology Needs Assessment (TNA) developed for Madagascar.⁷¹ The TNA identifies the following initiatives as having potential for paradigm shifting: optimization of water management and drip irrigation; adoption of landscape conservation techniques; technical and financial support for risk management; adoption of adapted farming techniques and stress-tolerant varieties; rehabilitation of irrigation infrastructure; and diffusion of drainage systems. Further assessments focus particularly on the promotion of smallholder agroforestry; and the development of resilient rice technology packages.⁷² DEFIS+ will also contribute to the regulatory framework for irrigation infrastructure through a focus on strengthening the capacities and standards of design offices and construction companies. The project is in line with the above-cited analyses, as it promotes the adoption of resilient agroforestry and landscape management systems, the adoption of Systems of Rice Intensification (SRI), the adoption of drought- and stress-tolerant, and better adapted to climate change seeds and the reinforcement of irrigation networks as key strategies to improve the climate resilience of agricultural production systems.

The project will shift agriculture in the country's target areas from its current situation of high vulnerability toward an alternative paradigm wherein better information and climate smart farming practices will significantly increase resilience to climate change. The reliance on state-of-the-art GIS tools will support strategic planning through the provision of more systematic data, which will act as a springboard for science-based decision-making processes. The project constitutes an important step in a longer-term process of transforming agriculture in Madagascar. The sheer magnitude of the required behavioural change, among a large and poor population highly dependent on agricultural output, is an enormous task that the government cannot achieve on its own.

This model can easily be applied to other geographic areas in Madagascar or other countries. The approach and the tools to be utilised constitute a "package" that could be easily replicated in other systems in the country. The successful implementation of this project will result in a new, replicable, long-term and sustainable funding model in other landscapes that could enhance climate resilience, improve conservation, reduce land degradation and boost economic development. Outside Madagascar, the proposed model of a land use transition investment fund is well suited for the most vulnerable countries on the Africa continent, where unsustainable agriculture takes up an increasing share of available land, resulting in deforestation, land degradation and an agricultural sector not resilient to climate change. Thus, this model could be replicated (but not exclusively) in other least developed countries in Africa that need large-scale climate investments in rural areas to finance the activities necessary to achieve the objectives set in their NDCs and in their national adaptation programs.

D.3. Sustainable development (max. 500 words, approximately 1 page)

DEFIS+ will be implemented by the Government of Madagascar, particularly MINAE, the Ministry of Environment and MEF in close synergy with other rural and agricultural development investments intervening in the southern regions of Madagascar. Like other IFAD-funded projects, its interventions will rely on using national systems at both the central and decentralised levels, and on strengthening the capacity of targeted poor smallholders and their local institutions such as Farmers' Organisations, in order to better empower the rural poor, with a particular focus on women and youth. When looking at ensuring sustainable development, climate information represents an essential element for increasing the capacity of vulnerable households to adapt to climate variability.

The DEFIS+ project includes important sustainable co-benefits:

⁷¹https://unfccc.int/ttclear/misc/_StaticFiles/gnwoerk_static/TNA_key_doc/c5df277319904ab1ae1150e48ff7670f/57d71de082f9453ab8e8a42241eb27cc.pdf

⁷²https://unfccc.int/ttclear/misc/_StaticFiles/gnwoerk_static/TNA_key_doc/eb3e65055e5240b6b609f62f8402e57f/a99addeebd3e4687aed5daa27cb2123f.pdf

Environmental co-benefits. DEFIS+ will promote sustainable and climate-resilient agricultural practices, technologies and infrastructures that contribute to improving the efficient management and use of water resources in agriculture, enhancing soil quality and restoring degraded lands. Its integrated watershed management approach will help increase moisture content in the soil, optimize the use of water and expand the vegetative cover of degraded lands, while reducing their wind- and rainfall-induced erosion. It will also help reduce the siltation of water bodies, hydro-agricultural infrastructure and irrigation canals. This will, in turn, lead to greater biodiversity in the target areas and surrounding landscapes, thereby maintaining key ecosystems services that are crucial for agricultural production, especially in the context of climate change.

Practices that increase soil organic matter also help increase carbon sequestration, reduce GHG emissions and improve soil nutrient availability and water-holding capacity, which in turn strengthens the resilience of farming systems and increases productivity. Sustainable soil fertility management will allow the mitigation of climate change due to the improved carbon sequestration capacity of soils rich in organic material. Agroforestry and silvopastoral systems can increase livestock productivity by reducing heat stress to animals. Thus, the application of good livestock and manure management practices results in a reduction of methane emissions compared with the current BAU. In addition, trees increase carbon storage in soils and biomass. Watershed rehabilitation increases carbon stored in the land and the use of renewable energy technologies will reduce carbon emissions. The process of manufacturing biogas through anaerobic digestion makes it possible to reduce the sources of methane production and to concentrate them in one place, in addition to the other advantages of valorisation of agricultural by-products, but also to reduce the use of wood, which is also a source of household energy. Improved stoves will also reduce pressure on forest products.

Social co-benefits including health impacts. Higher food security and nutrition will be among the key social co-benefits that will arise. The proportion of targeted households with sufficient food intake is expected to increase from 50 to 90%. In addition, 45,000 households will benefit from nutritional education, which will improve women's and infants' health. Better access to water, more effective watershed management and the use of off-farm technologies will reduce women's time burden and workload. Women's and youth's employment opportunities will increase, and women's participation in the rural economy will be further formalised and recognized. By increasing and diversifying income sources, as well as improving food security, the project will also contribute to reducing migration flows to urban areas.

Specific actions proposed for women and young people in vulnerable situations to facilitate their access to productive capital and to income-generating activities or jobs include: (i) inclusion of young people trained by FORMAPROD by reserving them plots within the newly developed perimeters; (ii) positioning of young people and women as rural entrepreneurs involved in the service professions linked to development poles; (iii) prioritisation of young people and women in capacity building actions and access to financing for producers. Young people will be trained not only on production but also on processing, storage, marketing, sale/distribution of agricultural inputs, management of irrigated areas and advisory services as well as maintenance of new technologies and equipment.

Economic co-benefits. DEFIS+ will contribute to reduce poverty by supporting the diversification of agricultural production, building the climate resilience of production systems and infrastructures and creating market linkages, which will reduce transaction costs and boost the local economy. The sustainable management of agricultural water within an integrated watershed approach, coupled with more climate and stress-tolerant seeds, will sustainably increase agricultural productivity. In addition, the dissemination of climate resilient practices and technologies will enhance the adaptive capacity of agricultural systems and the income of smallholder farmers and their households. The activities planned, based on previous experiences, that contribute to the increase in productivity are efficient irrigation, optimal organic fertilizer use, improvement of agricultural practices through FFS, use of improved and certified seeds that are climate resilient.

Owing to increased productivity and improved linkages to the market, the income of the targeted smallholders will increase. At least 53,900 small-scale farmers are expected to adopt more resilient agricultural technologies. Based on conservative assumptions, the Economic Internal Rate of Return (E-IRR) of DEFIS+ is predicted to be 24% and its Economic Net Present Value (E-NPV) is expected to reach USD 255.5 million (see Annex 3, Economic and Financial Analysis).

Mitigation co-benefits. The co-benefits of integrating a mitigation component into the DEFIS+ project are diverse and impactful. From environmental sustainability to operational efficiency, cost-effectiveness, stakeholder engagement, and social responsibility, climate change mitigation contributes to the holistic success and positive impact of this project. Co-benefits include the restoration of degraded and marginal areas and sustainable agroforestry practises that contribute to carbon sequestration and climate change mitigation. The Project also introduces innovative, low-cost, context-specific renewable energy and energy-efficient technologies for farming and processing (such as solar pumps and dryers) and domestic activities (such as improved cooking stoves to reduce the use of firewood), which contribute to reducing GHG emissions. Thus, climate change mitigation co-benefit

activities will contribute to NDC priorities, which include facilitating access to energy by strengthening existing systems and promoting renewable energy and green technologies.

Gender-sensitive development impacts DEFIS+ will implement a gender strategy to ensure women's specific needs are considered at all stages of implementation and will provide opportunities to improve their quality of life, reduce their vulnerability to climate change and increase their incomes. The activities proposed for women and youth will include training of farming households on gender issues including the use of the Gender action Learning System (GALS) and measures to increase their access to productive capital and income-generating activities, primarily access to land, with a view to positioning women as rural entrepreneurs and service providers in development hubs and facilitating their access to capacity-building and financial services. The number of green jobs created with this project will target women, heading 30% of the selected households, in addition to women in male headed households, thus moving a step closer to narrowing the gender gap. The project will also involve minorities, vulnerable people and groups through the social inclusion strategy.

The project is well placed to enable the achievement of the Sustainable Development Goals. The 2030 Agenda for Sustainable Development, adopted by all United Nations Member States in 2015, provides a shared blueprint for peace and prosperity for people and the planet, connecting all countries in a global partnership. Countries have recognised that ending poverty and other deprivations must go hand-in-hand with strategies that improve health and education, reduce inequality and spur economic growth – all while tackling climate change and working to preserve oceans and forests. The DEFIS+ project contributes to SDG 13 (Climate Action), SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 5 (Gender Equality), SDG 7 (Affordable and Clean Energy) and SDG 15 (Life on Land).

Social equity. The overall strategy of DEFIS+ is to ensure social equity and cohesion among all members of civil society. Thus, this project follows an inclusive targeting strategy to reach vulnerable groups and poor households headed by women and youth, so that these targeted rural populations, characterised by their vulnerability to the effects of climate change, will have equal opportunities to access factors of production (land, water, inputs, financing, etc.) in order to ensure fair and equitable results and enable sustainable impacts on living conditions.

For gender equity, DEFIS+ will implement a gender strategy based on the consideration of inequalities related to sex, age and status. Such a strategy is based on the national policy, GCF updated gender policy and action plan, IFAD's action plan in this area and on the experiences and best practices developed by previous projects in Madagascar. The objectives of this strategy are to: (i) remove the constraints limiting the inclusion, active participation and empowerment of women and young people (men and women) in the different linkages of the climate change agenda and the agricultural sectors in order to create opportunities to improve their living conditions and coping with the changing climate; (ii) strengthen their productive capacities in order to increase the productivity and profitability of their economic activities by facilitating women's and youth access to resources and production factors (land, water, inputs, financing/credit, vocational training, climate information, technologies, advisory support, etc.); and (iii) promote women's leadership and their participation in decision-making bodies at the household level, in Producer Organisations (WUAs, Cooperative etc) in platforms/frameworks for consultation of the sectors and in local communities in general.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

With a GNI per capita of USD 480 in 2020, Madagascar belongs to the group of least developed countries (LDCs). After real GDP growth of 4.4% in 2019, the country went into a recession in 2020, with real GDP declining by 4%.⁷³ The combined impact of global trade disruptions and domestic containment measures is estimated to have resulted in a GDP contraction of 4.2% in 2020, similar to that observed during the devastating 2009 political crisis.⁷⁴ Led by the industrial and service sectors, GDP growth increased between 2013 and 2017, from 2.3% to 4.2%, whereas between 2014 and 2017, the agricultural sector contracted an average of 0.8% per year. In a country where agriculture employs nearly 80% of the population, this indicates that the economic growth experienced in the past few years did not have a positive impact on the living conditions of the rural population. The limited performance of the agricultural sector is the result of limited means of production, low productivity levels, insecurity in rural areas, overexploitation of natural resources, vulnerability to climate change (cyclones, drought and floods) and constraints on access to markets particularly in the rural areas.

Madagascar's ranking on the Human Development Index is still very low 0.528 positioning it at 164th place (out of 189). In 2018 it went from 0.512 to 0.519, it fell three positions on the global index during this period, from 158th to 161st place (out of 188 countries). Social and geographic inequalities in the country are still very pronounced: more than 80% of the population lives below the poverty level, with USD 2 per day, and 80% of the population lives in rural zones, sometimes in very remote areas with limited access to basic infrastructure. Food insecurity and malnutrition

⁷³ <https://www.afdb.org/en/countries/southern-africa/madagascar/madagascar-economic-outlook>

⁷⁴ <https://www.worldbank.org/en/country/madagascar/publication/madagascar-economic-update-covid-19-increases-poverty-a-new-reform-momentum-is-needed-to-build-back-stronger>

exist in the country at high prevalence, affecting predominantly women of reproductive age and children, and creating huge socio-economic losses.

The country is among the poorest in the world and poverty increased significantly as a result of the 2009-2013 crisis. The poverty rate (at \$1.9/day) is estimated to rise to 77.4% in 2020, up from 74.3% in 2019, corresponding to an increase of 1.38 million people in one year.⁷⁵ The incidence of extreme poverty (less than USD 1.25 per day) rose to 78% in 2012. Poverty is clearly more severe in rural areas, where it affects 87.6% of the population, in comparison to 55.1% in urban areas. Extreme poverty is more pronounced in the south. The activities of DEFIS+ aim to restore and/or rehabilitate the soil and improve land use through a participatory approach with the local community. This approach will ensure that any potential land conflict is identified and resolved before any activities commence. For the rehabilitation of roads, a resettlement policy for beneficiaries who may be voluntarily displaced will have to be implemented before the works are initiated. Adherence to IFAD's Social, Environment and Climate Assessment Procedures (SECAP) also follows the principles of Free Prior and Informed Consent that ensures existing inequalities are not augmented. A Grievance Redress Mechanism is also articulated for the project to manage any potential conflicts. In addition, DEFIS+ fully takes into account the needs of the recipient, which are in line with the national priorities. Furthermore, continuous engagement with the NDA has resumed, and DEFIS+ documents have been shared with the NDA. Further information is available in Annex 2a.

Several studies have confirmed Madagascar's high vulnerability to climate change (see Section B.1 and Annex 2b). The growing frequency and intensity of droughts coupled with increasing rainfall variability pose significant challenges to the agricultural sector. This results in very high exposure and vulnerability of the rural sectors in southern regions to climate-driven changes in rainfall patterns, rising temperatures and consequent declines in agricultural water availability. Such changes will adversely impact crops that are important in economic and food security terms such as rice, maize, among others. Furthermore, aridification in the southernmost regions will lead to greater wind erosion, while more frequent flash floods will exacerbate problems such as soil erosion, siltation of irrigation infrastructure, and damage to rural roads. Rising temperatures will also contribute to the proliferation of pests and diseases and put even more pressure on scarce water resources for irrigation purposes.

The precarious economic situation of Madagascar entails that the implementation of the project's activities is conditioned by the availability of external financial support, hence this request for GCF resources. The GCF resources will supplement the limited national resources, while also ensuring the debt sustainability for Madagascar as a country that is highly vulnerable to the effects of climate change. In its 12th replenishment cycle (2022-2024) IFAD has increased its targets in relation to climate change, meaning that at least 40% of its investments have to be invested in climate change adaptation or mitigation measures. Recognising the high demand and need for climate finance and a better integration of climate change adaptation into project design and implementation, IFAD supports countries in accessing Global Climate Funds. In this context, the additional funds provided by the GCF will enable DEFIS+ to contribute to meeting such a target with a robust climate rationale, a sound climate risk analysis and a well-structured strategy to deal with the negative impacts of climate change in six target regions in Madagascar.

D.5. Country ownership (max. 500 words, approximately 1 page)

DEFIS+ activities will support Madagascar's national policy for climate change, environment and sustainable development and the national strategies, such as the sectoral program for agriculture, livestock and fisheries. DEFIS+ directly supports the National Policy for Action on Climate Change by strengthening climate change adaptation, while taking into account the country's needs. It will also promote the integration of climate change into all levels of policymaking, as well as research, development and technology transfer in this field. Madagascar is also at the early stages of designing a GCF country program (GCF/UNEP Readiness grant). DEFIS+ investments are fully aligned with key priority sectors identified in Madagascar's National Adaptation Plan of Action: infrastructure, water resource management, agriculture and livestock. According to the National Adaptation Plan, DEFIS+ activities are aligned and perfectly integrated with the following priorities: (a) Strategic priority for agriculture. Promote resilient agricultural systems. This will primarily involve supporting rural populations (men and women, including people with disabilities) in setting up an agriculture that is resilient to climate change, also helping to strengthen the fight against food and nutritional insecurity, with a particular focus on the most vulnerable groups and on agricultural activities differentiated between men and women. (b) Developing pilot initiatives for the application of Integrated Models of Resilient Agriculture. Madagascar has a wide range of agricultural systems and agroecological practices in place: sowing on permanent plant cover, intensive rice cultivation (SRI) and improved rice cultivation (SRA), random irrigation rice cultivation (RIA), integration of agriculture and livestock, agroforestry, combinations of agroecological techniques and systems, etc.

⁷⁵ Ibid

Madagascar's Nationally Determined Contributions aim to build national resilience to cyclones, floods and droughts and give priority to adaptation interventions by 2030, such as the widespread application of resilient agriculture models; sustainable and integrated water management, including the mobilisation of surface water resources for semi-arid zones and the upgrading of irrigation infrastructure to cope with the reduction in runoff during dry periods; reduce GHG emissions and increase carbon sequestration; improve rice farming systems (SRI/SREA) and the implementation of ecosystem-based adaptation good practices. Blue or green infrastructure solutions that integrate and interact positively with the rural environment which they are part of must be considered: alternatives to charcoal, fossil fuel, water abstraction systems, etc. The design of these infrastructures must consider durability and use of resilient materials. DEFIS+ will support measures listed in Madagascar's Third National Communication, such as improved soil management, the development of production systems adapted to climate change and optimal use of water resources. DEFIS+ will also contribute to the implementation of Madagascar's Nationally Appropriate Mitigation Actions, focusing on the multiplication of fodder seed production and outreach in pastoral areas and intensification of improved seeds production. It will contribute to priority actions defined by Madagascar's National Capacity Self-Assessment, including the involvement of regional/local authorities in decision-making on climate change, an awareness program to address climate change, training on good practices and solutions identified in its Technology Need Assessment, such as strengthening integrated water resources management and the capacity of technical officers in charge of water infrastructures.

Since 1979, IFAD has provided USD 404.8 million in financing to a total of 16 projects in Madagascar. IFAD's COSOP focuses on integrated management of natural resources and increasing vulnerable rural households' resilience. The evaluation of the Madagascar country program for the period 2000-2012 carried out by the Independent Evaluation Office of IFAD indicates satisfactory performance in terms of relevance, effectiveness, efficiency, sustainability and impact on rural poverty. Households receiving IFAD investments have experienced increased agricultural production and productivity, and a marked increase in their cash income. DEFIS+ will scale up successful results from IFAD projects (AD2M, FORMAPROD, AROPA, DEFIS, and PROSPERER). The Implementing Agency will be MINAE, with whom IFAD has been working for 40 years and which houses a permanent unit that oversees project implementation (CAPFIDA). During the program formulation process, there was continuous engagement with the National Designated Authority (NDA), which has endorsed DEFIS+ investments. The MEDD, which hosts the NDA, has confirmed that it is on board to support DEFIS+ and optimise synergies with other initiatives that it oversees.

DEFIS+ has been informed by rounds of discussions and consultations with regional and national stakeholders and related formulation missions, all led by the Madagascar government. The consultations included specific meetings with representatives from the Ministry of Women's Affaires and focus group discussions with women. In addition to MINAE and MEDD, several rounds of consultation were conducted with stakeholders such as MEF, FOFIFA, the National Chamber of Agriculture, Federation of the Chamber of Commerce and Industry, Ministry of Water and Energy, the Agriculture Development Fund, General Directorate for Meteorology (DGM), regional authorities, private sector entities, family farming communities, civil society and NGOs, WFP, FAO, UNDP, GIZ, CI, OFID and World Bank, and farmers as evidenced by the list of stakeholders meetings held in Annex 7. Furthermore, the engagement of Farmers' Organisations in the implementation of the existing project defines an inclusive stakeholder approach as FOs are representatives of the target group.

During the studies for the rehabilitation of irrigated areas and roads, community consultations must always be carried out following the current legislation in Madagascar. For any activity and according to local traditions, meetings with local authorities and representatives of local institutions/organizations (in the form of participatory rural appraisals) should always be led by service providers in a participatory and inclusive manner. The stakeholders' consultations on the CSA practices are no exception to this approach. Stakeholders are consulted on the current practices for adapting to climate change in their agricultural practices. The traditional knowledge is then built on with the scientific projections of climate change and potential solutions are then discussed with communities. Climate change adaptation measures can then be prioritised as a result of the discussions with communities. The discussions are facilitated by technicians such as extension officers or community leaders.

These sub-regional consultations also facilitate the development of a strong sense of local ownership over the project outcomes. They also occur throughout implementation as part of the M&E strategy in order to evaluate the program's progress toward reaching the expected results. It is important to mention that an MRV analysis was conducted for both MINAE and MEDD and it has shown that the Ministries have the necessary skills and capacities to fully ensure the lead in the technical implementation and monitoring of DEFIS+ activities, outputs and outcomes. Furthermore, both MINAE and MEDD have managed large grants from international donors, and in the case of MEDD, it has extensive experience successfully managing funds from the GCF. Finally, IFAD has been working in Madagascar with the GoM, MINAE, and MEDD for 20+ years, and the partnerships and high professional standards have been nurtured.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

The overall approach adopted by DEFIS+ is designed to increase efficiency and effectiveness by focusing on the set of structural investments associated with the introduction of improved technologies and climate resilient practices as well as the capacity building of local institutions that will leverage growth in targeted agricultural areas. The integrated approach and integration of science and engineering for the in-depth modelling and understanding of agricultural risk will inform the structural investment through the studies that will be undertaken. Furthermore, to move from low-yield subsistence farming to market-oriented agriculture, the project's theory of change is based on improving the productivity and adaptive capacity of production systems to climate change and beneficiaries' access to markets under favourable conditions. Lastly, as the theory of change requires significant investments in basic infrastructure, priority will be given to the development of irrigation infrastructure and rehabilitation of roads to reduce the risks and costs of accessing the nearest markets.

The main expected benefits for people in the DEFIS+ project area are: (i) create climate resilient rural communities; (ii) diversification of income sources of the target populations, especially young people and women and green jobs creation in the project area; (iii) reduced transportation costs through rural road rehabilitation, water storage infrastructure and market access; (iv) increased revenues for the rural poor people; (v) improved food security, diet quality, diversification and nutrition; and, above all, (vi) greater awareness of and protection from the climate change impacts negatively affecting the livelihoods of the rural population.

DEFIS+ will offer several interrelated direct socio-economic benefits to 447,200 households in its area of intervention who have access to improved climate smart technologies, farming practices and infrastructures. The total farm areas to be cultivated with climate efficient technologies and practices is estimated at 16,000 hectares and 8,100 hectares of land in sub-watersheds will be treated to reduce their vulnerability to soil-erosion processes. The promotion of resilient agroforestry systems and landscape management in rain-fed upland farming areas will concern 10,000 hectares. At least 150 water tanks/reservoirs will be installed and 540 km of feeder roads will be rehabilitated while strengthening their climate resilience.

The procurement of technologies will be done competitively as outlined in Annex 10 (Procurement Plan) led by the PMU and MINAE and overseen by IFAD through the no objections that will be provided to the bidding documents and draft contracts. The goods to be procured through national competitive bidding will be divided into lots and published in various media to encourage local suppliers including at the regional level to participate. The technologies will be distributed to the most vulnerable households in order to catalyse demand from other community members. Uptake by other farmers will be encouraged through FFS and demonstrations in the fields of the most vulnerable community members. Local entrepreneurs and enterprises will be encouraged to supply the technologies and strengthen the supply chain where this is weak. The cost structure will be grant based to the most vulnerable households while for the rest of the community members they will purchase the technologies. For technologies such as the solar energy, youth groups and associations will be engaged and trained in the operation and maintenance of the equipment as an income generating activity and support to their communities. Women and youth groups will also be trained in the manufacture of fuel efficient cookstoves using local materials. The groups can then sell the cook stoves among community members for income generation.

With the promotion of efficient irrigation systems, watershed management, and improved farming practices and varieties and planting materials that are stress tolerant (rice, maize, cassava, sorghum, groundnut, small ruminants, pulses, etc.), DEFIS+ contributes to the implementation of Madagascar's international climate commitments. It also aims to empower and strengthen the institutional capacities of smallholder farmers and their organisations who will be better equipped to undertake collective actions, such as managing and maintaining common infrastructures, or provide services to their members, and will benefit from the reduction in transaction and transportation costs derived from the rehabilitation of rural roads opening up remote villages. DEFIS+ with its innovative climate-oriented approach will benefit directly and indirectly 3,990,920 beneficiaries (1,995,460 females and 1,995,460 males).

The Economic Internal Rate of Return (E-IRR) of DEFIS+ is estimated at 24% with a Net Present Value (E-NPV) of USD 255.5 million (Table 8). These results indicate that based on an opportunity cost of capital of 8.5%, DEFIS+ will generate a satisfactory E-IRR and is more than justified on economic grounds. Further, details of relevant indicators are shown in Section E.

Table 8: Economic Internal Rate of Return (E-IRR) and Net Present Value (NPV)

Sensitivity Analysis	Base case	Costs Increase			Benefits Increase		Decrease of Benefits			Delay of Benefits	
		+10%	+20%	+50%	+10%	+20%	-10%	-20%	- 30%	1 year	2 years

IRR (%)	24%	23%	22%	18%	26%	27%	23%	21%	19%	21%	19%
NPV (USD 000)	255 526	245 483	235 440	209 006	291 121	326 717	219 931	184 335	148 740	221 565	189 526

E. LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

- ☐ Reduced emissions (mitigation)
- ☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Malagasy agriculture faces significant challenges, primarily marked by subsistence farming exhibiting low productivity per hectare. This is closely linked to the limited accessibility of services for producers, as well as poor soil fertility. The agricultural sector faces issues such as rural insecurity, the overexploitation of natural resources, and heightened susceptibility to natural disasters like cyclones, droughts, and floods. Furthermore, the remoteness of the project regions, coupled with deteriorating infrastructure, hampers	Medium	DEFIS+ will scale up successful results from IFAD projects (AD2M, FORMAPROD, AROPA, DEFIS, and PROSPERER) and transform the climate vulnerable, and unsustainable subsistence (mostly slash-and-burn) rainfed agricultural sector to a climate-resilient, market-oriented and profitable economic sector. The scale of IFAD productive investments for sustainable agricultural development and the focus on building local capacities constitute a major paradigm shift in southern Madagascar renowned for humanitarian and emergency assistance. Thus, the project will scale up well-tested approaches and technologies building on innovative	- Rural development policies and strategies are formulated, implemented, and monitored at both national and regional levels. - The negative climate change impact on agriculture, farmers, and production is mitigated by an efficient and effective agrometeorological system, resilient seed varieties and climate resilient infrastructures - Institutional and organisational capacities are strengthened in terms of sector coordination and monitoring of strategies for promoting priority climate and agricultural sectors and the delivery of services to farmers.

	vulnerability to climate change, making it difficult for farmers to engage with agricultural markets and secure rural finance. The cumulative impact of these factors not only jeopardizes the livelihoods of those involved in agriculture but also contributes to the overall climate vulnerability of the Malagasy population. Addressing these challenges is crucial not only for the well-being of local farmers but also for the resilience and sustainability against the negative impact of climate in Madagascar. The DEFIS+ project capitalises on past positive lessons from the DEFIS program and scales up experiences tried and tested by other IFAD projects and programs to transform agriculture into a more productive, sustainable, and market-oriented sector; promotion of climate-smart agriculture, mobilization of surface water resources promotion of intensive or improved rice-growing systems (SRI/SRA), upgrading of irrigation infrastructure to cope with reduced runoff in dry periods: to connect them (i) to support services for production and markets, irrigation and water control infrastructure; (ii) to agricultural advice; and (iii) to technical itineraries and improved technologies adapted to climate change.		approaches and lessons learnt from previous IFAD investments.	- Contractual relations between agricultural market players are strengthened. - The project will contribute to this agricultural transformation by enhancing conditions that support climate, environmental and social goals (e.g., improving climate resilience, land health and production through the adoption of sustainable practices promoted by training and facilitated by access to financial products).
Replicability	Agriculture in Madagascar is not presently resilient to climate change. Existing and pipeline programmes/projects on agriculture and rural development address Climate Change.	<u>High</u>	The DEFIS+ project activities are applied by IFAD to other geographic areas in Madagascar and in other countries. The project results in a new, replicable, long-term and sustainable funding model in other landscapes that enhances climate resilience, improves conservation, reduces land degradation and boosts economic development. Outside Madagascar, the proposed model of a land use transition investment fund is well suited for the	- A long-term financing model is in place which enhances climate resilience and boosts economic development. - A land use transition investment fund is established.

	Many good practices on climate change resilient and replicable agricultural techniques are already in place and replicable across DEFIS+ zones and respond to the country's national adaptation plan and agricultural sector strategies.		most vulnerable countries, where unsustainable agriculture takes up an increasing share of available land, resulting in deforestation, land degradation and an agricultural sector not resilient to climate change. The Investment Fund is replicated in other least developed countries in Africa that need large-scale climate investments in rural areas to finance the activities necessary to achieve the objectives set in their NDCs and in their national adaptation programs.	
Sustainability	The existence of numerous strategic framework documents conducive to the implementation of actions to combat climate change. Existence of a national action plan to combat climate change and a national strategy for the agricultural sector to combat climate change, but a low percentage of funds allocated to the agricultural sector. Little effective application of existing regulatory texts. Infrastructure management structures exist but are not dynamic and functional. Absence or weakness of technical agricultural advisory services. Limited use of technological innovations. Weak access to appropriate adaptation technologies and practices. Weak capacity to predict the impact of climate change in the project area. Weak organisation capacities of farmers and poor connection to production and market support services.	<u>Medium</u>	Government institutions continue to maintain and build climate smart infrastructure. Local communities adopt climate smart practices and manage land and water resources sustainably. Farmers have continued access to markets and continuously increase their agricultural production. Agricultural development institutions take ownership of activities and ensure the sustainability of the investments from an institutional standpoint. DEFIS+ feeds into national policy dialogues and strategies, particularly those pertaining to climate change, to ensure that the project's components are integrated into and can inform the country's legal and policy framework. Partnerships between producers and market organisations, as well as the inclusion of smallholder farmers in the formal economy will generate social and economic benefits that last well beyond the project's duration. Innovative climate prediction tools can reduce the impact of climate change and improve human well-being.	● The establishment of simple and efficient water harvesting systems that require minimal technical skills and that can be easily managed and maintained by beneficiaries beyond the project's lifetime. ● Beneficiaries adopt climate smart practices & efficient water use management. ● A maintenance programme for roads, storage and packaging infrastructures is established by the communities and financed through user contributions/ service payments and Regions budgets. ● Rural communities have access to better climate change predictions and, consequently, can adapt better to the negative impact of climate change. ● Community members are trained in operation and maintenance of small-scale infrastructure. ● Climate smart infrastructure and agricultural practices are integrated into the country's legal and policy framework. ● Strengthened sustainable business relationships between farmers' organisations and market operators. ● Households have diversified their sources of income. ● Small-scale renewable and efficient energy options, such as biogas production units and solar dryers' technologies and improved cooking stoves (as alternative to firewood and

				charcoal) are adopted and continue to be built using local materials and local expertise.
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E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷⁶	
Total number of direct and indirect beneficiaries	Core 2: Direct and indirect beneficiaries reached	Mid-term and Final Evaluation Reports; District Development and Economic Reports	0	Direct: 178 881 (Male: 89 440; Female: 89 441) Indirect: 1 417 488 (Male: 708 744; Female: 708 744)	Direct: 447 201 (Male: 223 601; Female: 223 600) Indirect: 3 543 719 (Female: 1 771 859; 1 771 860)	As outlined in Annex 18, direct beneficiaries are those receiving direct interventions from the GCF funded project. The 447,201 beneficiaries will be benefitting from the following interventions: i.) The development of efficient water management systems; ii.) Regular access to reliable-climate data; iii.) The diffusion and adoption of resilient agricultural inputs, new technologies and management practices; iv.) Improved climate resilience of basic rural infrastructure; v.) Training and capacity building services; vi.) Knowledge management on the adaptation of food production

⁷⁶ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

						systems to climate change and on carbon sequestration. Methodology: The project-level M&E Specialist will provide a registry for each DEFIS+ region to be filled in. The information will be consolidated and used to assess the level of achievement of the targets. These registers will be supplemented by surveys conducted every 2 years.
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Mid-term and Final Evaluation Reports; District Development and Economic reports	0	Direct: 178 881 (Male: 89 440; Female: 89 441) Indirect: 1 417 488 (Male: 708 744; Female: 708 744)	Direct: 447 201 (Male: 223 601 Female: 223 600) Indirect: 3 543 719 (Female: 1 771 859; Male: 1 771 860)	Stable political and macro-economic environment
ARA2 Health, well-being, food and water security	Core 2: Direct and indirect beneficiaries reached	Mid-term and Final Evaluation Reports; External data will be collected from Governmental reports, surveys from research institutions, UN reports (FAO, UNICEF), etc.	0	Direct: 178 881 (Male: 89 440; Female: 89 441) Indirect: 1 417 488 (Male: 708 744; Female: 708 744)	Direct: 447 201 (Male: 223 601 Female: 223 600) Indirect: 3 543 719 (Female: 1 771 859; Male: 1 771 860)	Agro-forestry and watershed management activities are effectively implemented
ARA3 Infrastructure and built environment	Core 2: Direct and indirect beneficiaries reached	Mid-term and Final Evaluation Reports; District Development and Economic reports	0	Direct: 178 881 (Male: 89 440; Female: 89 441) Indirect: 1 417 488 (Male: 708 744; Female: 708 744)	Direct: 447 201 (Male: 223 601 Female: 223 600) Indirect: 3 543 719	Farmers assisted with heavy rehabilitation works

				744; Female: 708 744)	(Female: 1 771 859; Male: 1 771 860)	
ARA 4: Ecosystems and ecosystem services	Core 2: Direct and indirect beneficiaries reached	Mid-term and Final Evaluation Reports; Annual Progress Reports; District Development and Economic reports	0	Direct: 178 881 (Male: 89 440; Female: 89 441) Indirect: 1 417 488 (Male: 708 744; Female: 708 744)	Direct: 447 201 (Male: 223 601 Female: 223 600) Indirect: 3 543 719 (Female: 1 771 859; Male: 1 771 860)	Adoption of climate resilience technologies
ARA1 Most vulnerable people and communities	Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options	Mid-term and Final Evaluation Reports; District Development and Economic reports	0	Direct: 178 881 (Male: 89 440; Female: 89 441) Indirect: 1 417 488 (Male: 708 744; Female: 708 744)	Direct: 447 201 (Male: 223 601 Female: 223 600) Indirect: 3 543 719 (Female: 1 771 859; Male: 1 771 860)	As outlined in annex 18, this includes the number of farmers trained in climate-resilient agricultural production technologies and practices and indicating that their knowledge of climate adaptive practices has increased (ARA1) Methodology: The project-level M&E Specialist will provide a registry for each DEFIS+ region to be filled in. The information will be consolidated and used to assess the level of achievement of the targets. These registers will be supplemented by surveys conducted every 2 years.

ARA2 Health, well-being, food and water security	Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security	Mid-term and Final Evaluation Reports; District Development and Economic reports; UN water reports	0	Direct: 107 328 (Male: 53 664; Female: 53 664) Indirect: 850 493 (Male: 425 246; Female: 425 247)	Direct: 268 321 (Male: 134 160; Female: 134 161) Indirect: 2 126 231 (Male: 1 063 116; Female: 1 063 115)	As outlined in annex 18, this refers to the number of farmers using renewable energy and energy efficient and water efficient technologies (ARA3) leading to improved and more reliable (year-round and year-to-year) access to safe, sufficient and affordable water to meet basic needs for drinking, sanitation and hygiene. Assumption: Water units owned by communities not HHs Methodology: The project-level M&E Specialist will provide a registry for each DEFIS+ region to be filled in. The information will be consolidated and used to assess the level of achievement of the targets. These registers will be supplemented by surveys conducted every 2 years.
ARA2 Health, well-being, food and water security	Supplementary 2.2: Beneficiaries (female/male) with improved food security	Mid- term and Final Evaluation Reports; District Development and Economic reports	0	Direct: 178 881 (Male: 89 440; Female: 89 441) Indirect: 1 417 488 (Male: 708 744; Female: 708 744)	Direct: 447 201 (Male: 223 601; Female: 223 600) Indirect: 3 543 719 (Female: 1 771 859; Male: 1 771 860)	As outlined in annex 18, this includes the number of farmers trained in climate-resilient agricultural production technologies and practices and indicating that their knowledge of climate adaptive practices has increased (ARA1)

						Assumption: Adoption of climate resilience technologies Methodology: The project-level M&E Specialist will provide a registry for each DEFIS+ region to be filled in. The information will be consolidated and used to assess the level of achievement of the targets. These registers will be supplemented by surveys conducted every 2 years.
ARA3 Infrastructure and built environment	Core 3: Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions	Annual Progress Reports; District Development and Economic reports	0	USD 10 215 196 8 500 ha of irrigated perimeter 300 km of roads rehabilitated	USD 18 573 066 15 000 ha of irrigated perimeters 540 km of roads rehabilitated	Assumption: Farmers assisted with heavy rehabilitation works. Trainers, actors and professionals will be trained on new standards (ARA3). Methodology: The project-level M&E Specialist will provide a registry for each DEFIS+ region to be filled in. The information will be consolidated and used to assess the level of achievement of the targets. These registers will be supplemented by surveys conducted every 2 years
ARA 4: Ecosystems and ecosystem services	Core 4: Hectares of natural resource areas brought under improved low emission and/or climate-resilient management practices	Mid-term and Final Evaluation Reports; GIS	0	50% of restored ecosystems	100% of restored ecosystems	Adoption of climate change adaptation activities with mitigation co-benefits and

						restoration of 1,500 ha of forest ecosystems.
ARA 4: Ecosystems and ecosystem services	4.1 Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal-marine areas brought under restoration and/or improved ecosystems (unit: hectares)	Annual Progress Reports; Mid- term and Final Evaluation Reports	0	6,000 ha Terrestrial forest: 2,100 ha Terrestrial non-forest: 3,900 ha	18,100 ha Terrestrial forest: 6,330 ha Terrestrial non-forest: 11,770 ha	Planned agro-forestry and watershed management activities are effectively implemented. Annex 18 also includes information on the number of farmers practicing agroforestry and adaptive practices (ARA4) Methodology: Information available in project implementation and supervision reports. These reports can be complemented with surveys and GIS mapping.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation</u>	Low-performing subsistence farming systems. Malagasy farmers are particularly vulnerable to any shocks affecting the agricultural system owing to their high livelihood dependence on the agriculture sector, high occurrence of climate related disasters, limited climate resilience investments and capacity, and lack of access to formal safety nets.	<u>low</u>	Malagasy farmers are enabled & supported in the adoption of more efficient and sustainable agricultural practices (e.g., new technologies and access to financial products to afford them) allowing them to increase their climate resilience, production and incomes.	● Capacity building & trainings ● Adoption of smart practices ● Increased financial access ● Increased access and use of climate information ● Improved land and water management ● Climate resilient infrastructure	<u>Multiple sub-national areas within a country</u>
<u>Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level</u>	Farmers sell very little produce due to i.) low production levels; ii.) low access to markets due to poor roads and iii.) low bargaining power & informal relations with market players.	<u>low</u>	Malagasy farmers are able to produce enough to meet their own consumption needs and to sell produce in markets. Farmers have access to these markets and the ability to assure a fair price.	● Increased access to markets ● Increased access to market information ● Contractual relations between agricultural market players are strengthened.	<u>Multiple sub-national areas within a country</u>

E.5. Project/programme specific indicators (project outcomes and outputs)

*This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the “**Project/programme co-benefit indicators**” section in table below. Add rows as needed. Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).*

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1: Strengthened climate resilience of agricultural production systems	Average yields of agricultural production	Monitoring and Supervision Reports	0	15%	30%	Adoption and implication of beneficiaries; No major changes in the country's Agricultural Policy
Output 1.1. Improved water management for sustainable agriculture	Area of land with improved water management systems	Monitoring and Supervision Reports; District Development and Economic reports; Basin Management Regulations	7096 ha of land is irrigated in the intervention area by other projects and programmes	8 500 ha	15 000 ha	Adoption and implication of communes; No major changes in the country's Agricultural policy
	Protected areas through the implementation of anti-erosion measures, protection of watersheds and reforestation	Monitoring and Supervision Reports; District Development and Economic reports; Basin Management Regulations	0	5 000 ha	8 100 ha	Adaptation by farmers and implication of commune and region; Development plans at different levels put in place, equipped with the means to implement them, and benefiting from guaranteed land security
	Number of farmers engaged in the effective management of connects sub-watersheds and	Monitoring and Supervision Reports; District Development and Economic reports	0	Direct: 107 328 (Male: 53 664; Female: 53 664)	Direct: 268 321 (Male: 134 160; Female: 134 161)	Engagement of farmers and incentives to adopt improved water management techniques

	irrigated perimeters and adopting improved water management techniques			Indirect: 850 493 (Male: 425 246; Female: 425 247)	Indirect: 2 126 231 (Male: 1 063 116; Female: 1 063 115)	
Output 1.2: Enhanced agro-climatic information systems, new technologies and initiatives	Number of farmers with access to agro-climate data	Monitoring and Supervision Reports; District Development and Economic reports; field surveys	0	Direct: 71 552 (Male: 35 776; Female: 35 776) Indirect: 566 995 (Male: 283 498; Female: 283 497)	Direct: 178 881 (Male: 89 440; Female: 89 441) Indirect: 1 417 488 (Male: 708 744; Female: 708 744)	Engagement of DGM, SRM and DRAEP
	Areas covered by climate smart agricultural and agroforestry systems	Monitoring and Supervision Reports; District Development and Economic Reports	0	6 000 ha	10 000 ha	Adoption by farmers of climate smart agricultural and agroforestry systems
	Number of farmers reporting the adoption of new/improved technologies, practices or inputs	Monitoring and supervision reports; District Development and Economic reports; Field surveys	0	Direct: 107 328 (Male: 53 664; Female: 53 664) Indirect: 850 493 (Male: 425 246; Female: 425 247)	Direct: 268 321 (Male: 134 160; Female: 134 161) Indirect: 2 126 231 (Male: 1 063 116; Female: 1 063 115)	Interests/incentives of the actors is maintained
	Number of farmers using renewable energy and energy efficient and water efficient technologies	Monitoring and supervision reports; District Development and Economic reports; Field surveys	0	Direct: 107 328 (Male: 53 664; Female: 53 664) Indirect: 850 493 (Male: 425 246; Female: 425 247)	Direct: 268 321 (Male: 134 160; Female: 134 161) Indirect: 2 126 231 (Male: 1 063 116; Female: 1 063 115)	Interests/incentives of the actors is maintained

					Female: 1 063 115)	
Outcome 2: Increased income generation through improved market accessibility	Average annual farm income per household	Monitoring and supervision reports; Household survey	838 100 ariary/190.447 USD	4 000 000 ariary/909.09 USD	4 500 000 ariary/1 022.72	Increase in the number of productions placed on the market and development of long-term relationships with market operators Exchange rate: 1 USD = 4 400 ariary
Output 2.1. Reinforced rural access roads	Kilometres of unpaved rural access roads in good condition all year	Monitoring and Supervision Reports; District Development and Economic reports	0	300 km (USD 27 million)	540km (USD 48.6 million)	Adoption and implication of communes
	% of ground water recharge at specific sites along the greened road	Monitoring and recording of hydrological data	0-1% (average) ⁷⁷	Increase in ground water recharge by at least 2.5% at specific sites	Increase in ground water recharge by 5% at specific sites ⁷⁸	Farm ponds and check dams are constructed, and community support ensures their sustainability and care
Output 2.2. Improved capacity to build and maintain rural infrastructure	Number of trainers, actors and professionals trained on new standards and whose maintenance capacity has improved as a result	Monitoring and Supervision Reports; Field Surveys; Training Reports; Follow up tests on trainings	0	Direct: 71 552 (Male: 35 776; Female: 35 776) Indirect: 850 493 (Male: 425 246; Female: 425 247)	Direct: 178 881 (Male: 89 440; Female: 89 441) Indirect: 1 417 488 (Male: 708 744; Female: 708 744)	Adoption and implication of the actors; The control and validation bodies remain operational
Outcome 3: Improved food and nutrition security	Households receiving targeted support to improve their nutrition	Monitoring and Supervision Reports; Field Surveys; Training Reports;	0	100 000	270 000	Adoption of good nutrition practices

⁷⁷The amount of ground water recharge is influenced by various factors which include topography, land use, land cover, among others.

⁷⁸This estimation is based on the globally estimated average annual recharge rates for semi-arid and arid regions (see Hoori Ajami, Groundwater Recharge in [Encyclopedia of Geology](#) (Second Edition), 2021)

Output 3.1 Knowledge on climate change and food security generated and shared	Number of studies carried out; Number of knowledge products produced; Number of awareness events; Number of exchange visits	Capitalization reports; Baseline study; Mid-term study	0	8 studies 10 thematic knowledge products 5 awareness events 15 exchange visits	20 studies 20 thematic knowledge products 10 awareness events 30 exchange visits	Expertise is available at the different levels
	Number of farmers trained in climate-resilient agricultural production technologies and practices and indicating that their knowledge of climate adaptive practices has increased	Monitoring Reports and Surveys; District Development and Economic reports; Follow up tests on trainings	0	Direct: 178 881 (Male: 89 440; Female: 89 441) Indirect: 1 417 488 (Male: 708 744; Female: 708 744)	Direct: 447 201 (Male: 223 601; Female: 223 600) Indirect: 3 543 719 (Female: 1 771 859; 1 771 860)	Expertise is available at different levels
Project/programme co-benefit indicators						
Improved health	Percentage of women of reproductive age in the targeted areas reporting a minimum dietary diversity (measured with the MDD-W indicator: Minimum-Dietary Diversity-Women)	DEFIS baseline, mid-term and end-line survey	7,1% of women have attained the MDD in the project area	50%	70%	Adoption of climate-resilient practices, improved incomes and adoption of good nutrition practices. No major events affects food availability.
Improved WASH	Number of people reporting improved access to basic sanitation	Government reports, UN agency reports (UNICEF, UNWATER, etc),	10% of the population have access to basic sanitation.	Linked to beneficiaries of Output 1.1. Direct: 107 328 (Male: 53 664;	Linked to beneficiaries of Output 1.1 Direct: 268 321 (Male: 134 160;	Improved water management lowers risks of water borne diseases and population is also facilitated in adopting improved sanitation practices

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.

Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.

Activities	Description	Sub-activities	Deliverables
Activity 1.1.1. Protection of irrigation schemes against climate change impacts	Protection of irrigated schemes from the destructive impacts of violent rain and climate impacts	Improve drainage networks, irrigation channels, de-silting systems, water retention works, etc.	15 000 ha of irrigated schemes strengthened and improved
Activity 1.1.2. Sustainable sub-watersheds (adjacent to irrigated schemes) management and restoration of landscape	Biological and mechanical protection of sub-watersheds, soil management, water infiltration management Planting fast grown trees and species resilient to climate change	Plant rows of herbaceous plants or shrubs along curves; build benches, terraces and mechanisms to prevent gullies from forming Establishment of nurseries and production plants Implementation of reforestation activities	8 100 ha of land improved on water infiltration with anti-erosion systems of which 5 500 ha of landscape restored and reforested 900 technicians are trained
Activity 1.1.3. Strengthening capacities of Water Users Associations to manage water and apply sustainable water management practices	Training on climate change adaptation, water resource preservation and management plans Organization of exchange visits between Water Users Association (WUA)	Information on and identification of leaders; organization of visits; facilitation of the evaluation of the visits' results	1 400 WUA members exchanging and learning good practises
Activity 1.1.4. Construction of climate smart water tanks for drip-irrigation and for small livestock	Build a surface water storage system to irrigate the terraces and reduce runoff	Identification of sites and user; technical feasibility study and construction. Training for farmers	150 water tanks in use
Activity 1.2.1. Installation of automated agro-meteorological stations and development of flood and drought monitoring and forecasting system	Acquisition and installation of automated agro-meteorological stations	Partnership with DGM; choice and acquisition of stations in collaboration with DGM; installation of the stations; training for technicians	6 automated agro-meteorological stations are operational
Activity 1.2.2. Capacity building of decentralised government (SRM & DRAEP)	Support for processing and distribution of agro-meteorological data	Partnership with SRM and DRAEP; elaboration of work plans; support for the execution of work plans,	24 DRAEP and SRM teams trained and involved in processing and

		especially data distribution; monitoring of the quality of the data distribution system	disseminating agro meteorological data 178 880 farmers use agro meteorological information in the implementation of the cultural campaign
Activity 1.2.3 Promotion and dissemination of meteorological and climate smart agriculture practices	Area restored with the adoption of intelligent agriculture	Farmers practicing Climate Smart Agriculture Practices	Adoption of intelligent agricultural on 10 000 ha
	New methods of integrated pest management disseminated	Training farmers on modern pest management control in the FFS	At least 3 new methods of integrated pest management techniques adapted to the context of FFS
	Quantity of additional seeds produced by the CPSA per year (tons)	Support the Agnarafaly Seed Production Center (CPSA) and Kianjavato Center in the multiplication and promotion of certified or quality declared seeds.	Several tons per year of certified or quality declared seeds
	Numbers of farmers benefiting from improved climate-resilient species and varieties of seeds	Training farmers on Climate Smart Agriculture Practices and resistant seed varieties	Annex 18 includes the number of people adopting improved and/or new climate-resilient livelihood options
	Number of REEPS built	Climate smart water tanks installed and made available for livestock, crops, and farmers. Two models: conventional water storage tanks (gravity water tanks) and buried sand-filled water tanks	150 units of climate smart water tanks are installed
	Area with resilient fodder production	Training to improve the quality and quantity of fodder. Production of high-quality resilient fodder for the livestock	1 500 ha of land will be improved to produce resilient fodder
Activity 1.2.4. Promotion and dissemination of sustainable home technologies and social initiatives	Number of farmers using renewable energy and energy efficient and water efficient technologies	Identification of farmers benefiting from green renewable technology	2 500 farmers trained on the utilisation of new technologies

Activity 2.1.1. Reinforcement of unpaved rural access roads to reduce climate change impacts and water collection along these roadways for agricultural use	KM of roads repaired (repairs to critical sections of rural access roadways, drainage, pavement; construction of underground water recharge structures)	Identification of roads in need of repair and the works that need to be done; completion of road work; support to communes for the establishment of a maintenance program	540 km of roads are repaired or improved to make them more resilient to climate change
Activity 2.1.2. Training on construction standards related to climate change	Number of technicians trained on construction standards related to climate change	Information on and identification of individuals to train; organisation of training sessions; monitoring of the appropriation of knowledge	178 880 regional and local technicians trained on construction standards related to climate change
Activity 3.1.1 Training rural development actors on climate change and food production systems for enhanced nutrition	Number of rural actors trained on climate change impacts and measures (training on climate change; training on nutrition and gender; farmer field schools (FFS) on climate change)	Building awareness, information, training sessions; monitoring of the appropriation of knowledge; monitoring of their use Development/update of training manuals, organisation of training sessions through FFS and community based SBCC activities Selection and awareness-building of trainees; organisation of farmer field schools; support for trained producers	447 200 trained rural actors 45 000 producers trained on improved management practices of which 50% are women and 30% are youth
Activity 3.1.2. Exchange visits on climate resilient food systems	Number of people involved in exchange visits and indicating that their knowledge of climate adaptive practices has increased	Identification of best practices, organising exchange visits, promoting peer mentoring	360 participants in exchange visits
Activity 3.1.3. Capitalization of best practices and knowledge on food and nutrition security measures	Number of best practices products	Identification of best practices and successful approaches, carrying studies and publishing knowledge products	20 studies carried and knowledge products produced Degradation baseline enriched

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

The monitoring, reporting and evaluation (M&E) system of DEFIS+ will focus on the results and outcomes identified in the logical framework and AWP/Bs. It follows the procedures clearly set out in the IFAD Monitoring Manual (third edition). This system is also used to assess the work of implementing partners in order to guarantee regular, systematic implementation of harmonised tools for data collection and management (databases, GIS) and reporting, as well as for knowledge management to evaluate knowledge capital and identify good practices.

A participatory M&E process will be used to ensure the establishment of a good understanding of stakeholders' expectations and a common approach since the beginning. This will facilitate the appropriation and adoption of the information and results of the analysis generated by the system. This approach seeks to engage all stakeholders in: establishing a deeper, common comprehension of the objectives, strategies and appropriate information needs; organising collective moments of critical reflection to adjust intervention strategies where needed; and using monitoring data as tools to support management and decision-making.

DEFIS+ M&E unit is led by an M&E officer, with the support of a specialist in charge of monitoring climate change adaptation that will be recruited, a national assistant and M&E officers based in the inter-regional coordination units. The M&E officer on climate change adaptation and mitigation, to be hired with GCF funds, will be fully responsible for all M&E and knowledge management activities financed by the project.

The M&E system will be used for: continuous, systematic internal monitoring of the implementation of planned activities and progress on different levels of outcomes, assessed on a biannual and annual basis; periodic internal evaluations specific thematic evaluations undertaken to respond to identified needs; and periodic external evaluations. The latter includes baseline surveys carried out at the beginning of the project and impact assessments at the project end to determine the benefits generated by the project activities.

The quantification of adaptation beneficiaries (direct) is based on two conditions. The first condition is that the individuals receive direct interventions from the GCF funded project. The 447,200 beneficiaries will benefit from the following interventions: i.) The development of efficient water management systems; ii.) Regular access to reliable climate data; iii.) The diffusion and adoption of resilient agricultural inputs, new technologies, and management practices; iv.) Improved climate resilience of basic rural infrastructure; v.) Training and capacity building services; vi.) Knowledge management on the adaptation of food production systems to climate change and on carbon sequestration.

The second condition is that these beneficiaries receive measurable adaptation benefits. All of these interventions aim to increase the beneficiaries' adaptive capacity, which is a combination of assets and abilities that can be used to "prepare for and undertake actions to reduce adverse impacts, moderate harm, or exploit beneficial opportunities" (IPCC, 2012). The above interventions will directly lead to measurable adaptation benefits, including increased resilience to climate change, increased food and water security, and improved livelihoods. The ToC describes the way in which the interventions will lead to these benefits. Food security, for example, is enhanced through improved agricultural production, while improved water security will be achieved through better water management practices. The quantification of adaptation beneficiaries (indirect) will be a combination of primary and secondary data collection and extrapolation. Primary data collection will consist of surveys, and secondary data collection will consist of reviewing and triangulating national census data and information gathered from government counterparts and other agencies in the project area. It is envisaged that the following beneficiaries will be considered indirect beneficiaries of adaptation interventions. Details are included in Annex 11 (page 1-3 and tables).

Twice a year, the Government (MINAE, MEF, MEDD, and NDA) and IFAD will organise supervision missions with the goal of assessing progress on activities and the level of funds disbursement, the observance of administrative and financial procedures and the ability of the project activities to contribute to the achievement of the project's objectives. DEFIS+ will publish on a biannual and annual basis a progress report that presents an overview of the activities carried out and their cost, the products and results obtained and the level of progress achieved on the objectives

defined in the AWP/Bs as well as the expected outputs and outcomes. These reports will be distributed to the line ministry, NSC members, the NDA and IFAD. DEFIS+ will promote knowledge and learning in all investment areas, including the design and development of resilient agricultural infrastructures and production systems and efficient water management techniques. The overall approach to knowledge and learning relies on learning-based approaches such as farmer field schools, exchange visits, communities of practice, learning routes, contributions to thematic networks, e-Library, training, and capacity support and on already proven knowledge management methods and tools. The goal is to generate a detailed repertoire of good practices in the area of climate change adaptation.

DEFIS+ Outcome 3 is specifically designed to identify paths to knowledge development and build on partnerships between public and private entities to generate and disseminate further learning. The DEFIS+ M&E officer will be in charge of supporting these knowledge management activities in close collaboration with the local IFAD M&E officer. Approaches to identifying and disseminating lessons learned will include the use of various tools, including websites, case studies, policy briefs, training sessions, knowledge sharing events and workshops, radios, etc.

DEFIS+ will create sustainable enabling conditions for both climate change adaptation and rural economy transformation by building the capacity of local actors and regional authorities to plan infrastructure (e.g., rural roads and water management infrastructures) and natural resource use in line with expected climate change impacts. Furthermore, through Outcome 3, DEFIS+ will support an enhanced policy dialogue and the development of evidence-based policies. The GCF investments will lift the structural barriers to resilient agriculture related to infrastructures, production systems and/or capacity, which will further contribute to create an enabling environment. Assistance to key rural stakeholders such as chambers of commerce, farmer associations and market organisations will also help lift institutional barriers and create an environment that is more conducive to a productive and climate resilient sector and economy. Annex 11 includes a full description of the Monitoring and Evaluation Plans. Annex 11 includes a full description of the Monitoring and Evaluation Plans.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

The DEFIS+ risk is assessed as part of project design for IFAD funded projects within the parameters of IFAD's risk framework. It takes into consideration the country environment, PEFA reports, lessons learned from other projects implemented by the EE and diagnostics from other International Financial Institutions. Risk assessments are updated on a continuing basis throughout project life, based on performance and audit findings.

Introducing and adopting a paradigm shift towards sustainable and climate-resilient farming practices is a major challenge for both rural development actors and smallholders. The main risks that DEFIS+ will face during implementation are linked to: (i) the limited number and capacity of public and private service providers in remote rural areas; (ii) smallholder farmers' lack of knowledge and resources to invest in more efficient and more climate-resilient agricultural production systems; and (iii) producers' low level of organisation and integration into more lucrative value chains, which prevents them from obtaining higher prices for their agricultural products.

Climate change risks, which have been increasing in recent years, raise the risk of poor harvests or crop failures and, as a result, of problems of food security and the erosion of rural households' productive assets. The project's main risk factors and mitigation measures are presented in detail below. Overall, considering the mitigation measures proposed below, the level of impact of the risks and the probability of their occurrence will be medium to low.

Selected Risk Factor 1

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>
Description		

There are few public and private service providers in remote rural areas and the ones that do exist have limited capacity to intervene and few qualified staff members. These weaknesses may reduce the quality of support for producers and their appropriation of the proposed climate change adaptation measures.

Mitigation Measure(s)

Carefully defined personnel requirements will be broadly publicized to ensure that the most qualified individuals are able to apply. Service providers will be selected based on their skills, expertise and geographical presence. Partnership agreements and service contracts will be results-based and renewable based on performance. The staff of inter-regional coordination units and the national office will support and monitor service providers regularly. DEFIS+ will also propose training for personnel responsible for giving support to rural producers. The FFS approach, which promotes peer to peer learning and the capacity building of local technicians and extension services will also ensure more local level expertise is available.

Selected Risk Factor 2

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>

Description

Smallholder farmers lack the training and resources needed to adequately manage the resources of their natural surroundings. Their level of organisation is insufficient to enable them to increase their productivity levels and seize market opportunities. The few farmers' organisations that exist are often not very professional, act only locally and do not have the capacity to expand their range of activities, connect to services networks or integrate into more developed value chains.

Mitigation Measure(s)

Meetings and information sessions will be held with rural communities on the support that the program proposes to offer. The goal is to clearly explain the kinds of support offered, the expected results, the selection process and the eligibility conditions for beneficiaries. The meetings will serve to clarify doubts, discuss the expectations and commitments of the different parties involved and build trust among group members and in the program. Mechanisms to strengthen communication and ensure transparent decision-making will follow IFAD guidelines. The program will support the organisation of the groups, training for their members and their access to resources to be used to develop services for their members. Small-scale seed producer organisations will be called on to play an important role in the dissemination of climate-resilient varieties and species. The organisational and technical training sessions offered in the field and exchange visits shall help producers to improve their knowledge and capacity to analyse the constraints they face and to manage farming activities.

Selected Risk Factor 3

Category	Probability	Impact
<u>Governance</u>	<u>Medium</u>	<u>Medium</u>

Description

Tensions and conflicts over access to developed land, particularly the plots along the irrigated perimeters, may arise among people/farmers/herders and more vulnerable households.

Mitigation Measure(s)

Meetings and information sessions will be held in advance, before investments are made. These meetings shall serve to eliminate all ambiguities and discuss potential problems or conflicts that could hinder or even block progress on development activities or the future use of developed plots of land. A preliminary study on the land tenure situation of the different plots will be carried out. Communities will be associated to the planning and management of the works based on participatory maps. In the planning phase of the works, coordination and management committees will be formed and officially registered as user associations. Traditional and administrative authorities and commune officials

will be fully involved in the planning and monitoring of the implementation of the investments since the beginning of the project.

Selected Risk Factor 4

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>

Description

Climate risks are severe, particularly in the DEFIS+ area of intervention, which is highly exposed to prolonged droughts and violent rains. The risks of sand accumulating in water retention or distribution works are high. Cyclones often have devastating impacts on infrastructures and crops.

Mitigation Measure(s)

The irrigated perimeters will be designed and built in accordance with the NIHYCRI and systems to protect areas from floods and erosion coming from the sub-watersheds will be strengthened. Infrastructures will be built in compliance with standards for protection against cyclones to limit damages. Training and support for WUAs will improve the management of water resources and irrigation networks and therefore, the performance of irrigated perimeters. Anti-erosion works and improvements to vegetation cover will allow farmers to preserve and improve the productivity potential of their fields. Strengthening community management capacities is fundamental to ensure the adequate use and preservation of watershed resources.

Selected Risk Factor 5

Category	Probability	Impact
<u>Governance</u>	<u>Medium</u>	<u>Medium</u>

Description

It is often very difficult for beneficiaries, the communes or state services to ensure the upkeep and good maintenance of rural infrastructures.

Mitigation Measure(s)

DEFIS+ will adopt an approach for the appropriation of the works by beneficiary WUAs that begins in the planning, monitoring and participation in the works and continues until the official handover of the works. At that point, a definitive transfer agreement will be signed, which defines the responsibilities of the different stakeholders after the program has ended. In training activities for WUAs, the management and maintenance of the irrigation systems will be important issues to address. Finally, MINAE technical departments and commune officials will be closely linked to the rehabilitation and monitoring of rural infrastructures.

Selected Risk Factor 6

Category	Probability	Impact
<u>Health</u>	<u>Low</u>	<u>Low</u>

Description

The WHO does not consider COVID-19 a pandemic disease anymore, however, the risk of new strains exists and it is real.

Mitigation Measure(s)

In case a new pandemic disease outbreak is declared by the WHO and the Government, DEFIS+ will put in place all necessary measures to mitigate the risk of spreading the disease.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

The Social, Environmental and Climate Assessment Procedures (SECAP) are IFAD's main instrument for enhancing the sustainability of its projects and programs and include a risk assessment process that recognizes the heterogeneity of responses that exist given the wide range of circumstances in different countries and communities. SECAP are geared towards evaluating the risk and exposure of IFAD interventions to avoid, minimize and mitigate potential negative social and environmental impacts. SECAP also focuses on climate change to assist project/program managers in assessing the potential damages that it will cause to IFAD investments and in connecting adaptation to risk management and poverty reduction.

The SECAP describes how to better mainstream environmental, social and climate change considerations into DEFIS+ project cycles and demonstrate IFAD's commitment to go beyond the "do no harm" principle and to maximize development gains. SECAP seeks to ensure that IFAD's policies, strategies and investments are designed to leave no one behind and achieve sustainable development for all, especially the poorest and most vulnerable to climate change. SECAP provides (i) an overview of the main environmental, natural resource management and social issues in the intervention areas, namely any significant impacts (positive and negative) and social concerns likely to be associated with the project investments and services and (ii) the specific adaptation measures, envisaged on the basis of the risk analysis associated to the investments planned. The main risks identified (and corresponding mitigation measures) for DEFIS+ are: (i) adverse impacts of rehabilitation/construction work (hence, the application of national environmental and social impact assessment legislation); (ii) inefficient water use in rehabilitated areas (hence, capacity-building, coordination activities for WUAs, dissemination of water efficient agronomic practices such as SRI and its variants); (iii) erosion (hence, dissemination of anti-erosion techniques such as planting rows of shrubs and trees, strip cropping, shelterbelts; integrating cover crops, shrubs, soil improvers or fodder crops into crop rotations); and (iv) limited access to project benefits for the most vulnerable farmers (hence, preliminary land tenure survey and proactive support to facilitate farmers access to DEFIS+ benefits). All activities identified in the Environment and Social Management Plan (Annex 6) are included in DEFIS+, according to the nature of the activities and their respective budgets.

DEFIS+ project is categorized by IFAD as category B, as it might result in adverse environmental and/or social impacts on human populations or environmentally significant areas, although the impacts are less adverse than those for category A, site-specific and can be readily remedied by appropriate preventive actions and/or mitigation measures. Category B projects require an Environmental and Social Management Plan (ESMP) to be developed, which is the case for DEFIS+. The SECAP also foresees the application of a national Environmental and Social Impact Assessment legislation (MECIE decrees 99-954 and 04-167) for infrastructures-related works. IFAD's categorization of the project is in line with GCF environmental and social risk category B, which includes activities with potential limited adverse environmental and/or social risks or impacts that are few, generally site-specific, largely reversible and readily addressed through mitigation measures.

In addition to SECAP procedures, DEFIS+ is fully compliant with the national regulations on environmental and social safeguards of the National Office of Environment (ONE). ONE is the Authority mandated to screen all investments and irrigation works, including rural road rehabilitation, in order to identify the specific regulatory requirements for each one of them, depending on their environmental, social and climate impacts. The Environmental and Social Impact Assessment will be validated by ONE, while the ESMP, will be validated by the MINAE. Over the years, the MINAE and its dedicated Environment, Climate and Emergency Response Unit (SECRU) have developed extensive expertise on national and international environmental and social norms, particularly with regard to: assessing environmental and social strategies; monitoring environmental and social management frameworks; carrying out environmental impact studies; and supporting environmental engagement programs.

The DEFIS+ Environmental and safeguards officer will be the person in charge of ESMP implementation and monitoring. As already mentioned above, the PCU will ensure ESMP compliance with IFAD's SECAP and GCF's Environmental and Social Policy during project implementation. The climate change operations manager, and the Climate Change and Environment Officer in each inter regional coordination committee, will also be responsible for ESMPs implementation and monitoring throughout the project's duration. For the protection of the banks at the level of the hydro-agricultural infrastructures, vetiver, phragmites and herbaceous varieties will be used. Madagascar has developed Directives for reforestation, therefore the species and varieties adopted will depend on the pedoclimatic conditions of the areas and perennial trees as well as fruit trees, which are accepted by the beneficiary populations. The infrastructure activities will be screened for environmental and social risks based on the GCF Environment and

Social Policy, SECAP and ONE regulations. Thereafter, site specific ESMPs will guide the risk management by the contractors and beneficiaries.

A description of the program-level grievance redress mechanism is provided in Annex 7, together with a summary of the multi-stakeholder consultations carried out to present and gather feedback on the DEFIS+'s ESMP. The stakeholder engagement plan, also in Annex 7, provides indications on how DEFIS+ will engage with stakeholders throughout project implementation. Finally, Annex 7 presents disclosure-related information in line with both IFAD and GCF policies on this matter.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

The targeting and social inclusion strategy is based on the selection of intervention regions with high poverty levels and where soil degradation and the impacts of climate change hit the population's productive capital the most. Furthermore, DEFIS+ intervention strategy excludes commercial farms and only targets individuals, especially those with a food deficit (type 3). The targeting strategy was further informed by numerous visits and interviews with the communities and actors in rural areas.

Gender considerations will be mainstreamed throughout DEFIS+' activities. As detailed in the Gender Assessment and Action Plan (Annex 8), operational measures have been identified to guarantee that, while executing all programme activities, special care is systematically taken to ensure that women and youth from the most vulnerable beneficiaries effectively engage. Direct targeting will be carried out in beneficiary communities that have been identified to ensure that type 3 food deficit households, women and youth are taken into account and have the opportunity to access the resources and initiatives mobilised by DEFIS+. Eligibility criteria – based, for example, on age, gender, vulnerability level, education level, and socioeconomic status – will be defined with the beneficiary communities through a participatory and consensus-based process. DEFIS+' **gender strategy**, is based on the need to address inequalities related to sex, age and status in the implementation of the project activities. Its objective is to: (i) eliminate the constraints that limit the inclusion, active participation and empowerment of women and youth (male and female) in the different links of agricultural chains; (ii) create opportunities to improve their living conditions; (iii) strengthen their productive capacities and the resilience of their economic activities to climate change; and (iv) promote the leadership of women and their participation in decision-making groups and processes.

The specific DEFIS+ actions proposed for women and youth in vulnerable situations have the goal of facilitating their access to productive capital through employment or their involvement in income-generating activities. These actions include: (i) integrating women and youth into agriculture by reserving plots of land for them in the newly developed perimeters; (ii) positioning youth and women as rural entrepreneurs in the services field in areas linked to the poles of development; (iii) giving women and youth priority in capacity-building activities and access to funding; (iv) promoting pro-youth and pro-women participation in green jobs.

As in the case of the targeting strategy, the responsibility for executing the gender strategy lies with the entire project coordination team, as well as the different regional partners and institutions in charge of project implementation. These actors will receive training on gender issues and on DEFIS+ strategies and tools for implementing, monitoring and evaluating the gender approach and ensuring the achievement of its objectives. With the support of the technical project manager, the entire team in charge of implementing DEFIS+ will ensure the integration of a gender dimension in the implementation of the activities. DEFIS+ will support the operationalization of gender strategy in relation to specific climate resilient activities with the aim of strengthening the empowerment of women and young people.

The monitoring and evaluation system will integrate the gender strategy into its operationalization. In the measure of products, effects and impacts, the monitoring and evaluation indicators will be disaggregated according to gender and age (men, women and young people). DEFIS+ will conduct specific studies on the impact of the project on gender throughout its implementation (at the start, mid-term and end). The report will include the results of specific activities carried out by DEFIS+ as well as the results of actions carried out in synergy with IFAD projects in the country in relation to gender. Information on the results of specific activities carried out by DEFIS+ will be provided by its monitoring and evaluation system.

The PCU will monitor gender mainstreaming, in particular focusing on: (i) how gender issues are taken into account in the project; (ii) the appropriateness of the gender strategy in consideration of the context and sociocultural reality of the intervention zones; (iii) the disaggregation of M&E indicators by sex and age. This thorough exercise will enhance the narrowing of the gender gap in the target regions, paving the way for a scaling up of the approach at the national level.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

MEF will represent the Government of Madagascar in accordance with Decree 2019-093 establishing the Ministry attributions and its general organization. MEF supervises the planning, coordination and monitoring-evaluation of the country's medium- and long-term development programmes at all levels and the coordination of cooperation at the national level. The Ministry also manages, monitors and assesses external funds and contributes to the harmonisation of cooperation with donors.

MINAE will be the Implementing Agency for the project and will provide technical oversight for DEFIS+. The NSC will provide overall guidance and validate AWP/Bs and evaluate progress made on activities and project results and their alignment with rural sector strategies. Regional guidance will be provided by the RSMCs, who will also monitor the regional progress of implementation. The relevant departments of the RSMCs are represented in the NSC and have real decision-making power with regard to the mobilisation of resources, compliance with planning and implementation of action plans.

Within MINAE, the PCU's main function is to execute the overall programming and budgeting and to take the lead in overall project implementation, which will include working with service providers, government ministries and departments, individual project participants and farmers' organisations in the target areas. Specifically, the PCU will assume responsibility for the timely preparation of the AWP/Bs and Procurement Plans to be submitted for review and approval (and subsequently to IFAD for its "no-objection"). The PCU will lead in the implementation of the project's three components and will work closely with MINAE, MEDD and other implementing partners, to ensure that project outcomes are achieved.

IFAD's operational model ensures continuous implementation support and supervision of project activities by providing MINAE and, more specifically, the PCU with the required technical and expert guidance for effective implementation of project activities and adaptive management, should any issues arise throughout the project's lifecycle. IFAD's No Objection will be required for the approval of AWP/Bs, which will be screened to ensure project activities build towards successful project completion in an effective and efficient manner. In compliance with IFAD's Handbook on Financial Reporting and Auditing, adequate financial management arrangements (staff, budget, funds flow, internal control, audits) will be secured for efficiently managing GCF funds, including regular Interim Financial Reporting and liquidity forecast. At least yearly, external audits of the project's account will be conducted, as will supervision missions for monitoring and evaluation. This will be done in collaboration with staff from MEDD/NDA. Finally, IFAD with support from MINAE and the PCU will lead a mid-term review to ensure alignment with defined targets or propose corrective measures, and elaborate completion reports and impact assessments. Specific requirements for implementation support and supervision missions are fully developed in the Program Implementation Manual (PIM), which will be amended to integrate GCF implementation requirements.

As per clause 14 of the Accreditation Master Agreement between IFAD and GCF, the procurement of goods and services for funded activities, whether by the Accredited Entity itself, the Executing Entity or a third party, shall be done in accordance with the rules, policies and procedures of the Accredited Entity to the extent and scope of its accreditation.

In accordance with IFAD Procurement Guidelines, each procurement plan will include IFAD's proposed contracts, procurement methods and related review procedures. During the design phase, IFAD undertook an assessment of the national procurement regulations to ensure their compatibility with IFAD Guidelines. It was agreed that national regulations can be adopted for this program as defined in Law 2016-055 on the Public Procurement Code of 25 January 2017, except for the international competitive procurement, which follows the rules defined in the IFAD Procurement Guidelines. The Procurement Plan (Annex 10) includes the details of the procurement to be undertaken under DEFIS+.

Specific applicable points will be clearly identified in the MPCs and the Procedural Manual. In accordance with paragraph 3 of Module F5 of the Loan Disbursement Handbook, International Competitive Bidding (ICB) will be the mandatory procurement method for the following contracts:

- Goods: contracts over USD 200,000;
- Work: contracts over USD 1,000,000;
- Consulting services: contracts of over USD 100,000 for firms and contracts of over USD 40,000 for individuals.

In accordance with IFAD Procurement Guidelines, the Fund will carry out a prior review of the above if the contracts are equivalent to or above the specified amounts.

DEFIS+ will receive a variety of internal audits to ensure high procurement performance: periodic internal control by the CAPFIDA; internal audit by the Unit of Internal Audit of MINAE; annual joint supervision mission (IFAD and Government); and posterior review by the National Public Procurement Commission from the MEF.

G.4. Disclosure of funding proposal

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES	
H.1. Mandatory annexes	
☒	Annex 1 NDA no-objection letter(s) (template provided)
☒	Annex 2a Feasibility study - Climate Change Analysis
☒	Annex 2b Feasibility study - Climate Modelling and Vulnerability Analysis
☒	Annex 2c Feasibility study - Infrastructure Analysis
☒	Annex 2d Feasibility study - Value Chain Analysis
☒	Annex 3 Economic and/or financial analyses in spreadsheet format
☒	Annex 4 Detailed budget plan (template provided)
☒	Annex 5 Implementation timetable including key project/programme milestones (template provided)
☒	Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): (ESS disclosure form provided) ☐ Environmental and Social Impact Assessment (ESIA) or ☐ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People’s Plan, Land Acquisition Plan, etc.)
☒	Annex 7 Summary of consultations and stakeholder engagement plan
☒	Annex 8 Gender assessment and project/programme-level action plan (template provided)
☒	Annex 9 Legal due diligence (regulation, taxation and insurance)
☒	Annex 10 Procurement plan (template provided)
☒	Annex 11 Monitoring and evaluation plan (template provided)
☒	Annex 12 AE fee request (template provided)
☒	Annex 13 Co-financing commitment letter, if applicable (template provided)
☐	Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule
H.2. Other annexes as applicable	
☐	Annex 15 Evidence of internal approval (template provided)
☒	Annex 16 Map(s) indicating the location of proposed interventions
☐	Annex 17 Multi-country project/programme information (template provided)
☐	Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity

- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☐ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)⁷⁹
- ☐ Annex X Other references

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

⁷⁹ Annex 22 is mandatory for mitigation and cross-cutting projects.

No-objection letter issued by the national designated authority(ies) or focal point(s)



To: The Green Climate Fund ("GCF")

Antananarivo, 20 FEB 2020

Re: Funding proposal for the GCF by International Fond for Agricultural Development (IFAD) regarding the project "Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme (DEFIS +)".

Dear Madam, Sir,

We refer to the project "Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme" in five (05) regions of Madagascar, as included in the Funding Proposal submitted by IFAD to us on November 25, 2019.

The undersigned is the duly authorized representative of the Ministry of Environment and Sustainable Development, the National Designated Authority/focal point of Madagascar.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme.

By communicating our no-objection, it is implied that:

- (a) The government of Madagascar has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Madagascar's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Name: Lovakanto Ravelomanana

Title: National Designated Authority/Focal Point of Madagascar

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme
Existence of subproject(s) to be identified after GCF Board approval	[Yes]
Sector (public or private)	Public
Accredited entity	International Fund for Agriculture Development (IFAD)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Madagascar - Amoron'i Mania, Haute Matsiatra, Vatovavy, Fitovinany, Atsimo-Atsinanana, and Ihorombe regions
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Friday, February 2, 2024
Language(s) of disclosure	English and French
Explanation on language	French is an official language of Madagascar
Link to disclosure	English: [https://www.ifad.org/documents/38711624/44264690/esmf-madagascar-2000001492_e.pdf/7e7754da-f9c0-2ae7-ff4a-2bd0e834cab5?t=1706862960873] French: [https://www.ifad.org/documents/38711624/44264690/esmf-madagascar-2000001492_f.pdf/22766d89-15d6-2b1e-2ce4-487f7d26f6f7?t=1706862966004]
Other link(s)	[The links are accessible from the SECAP documents page at https://www.ifad.org/en/secap-disclosed-documents (https://www.ifad.org/fr/secap-disclosed-documents)]
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the “Environmental and Social Management Framework”.]
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, February 2, 2024
Language(s) of disclosure	English and French
Explanation on language	French is an official language of Madagascar
Link to disclosure	English: [https://www.ifad.org/documents/38711624/44264690/esmf-madagascar-2000001492_e.pdf/7e7754da-f9c0-2ae7-ff4a-2bd0e834cab5?t=1706862960873] French:

	[https://www.ifad.org/documents/38711624/44264690/esmf-madagascar-2000001492_f.pdf/22766d89-15d6-2b1e-2ce4-487f7d26f6f7?t=1706862966004]
Other link(s)	[The links are accessible from the SECAP documents page at https://www.ifad.org/en/secap-disclosed-documents (https://www.ifad.org/fr/secap-disclosed-documents)]
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the “Environmental and Social Management Framework”.]
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, February 2, 2024
Place	Amoron’i Mania Regional Directorate of Agriculture and Livestock, Tamponiviny, Ambositra 306 Haute Matsiatra Regional Directorate of Agriculture and Livestock, Matsiatra Ambony, Tanambao, 301; Rural Engineering Office Kianjasoa Fianarantsoa 341 Inclusive Agricultural Value Chains Programme Office Vatovavy and Fitovinany Regional Directorate of Agriculture and Livestock Offices ex ODASE Tanakidy Manakara 316. Office DEFIS: Batiment TTMR Tanakidy, Manakara 316 Atsimo-Atsinanana Regional Directorate of Agriculture and Livestock Fenoarivo Avaratra Farafangana, Ihorombe Regional Administrative Offices Bâtiment Regional Directorate of Agriculture and Livestock Ihorombe-Ankilibe, Ihosy, 313
Date of Board meeting in which the FP is intended to be considered	

Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, March 4, 2024

Note: This form was prepared by the accredited entity stated above.

Secretariat assessment for FP227

Proposal name:	Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme (DEFIS+)
Accredited entity:	International Fund for Agricultural Development (IFAD)
Country:	Madagascar
Project/programme size:	Medium

I. Overall assessment of the Secretariat

- The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
This project seeks to alleviate the detrimental impacts of climate change by deploying sustainable adaptation measures. Leveraging valuable insights acquired from previous projects funded by the International Fund for Agricultural Development (IFAD) and the ongoing pro-poor agricultural investment initiative, Inclusive Agriculture Transformation Programme (DEFIS+), supported by IFAD, the OPEC Fund for International Development, and other contributors, the project aims to optimize positive experiences to augment its impact and ensure investments are resilient to climate challenges. The new GCF project, known as DEFIS+, specifically builds upon the existing program. The project targets vulnerable communities in one of regions of the world that is most affected by climate change.	Throughout the project, it is imperative to execute a sustainability strategy. This strategy should encompass a funding blueprint for replication, as well as a comprehensive plan to incorporate operational and maintenance considerations into GCF investments. Additionally, it is crucial to investigate potential replication opportunities within Africa and Madagascar.
This project will help to strengthen the resilience of the agricultural production system and vulnerable marginal communities to the impacts of climate change in drought-affected areas by improving water management and strengthening the resilience of roads by applying the concept of “green roads” for water-harvesting and nature-based solutions.	
IFAD will conduct technical assessments and feasibility studies to guide the detailed designs	

of climate-resilient infrastructure, such as irrigation systems, roads, water management schemes, and activities within the agricultural value chain.	
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2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XVII, titled “List of proposed conditions and recommendations”.

II. Summary of the Secretariat’s assessment

2.1 Project background

3. Madagascar has been ranked the fifth most vulnerable country to climate change in the world, and the Notre Dame Global Adaptation Initiative index, which ranks the climate adaptation performance of 181 countries, has classified Madagascar as one of the countries least adapted to cope with climate events (ranking 172 out of 181 countries¹). Madagascar belongs to the group of least developed countries, with more than 80 per cent of the population living below the poverty level (USD 2 per day) and 80 per cent of the population living in rural zones, most of them in very remote areas with limited access to basic infrastructure. The poverty rate was 74.3 per cent of the population in 2019, corresponding to an increase of 1.38 million people in just one year (2020), and it is expected to continue to rise. Madagascar experienced unprecedented extreme and consecutive droughts from 2017 to 2022, affecting crop production, food and nutrition security across the country. The south-west and the extreme south are characterized by semi-desert environments with erratic and low rainfall of less than 800 mm per year. A decrease in average annual precipitation of up to 8 per cent is projected by 2050. No significant change is expected in terms of average annual rainfall accumulation; however, it is predicted that there will be an increase in the amplitude of inter-annual rainfall variability, as well as frequency of extreme events. This is particularly true in the project area, which is formed of six regions, in the south of the country. The great variability of rainfall in the project area leads to either droughts or situations where the water balance is very much in surplus owing to floods and heavy rains. Temperatures in southern Madagascar are estimated to increase by between 1.2 and 2.1 °C by 2050. Under the Intergovernmental Panel on Climate Change representative concentration pathway (RCP) 4.5 scenario, the number of days subject to extreme heat events is projected to increase from 32.4 to 54.7 by 2050, posing extremely high risks to human lives.

4. The main climate-related changes include (i) a decrease in the amount of annual precipitation, with more marked and/or more frequent periods of drought; (ii) a delay in the arrival of the rains, which requires adaptation in terms of the agricultural calendar; (iii) increased intensity of extreme climatic phenomena (cyclones, droughts, floods, storms and strong winds); and (iv) a very high temporal and spatial variability in rainfall, which is unpredictable. Additionally, smallholder farms are frequently exposed to outbreaks of pests and disease which cause significant crop and income losses. Since 2010, the areas most affected by drought events are in the project area, particularly the Atsimo-Atsinanana, Vatovavy and Fitovinany regions.

¹ <https://gain.nd.edu/our-work/country-index/rankings/> Accessed on 26 September 2023.

5. The greenhouse gas (GHG) emissions in Madagascar come predominantly from the land-based sectors, namely agriculture, forestry and land-use change, which together represent 80.5 per cent of Madagascar's national emissions. The emissions are mainly from livestock (enteric fermentation and manure) and agriculture (methane from rice cultivation, poor soil management, deforestation for agricultural purposes and fertilizer application).
6. The proposed project, Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme (known as DEFIS+), will help to reduce the negative impact of climate change by applying adaptation measures. It will build on positive lessons learned from previous International Fund for Agricultural Development (IFAD) projects and from the ongoing pro-poor agricultural investment programme, the Inclusive Agriculture Transformation Programme (DEFIS), funded by IFAD and the OPEC Fund for International Development and other donors.
7. The project will also allow capitalization of the potential of agroforestry, reforestation and renewable energy practices to provide mitigation co-benefits.

2.2 Component-by-component analysis

8. The project will be implemented around the following three complementary outcomes, which have been strategically conceived to contribute to climate change adaptation, with mitigation co-benefits:
 - (a) **Outcome 1: Strengthened climate resilience of agricultural production systems** and the enhancement and maintenance of ecosystem services by improving access to efficient water management systems for sustainable agriculture, which includes enhancing the resilience of 15,000 ha irrigation schemes and 8,100 ha subwatersheds through mechanical protection and ecosystem-based approaches to restore 5,500 ha landscape. In addition, it includes reforestation of degraded and marginal areas and sustainable agroforestry practices; improving access to regular and appropriate agro-climate data and disaster risk reduction using agro-climatic information systems; improving the climate resilience of food production systems for 45,000 small family farms through the promotion of climate-resilient farming technologies and practices and integrated pest management techniques; and introducing innovative, low-cost, context-specific renewable energy and energy-efficient technologies for farming and processing (such as solar pumps and dryers) and domestic activities (such as improved cookstoves to reduce the use of firewood), which contribute to reducing GHG emissions. The economic and financial analysis highlights the climate and environmental benefits envisaged by the project ;
 - (b) **Outcome 2: Increased income generation through improved market accessibility.** This outcome revolves around promoting "green roads" (i.e. climate-resilient rural tertiary roads), supporting the climate resilience of rural infrastructure used for access to markets and utilizing the water-harvesting potential of roads. This includes (i) investing in reinforcement works, including well-designed cross-drainage systems on 540 km rural roads to protect the roads from damage due to climate hazards, such as floods and heavy rainfall, and to harvest water for drought resilience by efficiently managing the water around the roads; (ii) training 2,000 trainers, construction workers and professionals on construction and maintenance standards, which will be more sustainable than the business-as-usual scenario in the long term owing to the reliance on local green technologies; (iii) improving climate resilience by introducing water harvesting options for increased water availability for farming, soil moisture

improvement and aquifer recharge; and (iv) adopting well-designed climate resilience cross-drainage systems to keep the roads functional all year round and not, as at present, being open only during the dry season (eight to nine months a year) and closed during the rainy season; and

- (c) **Outcome 3: Improved food and nutrition security** through managing knowledge of food production systems. Local skills and knowledge will be leveraged, for instance, through the development of construction operation manuals, in order to scale up innovation. This outcome is inherently linked to outcomes 1 and 2, as it benefits from the introduction of climate-smart practices and from increased income generation, both of which result in enhanced food and nutrition security and mitigate the negative impact of extreme climate events. Outcome 3 will also promote non-timber forest products and neglected and underutilized climate-tolerant food species as a strategy for adapting to climate change, reducing GHG emissions, reducing environmental degradation and improving agricultural diversification, livelihoods and food security.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: Medium to high

9. DEFIS+ proposes a comprehensive approach to sustainably improving resilience to climate shocks and enhancing nutrition and food security for vulnerable households in the target areas. DEFIS+ will enhance the capacity of agricultural systems and vulnerable rural communities to adapt to the adverse impacts of climate change so that farmers are better able to confront the increasing frequency and intensity of droughts and changes in rainfall patterns that often result in floods. This requires bringing up to date current agricultural production systems that are dependent on rainfall and whose practices and infrastructures are no longer appropriate in the context of climate change. DEFIS+ proposes to reduce smallholder farmers' exposure to climate and economic shocks by "climate proofing" irrigation infrastructure and rural roads to increase agricultural water availability, reduce the risk of market access disruptions caused by extreme climate events and promote and disseminate climate-smart agricultural practices. DEFIS+ will benefit 447,201 people, 50 per cent of whom are women.

10. At the field level, introducing water-saving technologies such as the implementation of drip-irrigation systems and the system of rice intensification, will improve the availability of water, reduce vulnerability to water shortages and increase productivity per hectare. This will increase the area of recoverable crops with a view to increasing levels of production. The support provided by DEFIS+, through the dissemination of drought- and stress-tolerant crop varieties and the adoption of integrated pest management methods and other improved agronomic practices, will contribute to further reducing production risk and sustainably increasing productivity. The resilience of production systems will be strengthened by investing in ecosystem restoration, including the promotion of an integrated watershed and catchment management approach. This reduces the risk of erosion and will improve the sustained productivity of farms through the retention of fertile topsoils. DEFIS+ will also promote low-cost on- and off-farm renewable energy and energy-efficient technologies, such as improved energy-efficient cookstoves and handmade briquettes, which will contribute towards reducing forest degradation. Other technologies to be promoted include solar pumps for irrigation, biogas production along with livestock systems, and solar milling and dryers for product processing. These technologies will contribute to improving efficiency and production and lowering GHG emissions. They will also contribute to reducing women's workload and time burden (i.e. the

amount of time spent on unpaid domestic work such as fetching water), with a positive impact on the health, education and nutrition status of women and, indirectly, their young children.

11. Through these interventions, it is estimated that the project will directly improve the climate resilience of 447,201 individual beneficiaries and indirectly impact the lives of 3,543,719 beneficiaries in the six regions targeted by DEFIS+, improving their ability to sustainably combat the negative effects of climate change while also helping them to produce greater yields and gain access to markets. This results in greater income generation for rural communities and reduced dependence on the exploitation of natural resources in lean periods. With the associated ecosystem regeneration activities implemented under the project, this will establish a positive feedback loop that will result in a greater impact on ecosystem restoration, while additional increases in biomass will have an associated reduction in GHG emissions and an increase in the carbon storage capacity of the targeted ecosystems.

3.2 Paradigm shift potential

Scale: High

12. The project constitutes an important step in a longer-term process of transforming agriculture in Madagascar. The required behavioural changes among a large and poor population highly dependent on agricultural output constitute an enormous task that the Government cannot achieve on its own. Several existing projects in the south of Madagascar will benefit from the DEFIS+ project through its contribution to transforming the climate-vulnerable and unsustainable agriculture sector into a green and sustainable climate-resilient, market-oriented and profitable economic sector with a modern infrastructure, which also significantly contributes to mitigating climate shocks and improving food and nutrition, security, gender equality and empowerment. The structural and capacity transformation will be achieved by addressing vulnerabilities to climate change and supporting all stakeholders in improving their resilience, efficiency and links to markets. The paradigm shift of DEFIS+ includes the following:

- (a) Climate proofing the production- and market-related infrastructure to improve its resilience. The integrated watershed and catchment management approach will scale up national experiences in bottom-up participatory watershed and catchment management with the dual aims of protecting natural resources and increasing income. Community-driven sustainable land and water management plans and activities will be supported. The planning process, which includes social engineering for behavioural change and implementation of the plans, will also facilitate knowledge-sharing and joint learning among peers in the water users associations and the farmer field schools;
- (b) The partnerships mobilized for DEFIS+, the scale of productive investments planned for sustainable agricultural development and the focus on building local capacities, together constitute a major paradigm shift in regions of southern Madagascar where humanitarian and emergency assistance is needed. The focus on scaling up well-tested approaches and technologies, building on innovative approaches and lessons learned from previous IFAD investments (including DEFIS and other IFAD projects in the country), is an important feature of the proposed interventions;
- (c) Access to stress-tolerant (regarding drought, heat, flooding, salinity, etc.) and nutritionally enhanced varieties and planting materials developed by research centres. The production, multiplication and dissemination at the community level of these climate-resilient technologies will be promoted, as will the adoption of integrated pest management and other agroecological approaches. Supporting national research centres and the farmer field schools will create an enabling environment for improved seeds promotion and adoption by the smallholders;

- (d) Greening the road infrastructure will have an overall positive impact on agriculture, water availability and soil erosion associated with running surface water, while having a minimal environmental impact and delivering large benefits for civil society; and
- (e) The project will use a decision support system to understand risk (e.g. using hazard exposure and vulnerability to assess the dynamic nature of impact scenarios for crops, land use and agri-infrastructure), including gender-related considerations and applications such as Météo Madagascar² and others provided by the Regional Meteorological Directorate.

13. The project will shift agriculture in the country's target areas from its current situation of high vulnerability towards an alternative paradigm whereby better information and climate-smart farming practices will significantly increase resilience to climate change. The use of state-of-the-art geographic information system tools will support strategic planning through systematic data-collection and management, which will act as a springboard for science-based decision-making processes.

14. The approach and the tools to be utilized constitute a package that could be easily replicated in other systems and geographic areas in Madagascar or other countries. The successful implementation of this project will result in a new, replicable, long-term and sustainable funding model that could enhance climate resilience, improve conservation, reduce land degradation and boost economic development. Outside Madagascar, the proposed model of a land-use transition investment fund is suitable for the most vulnerable African countries, where unsustainable agriculture takes up an increasing share of available land and results in deforestation, land degradation and an agriculture sector that is not resilient to climate change. Thus, this model could be replicated in (but not limited to) other least developed countries in Africa that need large-scale climate investments in rural areas in order to finance the activities necessary to achieve the objectives set in their nationally determined contributions and in their national adaptation programmes.

3.3 Sustainable development potential

Scale: High

15. The project will be implemented by the Government of Madagascar, particularly the Ministry of Agriculture and Livestock (MINAE) and the Ministry of Economy and Finance (MEF) in close synergy with other rural and agricultural development investment projects operating in the southern regions of Madagascar. Similar to other IFAD-funded projects, the interventions will rest on using national systems at both the central and the decentralized level, and on strengthening the capacity of targeted poor smallholders and their local institutions, such as farmers' organizations, in order to better empower the rural poor, with a particular focus on women and youth. When looking at ensuring sustainable development, climate information represents an essential element for increasing the capacity of vulnerable households to adapt to climate variability.

16. The sustainable development benefits include the following:

- (a) **Environmental co-benefits.** DEFIS+ will promote sustainable and climate-resilient agricultural practices, technologies and infrastructures that contribute to improving the efficient management and use of water resources in agriculture, enhancing soil quality and restoring degraded lands. Its integrated watershed management approach will help

² Available at, <https://play.google.com/store/apps/details?id=mg.dgm.gizgp&hl=en&gl=US&pli=1>.

to increase moisture content in the soil, optimize the use of water and expand the vegetative cover of degraded lands, while reducing their wind- and rainfall-induced erosion;

- (b) **Social co-benefits, including health impacts.** Improved food security and nutrition will be among the key social co-benefits that will arise. The proportion of targeted households with sufficient food intake is expected to increase from 50 to 90 per cent by the end of the project. In addition, 170,000 households will benefit from nutritional education, which will improve the health of women and infants. Better access to water, more effective watershed management and the use of off-farm technologies will reduce women's workload and time burden. Employment and employability of women and youth will increase, and women's participation in the rural economy will be formalized and recognized;
- (c) **Economic co-benefits.** DEFIS+ will contribute to reducing poverty by supporting the diversification of agricultural production, building the climate resilience of production systems and infrastructures and creating market linkages, which will reduce transaction costs and boost the local economy. The sustainable management of agricultural water within an integrated watershed approach, coupled with more climate- and stress-resilient seeds, will sustainably increase agricultural productivity;
- (d) **Gender-sensitive development impact.** DEFIS+ will implement a gender strategy to ensure that women's specific needs are taken into account at all stages and will provide opportunities to improve their quality of life, reduce their vulnerability to climate change and increase their incomes. The activities proposed for women and youth will include training farming households on gender issues, and measures to increase their access to productive capital and income-generating activities (primarily access to land), with a view to positioning women as rural entrepreneurs and service providers in development hubs and facilitating their access to capacity-building and financial services. The number of green jobs created through this project will have a target women headed households. The project will also involve minorities, vulnerable people and people of lesbian, gay, bisexual, transgender or other identities; and
- (e) **Social equity.** The overall strategy of DEFIS+ is to ensure social equity and cohesion among all members of civil society. Thus, this project follows an inclusive targeting strategy to reach vulnerable groups and poor households headed by women and youth, so that these targeted rural populations, characterized by their vulnerability to the effects of climate change, will have equal opportunities to access factors of production (land, water, inputs, financing, etc.) in order to ensure fair and equitable results and enable sustainable impacts on living conditions.

3.4 Needs of the recipient

Scale: High

17. Madagascar is among the poorest countries in the world. Poverty in Madagascar has increased significantly over the past decade. The poverty rate was estimated to rise to 77.4 per cent of the population in 2020, up from 74.3 per cent in 2019, corresponding to an increase in the population of 1.38 million in one year. . Poverty is clearly more severe in rural areas, where it affects 87.6 per cent of the population, in comparison with 55.1 per cent in urban areas. Extreme poverty is more pronounced in the south. The activities of DEFIS+ aim to restore and/or rehabilitate the soil and improve land use through a participatory approach with the local community. This approach will ensure that any potential land conflict is identified and resolved before any activities commence. For the rehabilitation of roads, in the exceptional

circumstances of beneficiaries who may be voluntarily displaced, the project will have a resettlement policy to be implemented before the works are initiated. Adherence to the IFAD Social, Environmental and Climate Assessment Procedures follows the principles of free, prior and informed consent and ensures that existing inequalities are not exacerbated. A grievance redress mechanism is also articulated for the project to manage any potential conflicts. In addition, DEFIS+ fully takes into account the needs of the recipient, which are in line with the national priorities.

3.5 Country ownership

Scale: Medium to high

18. This project is set to provide substantial support to the national policies concerning climate change, environmental conservation and sustainable development, along with aligning with national strategies (e.g. the Sectoral Program for Agriculture, Livestock and Fisheries) in a direct manner, and to reinforce the National Policy for Action on Climate Change by enhancing climate change adaptation efforts, while taking the country's unique requirements into consideration. Additionally, it will actively facilitate the integration of climate change considerations into all tiers of policymaking, as well as within research, development and technology transfer in this particular domain.

19. Madagascar is in the early stages of conceptualizing a GCF country programme using a GCF/United Nations Environment Programme readiness grant. Notably, DEFIS+ investments harmonize completely with the critical priority sectors outlined in Madagascar's national adaptation programme of action, namely infrastructure, water resource management, agriculture and livestock. These investments will play a direct role in advancing Madagascar's nationally determined contribution, which primarily seeks to bolster the country's resilience against cyclones, floods and droughts. The project places significant emphasis on adaptation interventions by 2030, encompassing initiatives such as the widespread adoption of robust agricultural models and sustainable and integrated water management (including the utilization of surface water resources in semi-arid zones) and the enhancement of the irrigation infrastructure to cope with diminished run-off during dry periods.

20. Furthermore, the project aims to decrease GHG emissions and enhance carbon sequestration, while also focusing on enhancing rice farming systems (specifically "system of rice intensification" (SRI) techniques) and executing ecosystem-based practices for adaptation. These solutions encompass, among other things, alternatives to traditional charcoal use, fossil fuels and water abstraction systems.

21. The national designated authority of Madagascar, the Ministry of Environment, Ecology and Forests, also fully supports the project and has already issued a no-objection letter for this project.

3.6 Efficiency and effectiveness

Scale: Medium to high

22. The overall approach adopted by DEFIS+ is designed to increase efficiency and effectiveness by focusing on the set of structural investments associated with the introduction of improved technologies and climate-resilient practices as well as capacity-building of local institutions that will leverage growth in targeted agricultural areas. This approach, where science and engineering are integrated into in-depth modelling and understanding of agricultural risk, will inform the structural investments through the studies that will be undertaken. Furthermore, in order to move from low-yield subsistence farming to market-

oriented agriculture, the project's theory of change is based on improving the productivity and adaptive capacity of production systems to climate change and beneficiaries' access to markets under favourable conditions. In this regard, the theory of change requires significant investments in basic infrastructure, so priority will be given to the development of irrigation infrastructure and rehabilitation of roads to reduce the risks and costs of accessing the nearest markets.

23. The main benefits expected for people in the DEFIS+ project area are (i) creation of climate-resilient rural communities; (ii) diversification of income sources for the target populations, especially young people and women, and creation of green jobs in the project area; (iii) reduced transportation costs through rural road rehabilitation, water storage infrastructure and market access; (iv) increased revenues for poor people; (v) improved food security, diet quality and diversification and nutrition; and, above all (vi) greater awareness of and protection from the climate change impacts negatively affecting the livelihoods of the rural population.

24. IFAD conducted an economic and financial analysis, with ten proposed interventions being modelled and appraised.

25. The analysis found that aggregate incremental results of the ten interventions combined is a financial net present value (NPV) of USD 31.07 million (over 20 years at 8.5 per cent financial discount rate) whereas the financial internal rate of return (IRR) and the modified IRR are 19 and 9 per cent, respectively. These incremental aggregate results point towards the financial viability of the proposed interventions because the incremental financial NPV is positive and the incremental financial IRR remains above the chosen financial discount rate of 8.5 per cent,

26. The aggregate economic incremental results are also promising, with the aggregate incremental economic NPV estimated at USD 255.57 (over 20 years at 6 per cent discount rate) and the economic IRR and modified IRR at 24 and 11 per cent, respectively. Further detailed estimates on a per-model basis are presented in the table below.

27. The sensitivity analysis results suggest that the obtained economic NPV and economic IRR remain robust and relatively insensitive to cost increase and benefit decrease. Consequently, it is expected that the proposed interventions will provide measurable benefits to the entire economy of Madagascar, improving the well-being of the environment and intended beneficiaries.

Detailed results of the economic and financial analysis per individual model

	Total investment (USD)	Revenues generated (USD)	Value of externalities (USD)	NPV		IRR		Benefit to cost ratio		Payback period (years)		Debt service coverage ratio
				(Economic)	(Financial)	(Economic)	(Financial)	(Economic)	(Financial)	(Economic)	(Financial)	
IT1	4 880 000	488 919 105	15 042 600	40 859 197	24 250 903	40	33	1.2	1.17	4.95	5.44	8.90
IT2	6 201 000	727 772 089	28 610 033	69 979 425	38 564 851	47	36	1.2	1.18	5.55	5.19	8.88
IT3	872 500	137 472 857	4 303 171	11 921 717	6 263 986	44	33	1.2	1.15	4.92	5.90	1.21
IT4	4 283 615	630 517 226	1 271 376	50 901 433	25 179 757	40	29	1.2	1.14	4.97	5.96	8.52
IT5	3 460 551	27 232 269	106 962	1 157 864	138 251	11	9	1.2	1.15	6.93	11.33	9.54
IT6	5 821 596	1 274 774 600	10 844 972	106 522 557	56 344 122	44	33	1.2	1.15	4.63	5.36	7.94
IT7	1 377 500	53 587 041	12 361 981	8 577 576	Not calculated	64	Not calculated	1.5	Not calculated	5.24	Not calculated	Not calculated
IT8	80 000	3 768 330	483 918	442 702	Not calculated	46	Not calculated	1.3	Not calculated	4.95	Not calculated	Not calculated
IT9	1 080 000	123 782 634	123 406 720	63 466 963	Not calculated	-	Not calculated	50.2	Not calculated	2.13	Not calculated	Not calculated
IT10	1 607 800	23 752 840	7 503 067	4 542 077	Not calculated	40	Not calculated	1.6	Not calculated	6.32	Not calculated	Not calculated

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

28. **Project background.** The project aims to strengthen the resilience of agricultural production systems and vulnerable communities to the impacts of climate change while contributing to emission reduction in six regions in Madagascar through the dissemination of climate-friendly agricultural and livestock management practices, the rehabilitation of road and irrigation infrastructure and watersheds, and the introduction of energy-efficient technologies while strengthening climate information systems and capacities of institutions through the adoption of best practices and knowledge on climate change adaptation. The environmental and social co-benefits include the improvement of agricultural water resources, enhanced soil quality and restored degraded lands. The project is also expected to increase soil moisture content and expand vegetative cover, reducing soil erosion and siltation of water bodies such as hydro-agricultural and irrigation canal infrastructure. It is envisaged that the project will increase biodiversity in the target areas and surrounding landscapes and maintain the important ecological services that it will provide. The social co-benefits include improved health and nutrition as a result of better access to water and increased and diversified income sources.

29. **Environmental and social risk category and safeguard instruments.** The accredited entity (AE), IFAD, has assigned a category B environmental and social risk rating to the project, given that it will have limited adverse environmental and social risks and/or impacts that are site-specific and can be readily mitigated. The Secretariat agrees with the risk categorization and it is within the level for which the AE is accredited. The AE has prepared an environmental and social management framework (ESMF) for the project that describes the potential environmental and/or social risks/impacts of the project and the measures to address them. The ESMF also outlines how different subproject activities would be screened for environmental and social risks/impacts during project implementation, which mitigation measures would be applied based on the risk level, and whether site-specific assessments and management plans would be required for these subactivities.

30. **Compliance with GCF environmental and social safeguards (ESS) standards.** The environmental and social risks and/or impacts of the project and the proposed mitigation measures are described below.

31. **ESS 1: Assessment and management of environmental and social risks and impacts.** The ESMF describes the baseline conditions for the targeted areas where project activities will take place to meet the project objective of increasing the resilience of rural communities to the impacts of climate change. The ESMF describes all subproject activities in detail and lists the potential direct and indirect impacts of each of the activities. It also proposes generic, standardized mitigation measures to avoid, minimize and mitigate such impacts. Most of the risks and impacts will be moderate in scale and can be readily managed and mitigated through the appropriate procedures and plans described in the ESMF. Site-specific, tailored and more detailed mitigation measures will be developed for subproject activities with a moderate risk, which will require site-specific assessment and plans for these activities to be prepared during implementation. The ESMF also includes a monitoring protocol and a budget for its implementation.

32. **ESS 2: Labour and working conditions.** The ESMF covers the impacts on labour and working conditions. It includes discussions on the typology of the workers who will be hired under the project and the key labour risks and impacts for these workers, including working conditions and terms of employment. Occupational health and safety concerns will be managed by equipping workers with personal protective equipment adapted to the types of work in which they are engaged. Staff will also be trained and educated on the risks and dangers of construction sites. Given that the project will use solar photovoltaic systems in its activities, the

AE will ensure that there will be no use of forced labour/child labour in the primary supply chain of such systems. Labour and working conditions will be aligned to the AE's Social, Environmental and Climate Assessment Procedures, which set out labour management principles and the grievance mechanism. Contract documents will also include clauses on labour conditions, including zero tolerance of sexual exploitation, abuse and harassment (SEAH), and the bidding documents for the solar photovoltaic component will include clauses that ensure that goods are procured from ethical sources.

33. **ESS 3: Resource efficiency and pollution prevention.** The ESMF identifies the activities that generate emissions to air and discharges to water, including possible contamination of land and generation of associated wastes, hazardous materials and pesticides, as well as activities expected to utilize natural resources, including water and energy. Atmospheric pollution, dust emissions and noise pollution from traffic (e.g. construction machinery and vehicles) are expected. Soil compaction and increased risk of erosion due to clearing of vegetation for the construction facility, as well as soil pollution by infiltration or leaching of pollutants (e.g. fuel storage), may occur. Alteration of the quality of water resources through wastewater discharges from the construction site, sedimentation in irrigation canals and pollution of water bodies through the use of agricultural inputs may occur. To mitigate these, several measures are planned, such as effluent treatment prior to discharge, proper storage of fuels and carrying out of proper maintenance of equipment and machinery. Implementation of a site waste management plan is also envisaged. Screening on resource efficiency and pollution prevention to assess these risks and impacts in a site-specific manner will be conducted during implementation and site-specific management plans will be prepared that are tailored for subproject activities.

34. **ESS 4: Community health, safety and security.** The ESMF describes the activities that will generate risks and impacts to the health and safety of communities. These are mainly associated with construction activities, which include risk of accidents to local populations associated with the use of vehicles and construction machinery, disturbance due to dust emissions, and nuisance noise and disruption of rural activities. The measures contemplated include carrying out regular maintenance of equipment, vehicles and machinery, ensuring that base camps and project activities are situated far from the villages and work is done during the daytime only, and conducting appropriate information and awareness campaigns for local populations on the duration of the activities to be carried out in the area. Existing social conflicts regarding use of agricultural land, water use and infrastructure maintenance activities between water user associations are expected to be reduced by implementing the activities of the project (e.g. outcome 1), which encompass improvements to agricultural land and food production and water resource management systems through participatory consultation, better procedures and regulations, an established maintenance plan and enhanced institutional capacities (such as adoption of a conflict prevention and resolution mechanism).

35. **ESS 5: Land acquisition and involuntary resettlement.** The project will avoid or minimize any resettlement or economic displacement that could have a negative impact on the populations concerned. The AE has also assessed the project as having a low probability of resettlement and economic displacement. However, should people lose their land and means of livelihood, compensation will be provided. Eligibility criteria, monetary, crop, land replacement and compensation scales, and an institutional arrangement for implementation of an action plan will be drawn up. There will also be a mechanism for consultation with and participation of beneficiaries in the planning, information dissemination, execution and monitoring, follow-up and implementation of the compensation and mitigation plan (i.e. abbreviated resettlement action plan), which could be prepared and implemented in accordance with the provisions of the AE's Social, Environmental and Climate Assessment Procedures.

36. **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** The project's construction and rehabilitation activities could result in loss of vegetation and disturbance of wildlife through the destruction of natural habitats, and pose a risk of possible encroachment on sensitive areas. The large-scale distribution of certain varieties of seeds can also lead to a reduction in floristic biodiversity. Furthermore, there is a risk of proliferation of invasive plants in the watershed protection-related activities of the project. To mitigate these, the project intends to prioritize the use of indigenous species, comply with forestry legislation and limit disturbance to the areas that are required for the implementation of the subprojects (such as avoiding infringing on protected natural sites and critical habitats or areas with significant biodiversity). The project activities will also involve conducting preliminary studies to analyse each variety to be introduced to anticipate their responses in the field based on the type of environmental/pedoclimatic condition, and conducting quarantine checks and follow national guidelines in the introduction of new germplasm. Reforestation of different species to promote biodiversity will also be implemented. The project will not support activities that are prohibited by host country legislation or international conventions relating to the protection of biodiversity resources, cultural heritage or other legally protected areas.
37. **Indigenous Peoples Policy and ESS 7: Indigenous Peoples.** The AE confirms that there are no Indigenous Peoples in the areas targeted by the project; however, it notes that good practices, including recognizing cultural heritage and identity as assets, obtaining free, prior and informed consent, strengthening community-driven development, promoting equitable access to lands, territories and resources and valuing Indigenous knowledge, will be developed through the project. Stakeholder consultations have also echoed the need for valorizing traditional knowledge, a recommendation in line with the GCF Indigenous Peoples Policy and on which the AE can consult with the GCF Indigenous Peoples Advisory Group, as part of its role and function.
38. **ESS 8: Cultural heritage.** Although adverse impacts are not expected on physical cultural heritage, given that the project will only renovate existing roads and irrigation structures and work with farmers/local governments to support climate-resilient agricultural practices, the construction works on such infrastructure may potentially affect culturally relevant sites through accidental encounters. In such cases, the project will implement the "chance find" procedures contained in the ESMF and comply with the national ordinance on protection and conservation of national heritage.
39. **Implementation arrangements.** The ESMF outlines the basic responsibilities for implementing arrangements and divides responsibility between the project management unit at MINAE, the ESS team in the project management unit, and that of the contractors/implementing parties. It specifies the implementation and monitoring of mitigation measures for the potential impacts of the project, including the frequency of monitoring of performance indicators. The ESS team will be responsible for monitoring the environmental and social aspects of the project during its implementation. Staff training and capacity-building of its project partners on environmental and social issues, as well as the verification of compliance of the project with environmental and social performance indicators, will be implemented and monitored.
40. **Stakeholder engagement and grievance redress mechanism.** Public consultations have been held with stakeholders in the target regions, including administrative authorities, various local partners and representatives of the beneficiaries and local communities to be affected by the project. The stakeholder engagement plan provided a description of how stakeholder engagement will take place during the subproject planning, implementation monitoring and evaluation phases. The stakeholder engagement plan also describes how project information will be disclosed by the AE and the Government of Madagascar through MEF and

MINAE, the project's executing entity (EE), during the preparation and implementation phases, using differentiated mechanisms to reach different stakeholders.

41. The project proposal also describes two grievance redress mechanisms: the institutional grievance redress mechanism of the AE and the project-level grievance redress mechanism. Both have various channels through which grievances may be submitted and general processes for managing grievances. At the AE level, grievances on non-compliance with environmental and social policies can be submitted by email or post and will be managed by the IFAD Regional Division. Complainants not satisfied with the response can appeal to the Vice-President of IFAD. At the project level, grievances may be submitted by email or in person and will be handled by the project management unit. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples focal point will be available for assistance at any stage, including before a claim has been made.

42. **SEAH.** The revised GCF Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from SEAH in GCF-financed activities. The ESMF of the DEFIS+ project outlines SEAH risks areas and proposes mitigation measures, linking the addressing of SEAH with the gender-based violence (GBV) risks identified in the gender assessment and its gender activity plan. In order to address SEAH, the project will, among other things, strengthen awareness of GBV and SEAH issues among staff and project beneficiaries; develop an information/awareness protocol on SEAH; strengthen GBV/SEAH operational procedures for projects, including a code of conduct for public works contractors; conduct various information and awareness-raising activities for direct and indirect project beneficiaries; and ensure that contractor contracts and collaboration agreements have relevant SEAH clauses. The AE has also proposed changes to the SEAH grievance redress mechanism in alignment with the GBV code of conduct, and various reporting platforms, namely online or by telephone or mail, and in-person reporting. The Secretariat recommends that the AE ensure that in the event that SEAH occurs, the reporting and SEAH complaints handling is survivor centred and as gender responsive as possible. Overall, the SEAH safeguarding measures presented by the AE are in alignment with GCF SEAH provisions.

4.2 Gender policy

43. The AE has provided a gender assessment and gender action plan, thereby complying with the requirements of the GCF Gender Policy.

44. The gender assessment describes the existing enabling environment that provides the space to address gender inequality and work towards the increased empowerment of women. The constitution of Madagascar enshrines equal rights between women and men; subsequent amendments to legal and regulatory systems were made to ensure that equality takes root in society. Furthermore, Madagascar has signed and ratified the Convention on the Elimination of All Forms of Discrimination Against Women and other international treaties and many of its national laws and existing national policies and legislation support the equality of women. Madagascar has developed a National Policy for the Promotion of Women, which aims at addressing gender inequality in incomes and economy, education and training gaps, rights to health, reproductive rights and decision-making powers of women while also improving institutional mechanisms. Despite the intent and span of effort, in practice ensuring equality has been and remains a challenge. Land-related issues, particularly ownership and inheritance, pose a challenge for women, while access to productive resources and assets is equally challenging for women because of inherent restrictive norms and practices.

45. The challenge to translate policies and legal intent into practice has resulted in women in Madagascar facing an added burden with respect to the impacts of climate change. The gender assessment, which was undertaken with stakeholder consultations and a desk review, describes the inequality felt by women. Regarding land and factors of production, women do not have the right to own and make decisions on land; neither are they allowed in most cases to inherit it. If they need to have access to land, it is only available in the form of renting or if they are part of a women's group. Access to land in such a manner does not give women the right to decide on long-term investments, even to plant trees. Female-headed households might own land but is generally marginal, less productive land. Limited access to inputs such as seeds, finance, fertilizers and irrigation infrastructure and low levels of skills restricts women's ability to meet their daily needs or mitigate against these limitations as well as the compounded effects of climate change induced realities. These challenges affect women's ability to generate lucrative activities and revenue/incomes. This does not, however, mean that women do not contribute to the agricultural economy; they do, but do so within the confines of the realities expressed above. The situation of women is similar regarding the use of forest and other natural resources, upon which they depend a great deal for making traditional medicine and fulfilling other needs. Men use forest resources in the same way, including for cooking and timber for construction. The impacts of overuse and misuse affect all. Decisions on the use of these resources are made by men, with a limited role for women in the decision-making structures. The burden on women is exacerbated by climate change in that it contributes to an increased workload, food insecurity, longer fuelwood and water collection times, with more time needed by women to fulfil their roles. In addition, the assessment indicates that violence against women is a challenge in the project area, and notes the importance of keeping this in mind during the course of project implementation to ensure that project interventions do not allow the perpetuation of GBV within the programming area and possibly beyond.

46. The AE has provided a gender action plan that is targeted at addressing the gender issues identified in the gender assessment. The gender action plan contains a set of activities with corresponding targets and indicators. The AE has also provided a budget to ensure the effective implementation of the actions identified and a timeline within which each of the activities will be undertaken.

47. With a view to building on existing capabilities of women and providing support to address identified gaps the project will target the following four areas of vulnerability for women (including female-headed household and youth): (i) limited inclusion and participation of women; (ii) limited empowerment of women and youth in the agricultural value chain; (iii) limited opportunities to improve their conditions within the community; and (iv) limited capacity-building and development opportunities rendering women less resilient and affecting their role and decision-making power. The gender issues identified will be addressed through various project components, such as in the reconstruction of irrigation schemes which will have benefits to women in terms of food diversification, bringing sufficient quantities of good-quality water closer to their homes (after consultative processes); engagement in water user associations; and addressing underlying social norms using community-led discussions to equal access and opportunities for women and girls. Community-led discussions, besides being expected to improve gender relations, are expected to be a platform that will encourage awareness-raising and adoption of good climate change practices. The project will also provide opportunities for women to develop leadership skills to ensure that they are able to engage and have a say in decisions that affect their life and livelihoods, such as in water user associations. To ensure women's participation and inclusion in the various activities, opportunities will be taken to organize women-only events, to institute quota systems and to provide targeted training for women in, for example, functional literacy and leadership skills. Furthermore, there will be ample opportunities for women to engage in and benefit from the activities of the project which focus on sustainable watershed management and restoration of the landscape, including

in increasing their representation at the leadership and decision-making level. The project will work to ensure that women and youth get access to and benefit from tailored knowledge (based on information on needs and preference in accessing), productive capital, climate insurance, green jobs (in the reconstruction and maintenance works through mandatory recruitment procedures with equal pay for equal work, while also encouraging women-owned construction enterprises coming into the workstream) and engagement in income-generating activities. Access to land, which is a major barrier to women, will be addressed through allocation and quota systems for women and youth. This will be one of the lessons to be learned and one of the main tasks of DEFIS+, which will facilitate access to land for women and youth through the development of new irrigated areas; while for existing areas, the project will set out the inclusion of women and youth as a condition for access to funding. At the same time, the project will invest in awareness-raising and mobilization of traditional authorities, to address and overcome certain practices that prevent access to land, especially for women heads of household.

48. The project's national steering committee (NSC) will include the Ministry of Social Protection and Gender and will have a gender expert to monitor the day-to-day implementation, monitoring and evaluation of the project, while the project management unit and the project monitoring and evaluation systems will be charged with ensuring the monitoring of the implementation of the gender action plan.

4.3 Risks

4.3.1. Overall project assessment (medium risk)

49. This funding proposal is a 100 per cent adaptation project to strengthen the climate resilience of the agricultural production system and vulnerable communities in Madagascar. The project will implement land-based activities related to agroforestry, reforestation and renewable energy practices, which will be measured as co-benefits.

50. GCF is considering a loan of USD 12.8 million and grant of USD 41.0 million together with co-financing of USD 127.5 million in loan and grant from IFAD. The co-financing ratio is 1:2.4 and accounts for 70 per cent of total financing.

4.3.2. Accredited entity/executing entity capability to execute the current project (medium risk)

51. IFAD is an international AE with over 40 years of development work in global agriculture and food security. To date, IFAD has provided over USD 23 billion in grants and low interest rate loans. The Government of Madagascar, through MEF and MINAE, is the project EE and the funding agreement will be implemented through both MEF and MINAE. MEF will manage, control and evaluate external funds while MINAE will enter into contracts for the procurement of services for project activities. IFAD has an extensive history of working with MINAE in implementing projects and programmes.

4.3.3. Project-specific execution risks (medium risk)

52. The funding proposal will expose GCF to several country-related risk issues in Madagascar and its low sovereign rating of B- (Standard & Poor's). Madagascar's rating is due in

part to the government's large budget deficit, low-income levels and the low level of economic development. The country had a reported gross domestic product per capita of USD 453 in 2022. Madagascar also reported a sharp economic contraction following the onset of the coronavirus disease pandemic in 2020. Flooding, drought and other climate change induced natural disasters have also negatively affected the country's economy. However, Madagascar enjoys strong relations with multilateral institutions that provide concessional and grant funding support, including support for infrastructure projects to address energy and transport development and donations for food insecurity. In addition, an International Monetary Fund extended credit facility should continue to help to boost economic growth by increasing investment and social spending as well as provide policy support to strengthen governance and anti-corruption efforts.

4.3.4. Project viability and concessionality

53. The proposed project will build on the IFAD existing agricultural investment programme, DEFIS, which was approved in 2017 and funded by IFAD, the OPEC Fund for International Development and other donors. To strengthen the climate resilience of small farmers, the project will, among other activities, strengthen irrigation infrastructure to increase water availability, build the resilience of tertiary rural roads to reduce market access disruption, and allow access to basic services during periods of excessive rainfall. The requested participation from GCF will represent approximately 30 per cent of total programme size in a partly loan- and grant-funded project. The level of concessionality is reasonable given the co-financing of 70 per cent from IFAD. It is also noted that the larger proportion of grant financing compared with loans in GCF funding is justified in view of Madagascar's weak sovereign rating. The country has been heavily reliant on support from multilateral institutions, and it will be appropriate for GCF to align its participation to the borrower's credit condition.

4.3.5. GCF portfolio concentration risk (low risk)

54. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of results area and single proposal is not material.

4.3.6. Compliance risk (medium risk)

55. This programme will mainly address the agriculture sector. In doing so, there is likely to be significant expenditure at the subnational or local level and among a wide variety of beneficiaries. While the expenditure that will benefit individuals in terms of technical and other assistance does not appear to be unusually large, there will still need to be monitoring and assurances that beneficiaries receiving such assistance, particularly in terms of financial benefits, are duly qualified and entitled to receive such benefits. The AE will need to exercise its policies and procedures to ensure that benefits received or used in the project are not used for illicit purposes such as money laundering, terrorist financing or other illicit purposes. The AE has such policies and procedures in place and will need to ensure that its implementing partners also apply those policies and procedures during the course of the project. However, in reviewing the proposal, it does not appear that the activities will constitute an unusually high risk and that, in view of the commitments of the AE, the compliance risk is determined to be medium.

4.3.7. Recommendation

56. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment		Rationale
Overall project	Medium	See 4.3.1
Accredited entity/executing entity capability	Medium	See 4.3.2
Project-specific execution	Medium	See. 4.3.4
GCF portfolio concentration	Low	See 4.3.5
Compliance	Medium	Compliance risk rating is based on relatively low risk of over money laundering/terrorist financing risks and capacity and commitment of AE to monitor and enforce policies and procedures in project implementation. See 4.3.6 for further details.

4.4 Fiduciary

57. The AE for the project is IFAD and, as such, the IFAD operational model ensures continuous implementation support and supervision of project activities. The IFAD Manual of Administrative, Financial and Accounting Procedures, which has been validated by the Government of Madagascar and IFAD, will be complied with. The EE for the project is the Government of Madagascar through MEF and MINAE.

58. As AE, IFAD will enter into a funded activity agreement with GCF on the contribution of the GCF loan and grant proceeds to IFAD, for which funds will flow into a dedicated bank account held by IFAD, while in parallel a subsequent subsidiary agreement between IFAD and the Government of Madagascar will outline the basis on which GCF proceeds (i) will be provided (on-lent/on-granted) to Madagascar in the agreed currency (i.e. United States dollars) and (ii) will flow from IFAD to the borrower (MEF) into a project designated account in United States dollars opened at the Central Bank of Madagascar.

59. As for the EE, a legally binding financing agreement (subsidiary agreement) will detail the specific roles and responsibilities of the relevant government ministries, MEF and MINAE, with regard to the implementation of the project. The subsidiary agreement will establish the terms for the financing and implementation of outcomes, outputs and activities, and include the relevant obligations of the EE in respect of the implementation of the overall funded activity, including compliance with the eligibility criteria and the project implementation manual. The EE through MINAE will select the beneficiaries of all activities of the project and procure service providers for specific project activities in accordance with the eligibility criteria and the project implementation manual. The EE through MINAE will enter into contracts for procured services with the selected service providers.

60. The project will have an NSC to oversee the activities of DEFIS+ at the national level. The NSC will meet twice a year and its main role is to validate DEFIS+ annual workplans and budget; evaluate progress made on project activities, their results and their alignment with strategies for rural areas; and manage project-related grievances that cannot be resolved at the regional level. Complementing the NSC, the project will have a project coordination unit to coordinate and manage DEFIS+ resources and activities in accordance with the financing agreement, project manuals on operations, financial management and administration procedures, monitoring and evaluation, and annual workplans and budgets, under the supervision of regional steering and monitoring committees, the NSC and IFAD. The project coordination unit

will be responsible for (i) managing the project's human, financial and physical resources; (ii) preparing annual workplans and budgets and project procurement plans; (iii) coordinating the implementation of the activities; (iv) monitoring and evaluation of the project's activities and results; (v) coordinating and monitoring of the work/activities of partners and service providers; (vi) reporting and submitting activity and financial reports to MEF, MINAE and IFAD on a regular basis (minimum of twice a year in the case of IFAD so as to fulfil its semi-annual reporting obligation vis-à-vis GCF); (vii) liaising regularly with regional government authorities and decentralized services and other actors in the DEFIS+ intervention zones; and (ix) ensuring compliance with IFAD safeguards (i.e. environmental and social management plan, gender action plan) during project implementation activities. Planning and coordination of activities at the regional level will be provided by two interregional coordination units located in Fianarantsoa for the Ihorombe, Haute Matsiatra and Amoron'i Mania regions; and Manakara for the Vavovavy, Fitovinany and Atsimo-Atsinanana regions.

61. Joint supervision of the project will be provided by the Government of Madagascar (Ministry of Agriculture and Livestock (MINAE), Ministry of Economic and Finance (MEF), National Designated Authority (MEDD NDA) and ONE) and IFAD. Supervision missions will be carried out twice a year and their main goals will be to (i) assess progress on the implementation of each DEFIS+ component; (ii) evaluate progress on the expected outcomes and outputs for each project component in comparison with what had been established in the concept note or other project documents; (iii) assess the quality of targeting and gender equality; (iv) analyse the different fiduciary aspects of the project (funding allocations, payment and disbursement commitments, conformity of expenditures with annual workplans and budgets and concept note, accounting records) and review outcomes of the external audits conducted in accordance with the IFAD Handbook for Financial Reporting and Auditing for IFAD-Financed Projects. A mid-term review will be held to evaluate the actions taken in relation to the project's objectives, analyse the effectiveness and appropriateness of the methodologies used, verify the level of disbursement of resources and determine the need to reallocate available resources.

62. A no-objection letter from IFAD will be required for the approval of annual workplans and budgets, which will be screened to ensure that project activities build towards successful project completion in an effective and efficient manner. GCF funds will flow into a dedicated bank account held by IFAD that will be opened for DEFIS+ resources. This GCF account will hold GCF proceeds in trust and will be United States dollars denominated and interest bearing. On-lent loans will be in the same currency of disbursement and repaid in such currency, which will be United States dollars.

63. The project's accounts will be verified annually through external audits conducted by independent and qualified auditing firms in accordance with the IFAD Handbook for Financial Reporting and Auditing for IFAD-Financed Projects and in compliance with the applicable GCF requirements (i.e. expenditure eligibility, categorization vis-à-vis GCF budget categories, etc.). A certified copy of the auditor's report will be submitted to IFAD a few weeks before its deadline for sending such a report to GCF.

64. It is recommended that conditions relating to the provision of a capacity assessment of the EE in form and substance that is satisfactory to the Secretariat be reflected in the term sheet and funded activity agreement as a placeholder.

4.5 Results monitoring and reporting

65. As a 100 per cent adaptation initiative, the project is expected to benefit 447,201 direct beneficiaries and 3,543,719 indirect beneficiaries. The proposal included a methodology and assumptions for defining both direct and indirect beneficiaries.

66. The theory of change diagram went through a few rounds of revision and now adequately captures different levels of expected changes. Adaptation outcomes are shown to clearly represent outcomes. The theory of change diagram identifies the co-benefits that can be achieved and captures the causal relationships between outcomes and co-benefits. The barriers and risks that may hinder the achievement of these changes are identified and linked to pertinent activities, which will tackle them accordingly.

67. The logical framework has been well designed in accordance with the GCF Integrated Results Management Framework. The paradigm shift potential is expected to be achieved with the potential target scenarios. Appropriate Integrated Results Management Framework core and supplementary indicators have been selected for monitoring and reporting the project results in respective GCF results areas and, for each indicator, baseline and targets (mid-term and final) have been set with different means of verification. However, some indicators could have better methodologies to measure their impacts (e.g. food security indicator) and units of measurement need to be consistent among baseline, mid-term and final targets.

68. At the GCF outcome level of enabling environment, two GCF core indicators are identified: technology deployment and dissemination, and market development. Given the low level of the baseline for each indicator, the project is expected to build a comprehensive enabling environment in Madagascar.

69. The theory of change clearly shows the logical pathways for generating results via project activities, and how those contribute to the outcomes and impacts of the project. Relevant assumptions have been also made and an explanation of the ways in which barriers are addressed has been provided.

70. The logical framework was designed in accordance with the GCF results management framework/performance measurement framework and the programme theory of change. The logical framework has selected relevant impact, outcome and output indicators aligned with the results areas to which the project is contributing, as well as the project's specific activities.

71. The implementation table is contained in annex 5 and the monitoring and evaluation plan in annex 11 to the funding proposal.

4.6 Legal assessment

72. The Accreditation Master Agreement was signed with the Accredited Entity on 24 September 2018 (the “**AMA**”), and it became effective on 9 November 2018. The Accredited Entity's five-year accreditation term lapsed on 8 November 2023 but was extended by Decision B.37/18, paragraph (q) until the earlier of three years from the date of lapse of the five-year accreditation term or the date on which a revised accreditation framework is adopted by the Board. The Board approved the re-accreditation of the Accredited Entity pursuant to Decision B.37/18, and negotiation of the Amended and Restated Accreditation Master Agreement is ongoing.

73. The Accredited Entity has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

74. The proposed project will be implemented in the Republic of Madagascar, a country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The GCF provided an updated draft agreement on privileges and immunities to the Government of the Republic of Madagascar on 13 May 2019.

75. The Heads of the Independent Redress Mechanism (IRM) and Independent Integrity Unit (IIU) have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities.

76. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Submission by the Accredited Entity to the Fund of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 120 days after Board approval, confirming that the Accredited Entity has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed project;
- (b) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the Accredited Entity has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and
- (c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP227

Proposal name:	Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme (DEFIS+)
Accredited entity:	International Fund for Agricultural Development (IFAD)
Country:	Madagascar
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1. This is the second submission of the funding proposal on “Increase Resilience to Climate Change of Smallholders Receiving the Services of Inclusive Agricultural Value Chains Programme” (known as DEFIS+), submitted to the independent Technical Advisory Panel (iTAP) under the standard proposal approval process. The accredited entity (AE) is the International Fund for Agricultural Development (IFAD). The executing entity (EE) is the Republic of Madagascar through its Ministry of Economy and Finance and its Ministry of Agriculture and Livestock.

2. Madagascar is one of the world’s least developed countries (LDCs), with 81 per cent of the population below the poverty line. In 2021, the Multidimensional Poverty Index (developed by the United Nations Development Programme) estimated that 68.4 per cent of the population of Madagascar is multidimensionally poor, and an additional 15.4 per cent was classified as vulnerable to multidimensional poverty.¹ Although in 2021 the share of the per capita global greenhouse gas emissions of Madagascar was only 0.01 per cent,² the country is highly vulnerable to climate change, with an Notre Dame Global Adaptation Initiative index score of 35.3 (ranked 172 out of 185).³ These figures clearly illustrate the high level of urgency of reducing the vulnerability of Madagascar to climate change.

3. The funding proposal states that the proposed project, DEFIS+, has the ambition of reducing the negative impact of climate change by applying sustainable and nature-based adaptation measures with mitigation co-benefits. The funding proposal is structured around three outcomes: (i) strengthened climate resilience of agricultural production systems; (ii) increased income generation through improved market accessibility; and (iii) improved food and nutrition security.

4. The funding proposal focuses on adaptation to climate change, with mitigation as a co-benefit. The project covers six regions in the south-central and south-eastern areas of Madagascar, and expects to reach 447,201 direct and 3,543,719 indirect beneficiaries (1.6 per cent and 12.7 per cent of the total population, respectively). The funding proposal states that the project will contribute to all four GCF adaptation results areas as follows: enhanced resilience of most vulnerable people and communities (30 per cent); health and well-being, and food and

¹ The Multidimensional Poverty Index (MPI) value is the share of the population that is multidimensionally poor adjusted by the intensity of the deprivations. See <https://hdr.undp.org/sites/default/files/Country-Profiles/MPI/MDG.pdf>.

² See <https://www.worldometers.info/co2-emissions/co2-emissions-by-country/> or <https://ourworldindata.org/co2/country/madagascar>.

³ See <https://gain-new.crc.nd.edu/ranking>.

water security (25 per cent); infrastructure and built environment (20 per cent); and ecosystems and ecosystem services (25 per cent).

5. IFAD requests USD 53,861,003 from GCF in the form of a grant of USD 41,045,298 (74 per cent) and a loan of USD 12,815,705 (26 per cent). In the funding proposal IFAD states that it will provide co-financing of USD 96,986,441 for a total project budget of USD 150,847,444.

6. During the assessment the iTAP considered all funding proposal documents as submitted on 15 December 2023, the written answers to the iTAP as received on 10 January 2024 and further clarifications during the online meeting on 11 January 2024.

II. Impact potential

Scale: High

2.1 Adaptation potential

7. The funding proposal sets out clearly the climate hazards (current and future) in the two covered areas of Madagascar and explains the exposure of farming communities to these hazards, including socioeconomic aspects of climate risk. Based on the understanding of these climate risks the funding proposal introduces corresponding adaptation measures and means to monitor and evaluate progress over time. The rationale, consistency and justification of the adaptation measures was improved following the recommendations made by the iTAP for the first submission of this funding proposal.

8. Further details about how the revised funding proposal fulfils the principles for demonstrating adaptation impact according to decision B.33/12 is provided below.

9. The six regions included in DEFIS+ can be grouped into two climatic zones: Vatovavy, Fitovinany and Atsimo-Atsinanana (south-east zone) and Amoron'i Mania, Haute Matsiatra and Ihorombe (central zone).

10. The south-east zone, which is also commonly referred to as "Great South East", is located in the climatic zone of the eastern slope of Madagascar, where the annual rainfall ranges from 1,500 to 2,500 mm. The climate is characterized by a hot rainy season (southern summer), with high rainfall distributed unevenly between November and April, and a cooler dry season from May to October (southern winter).

11. In the central zone, the Amoron'i Mania region and the Haute Matsiatra region (central highlands) benefit from a tropical climate of altitude in the highlands, while the Ihorombe region is a transition zone between the semi-arid climate of the south-east zone and the tropical climate of altitude characteristic of the highlands.

12. Using secondary information, the proposal shows an increment in temperature of around 1 °C in the period 1970 - 2020 and an upward trend in related extreme events. Projected temperature increases in the project area are estimated to be between +1.2 and +2.1 °C by 2050. In the near future under the mid-warming scenario (Representative Concentration Pathway (RCP) 4.5) the multi-model mean temperature is projected to increase from 0.5 °C to 0.7°C (high confidence) in the dry season. By mid-century, the average temperature could increase from 1.1 °C (HC) to 1.4 °C (high confidence) in the dry season and from 1.2 °C (high confidence) to 1.4 °C (high confidence) in the rainy season. Furthermore, the number of extreme heat events is likely to continue increasing.

13. With regard to rain patterns, the funding proposal shows that even if there are no major changes in average annual precipitation, rainfall has changed significantly across regions and within regions, and an increase in the amplitude of inter-annual rainfall variability is predicted, which increases the probability of extreme events such as floods (annex 2a).

14. Madagascar experienced famine in 2016 and again in 2021, following an unprecedented period of drought. This extended period of drought, in addition to non-climatic variables (especially the coronavirus disease (COVID-19) pandemic in 2020) led to a dramatic fall in productivity in the agricultural sector that affected food security severely.
15. The funding proposal further documents increases in both the frequency and intensity of cyclones affecting the project regions on the south-east coast. According to the proposal, the intensity of cyclones is projected to increase along with the risk of associated floods. The combination of poor land management practices and the loss of the protective function of the forest due to deforestation means that the local population on the eastern coast is increasingly vulnerable to cyclones.
16. Small-scale farmers are exposed to increasing climate hazards, which directly reduce farm productivity, thus reducing the main source of farmers' income. This situation is exacerbated by the damages that extreme events have caused to an already poorly developed and badly managed road network. This reduces access to markets for small-scale farmers. As the target population is already vulnerable owing to high poverty levels and food and nutrition insecurity, the negative impacts of increasing climate variability will increase climate risks in future.
17. This FP proposes a systemic response to the increasing climate risk around three components, involving a combination of grants to government with GCF funds, and loans to government with blended GCF and IFAD funds, some of which will be on-lent to microfinance institutions and from there to farmer organizations and cooperatives. The three components are described as follows:
 - (a) Component 1: strengthening climate resilience of agricultural production systems through improved water management for sustainable agriculture (better irrigation systems and strengthening the capacity of Water User Associations -WUAs-) and enhanced agro-climatic information systems, including better information systems, the dissemination of climate-smart agriculture practices and the promotion of sustainable home technologies (e.g. solar pumps for irrigation or efficient stoves);
 - (b) Component 2: improving the road network in order to secure market access through the introduction of practices related to the Green Roads for Water approach (GR4W)⁴ This component includes both the improvement of local competences for designing, building and maintenance of roads and the rehabilitation or improvement of 540 km of roads during the project implementation, to increase resilience to extreme events and to promote water infiltration in support of agriculture; and
 - (c) Component 3: improving food and nutrition by providing training on food production, nutrition and climate change that will be reinforced by exchange visits. Lessons from this component will be captured and shared as best practices and for knowledge transfer on food, nutrition and climate change.
18. Following the recommendations from iTAP, this second submission includes a framework for selecting adaptation measures (annex 23). This framework uses the following selection criteria: technical feasibility, cost-benefit ratio, response to climate risk, accessibility for smallholders, flexibility (i.e. avoids lock-in), mitigation co-benefits, transformative potential and complementarity with IFAD themes. The annex further documents the scoring process of a long list of adaptation options, which clarifies the process and substantiates the selection of the adaptation measures included in the funding proposal.

⁴ Following the guidance for GW4R provided in the World Bank document *Green Roads for Water: Guidelines for Road Infrastructure in Support of Water Management and Climate Resilience*, available at <https://documents.worldbank.org/en/publication/documents-reports>.

19. Regarding the third principle for demonstrating adaptation potential, the iTAP considers that the adaptation response proposed in this funding proposal is aligned with the priorities and policies from Madagascar (see section 1.5 country ownership, below).
20. With regard to the link to the national monitoring and evaluation system for adaptation at the national level, IFAD clarified that visits by government ministries and agencies (e.g. the National Designated Authority - NDA) will provide opportunities for information exchange and data collection. During these visits, data will be collected on project activities, achievements, challenges and adaptation outcomes. This will include quantitative indicators (e.g. number of farmers with access to agro-climatic data) and qualitative insights (e.g. success stories, lessons learned). A variety of methods will be used, including interviews, surveys, field visits and a review of project documents and reports (annex 11 of the proposal). The information collected during these visits and through the project's monitoring and evaluation system will be structured to support Madagascar in its reporting obligations to the United Nations Framework Convention on Climate Change.
21. The climate change mitigation potential of some of the activities included in this funding proposal (e.g. landscape restoration or agroforestry) is considered as a co-benefit and will be discussed in the section on sustainable development (section 1.3).
22. The iTAP recognizes the improvements made to this funding proposal regarding the adaptation response and the monitoring and evaluation system, compared with the first submission.
23. However, there are still two concerns. The first is on the detailed planning of the landscape rehabilitation and climate-smart agriculture interventions in relation to the specific site conditions, which has not yet been done. The iTAP considers that this can be solved through the preparation of the Project Implementation Manual (PIM) prior the first disbursement.
24. The second concern relates to the long-term financial viability of the irrigation and road infrastructure. This will be discussed in the corresponding section on efficiency and effectiveness (section 1.6).
25. Considering the above, the iTAP assesses the adaptation impact potential of this funding proposal as high.

2.2 Paradigm shift potential

Scale: Medium to High

26. The iTAP recognizes that some aspects of this proposal can contribute to a paradigm shift: in particular, the project's use of an innovative systemic approach that allows understanding of the links between climate change, road networks for accessibility to agricultural markets, and climate-resilient agricultural production.
27. The project has the potential to address the vulnerabilities of smallholder farmers in two regions of Madagascar by improving climate information, promoting resilient agricultural practices through climate-smart agriculture (CSA), promoting water and watershed management through the improvement of water tanks and subwatershed management, and facilitating access to markets (as well as the resilience of roads and groundwater recharge) by using the Green Roads for Water (GR4W) approach.
28. In this way, the project has the potential to achieve progress towards a more sustainable and resilient agricultural sector in the central and south-east zones of Madagascar at the subnational and local levels, as well as improving food access and availability, as well as nutrition.
29. The proposal is well positioned to upscale previous experiences from IFAD with regard to CSA and to use synergies with the GR4W approach. At the same time IFAD, as an international

agency, has the opportunity to promote knowledge exchange at the national and regional levels and to disseminate lessons learned from this project. This is especially relevant for scaling up the systemic approach used in the project and for documenting and sharing the lessons about the benefits of the GR4W approach as an opportunity to place road building and maintenance in the service of water and landscape management and climate resilience while securing their transport functions. Such sharing of experiences and knowledge transfer will be useful for other ongoing activities in Madagascar, especially those led by the Ministry of Public Works and Transport, which is responsible for road construction and maintenance. However, the project activities under output 2.1 only cover roadworks and training of technicians, and do not cover structured engagement planned with the Ministry or with its financial and technical partners, such as the World Bank, who are together engaged in major road construction projects, and have potential to apply more climate risk-informed approaches such as the GR4W approach.

30. The paradigm shift potential is further limited by the fact that CSA practices, and practices aimed at reducing climate vulnerability at the sub-watershed level, are highly context-dependent, and therefore lessons learned cannot be used in other regions without contextualizing the climate risks and the practices and techniques that can actually be considered as adaptive.

31. The level of consistency of the proposal with the adaptation strategies and plans in Madagascar is high (see section 1.5 on country ownership), but the potential for strengthening the regulatory frameworks and policies at the subnational and local levels is unclear. This is particularly relevant regarding the extent to which the project can effectively secure access to land tenure and land-use rights for women and youth, or how the project will align with or support the (new) decentralization plan launched in 2019. In addition to the limitations of output 2.1 (see para. 29), the iTAP notes that in failing to address the wider enabling environment, component 1 includes engagement with national and regional level government agencies only on climate information and misses opportunities for other engagement, for example, on agricultural extension.

32. Considering the above the iTAP assesses the paradigm shift potential as medium to high. The iTAP considers it paramount that IFAD clarifies the concrete measures regarding land tenure and land-use rights to be implemented by this project and the measurable changes to be achieved, with especial consideration to rights for women and youth. The steps towards defining and implementing measures on specific land tenure and land-use rights in the two project regions should therefore be included in the Project Implementation Manual, which, according to the answers provided by IFAD, will be prepared prior to the first disbursement of the GCF funding.

33. Furthermore, and due to the important learning potential of the systemic approach used in this proposal, the iTAP recommends that the AE and the EEs participate in or, if not yet existent, facilitate a learning mechanism that enables the lessons regarding GR4W to be shared with the actors responsible for road building and maintenance in Madagascar, especially the Ministry of Public Works and Transport.

2.3 Sustainable development potential

Scale: Medium to High

34. This funding proposal can contribute to achieving several Sustainable Development Goals (SDGs) in addition to SDG 13 (climate action), including SDG 1 (no poverty), SDG 2 (no hunger) and, to some extent, SDG 7 (affordable energy) and SDG 15 (life on land).

35. The project will have a mitigation co-benefit resulting from activities on CSA, agroforestry, landscape restoration and watershed management (annex 17), as quantified in the proposal. More importantly, as the AE, IFAD is committed to providing information about progress in mitigating climate change to the reporting system in Madagascar, facilitating

reporting of the project's mitigation co-benefits as part of the country's reporting under the UN Framework Convention on Climate Change.

36. Furthermore, important environmental co-benefits can be the improvement of soil conditions through agroecology or the provision of ecosystem services through agroforestry and forest restoration in sub-watersheds.

37. With regard to socioeconomic co-benefits, the project aims to improve nutrition through capacity-building on the management and use of new crops, as well as through the reduction of yield loss, and it will also contribute to increasing the income of local farmers through better access to markets and the possibility to participate in public-private-producer partnerships (known as 4P). These co-benefits can be further improved by securing the participation of small-scale holders from DEFIS+ in the entrepreneurship-related activities from the wider ongoing DEFIS programme.

38. Annex 8 of the funding proposal contains a gender assessment and action plan covering issues such as heads of households, agriculture, employment, use of time, decision-making, education and literacy, inequality in access to land and production factors, gender and natural resource management, and gender and climate change. It also includes an overview of the legal and administrative framework in Madagascar as well as strategy and specific actions aimed at improving the situation of women and youth. However, the lack of clarity about the concrete steps regarding access to land and securing land tenure for women and youth discussed in section 1.2 are a development concern. In addition, some important activities of the Gender Action Plan are not reflected in the detailed budget.

39. Considering the above, the iTAP assesses that the potential contribution to sustainable development in the central and south-east zones can be medium to high, depending upon the degree to which land tenure and land-use rights are properly addressed and how gender-related activities will be funded during implementation.

2.4 Needs of the recipient

Scale: High

40. Several indicators mentioned at the beginning of this assessment show the urgent situation of Madagascar. It is a country affected by climate risks and multidimensional poverty, with high levels of social inequality including gender inequality, and where individual and collective opportunities for sustainable development are strongly reduced.

41. According to a 2023 report on Madagascar by the International Monetary Fund (IMF), the COVID-19 pandemic, the high level of poverty and extreme weather events have aggravated the fragility of Madagascar. After a stronger-than-expected recovery in 2021, growth is estimated to have decelerated in 2022 mostly owing to climate shocks and a worsening global economic environment. Fiscal performance has deteriorated, with weak revenue performance and increasing cross liabilities with oil distributors. By 2023 the IMF considered that although fiscal and external deficits have widened, fiscal and external sustainability are preserved in the medium term.⁵

42. Multiple indicators show the high level of vulnerability to climate change. Furthermore, other factors, such as low income, food insecurity, lack of infrastructure or high levels of illiteracy, reduce the coping capacity of the farmers in Madagascar to respond to evolving climate risks.

43. The current proposal is aimed at reducing vulnerability to climate change while securing food production and opening development opportunities via physical access to the market.

⁵ IMF Country Report No. 23/239 (2023).

44. Thus, the iTAP considers the response to the needs of the recipient as high.

2.5 Country ownership

Scale: Medium to High

45. Annex 1 to the funding proposal includes a letter of no objection from the Government of Madagascar dated 2019. The funding proposal also demonstrates alignment with and contribution to several strategic areas included in the national adaptation plan (NAP) for Madagascar (2021);⁶ specifically agriculture, water management, improvement of coastal areas and management of infrastructure.

46. Further, the funding proposal includes a stakeholders' engagement plan (annex 7) that presents a list of stakeholders and mechanisms aiming at enabling the participation of disadvantaged and vulnerable groups. It also includes a list of activities mentioning the stakeholders, topics and frequency of the activities proposed and explains the need to have consultation processes. However, there is no budget included in the annex, and the detailed budget of the project (see annex 4) does not specifically consider consultations with stakeholders. The iTAP recognizes that the proposal includes activities aimed at "training, workshops and conferences" and that it is possible that consultation processes have been included in these activities. However, at the current stage is not clear if consultations on, for example, the implementation of different types of CSA practices, will be undertaken by the project. Thus, the iTAP considers it paramount that information on the participatory processes for designing the CSA practices and the techniques and management activities regarding the application of the GR4W approach is included in the Project Implementation Manual prior to the first disbursement.

47. Considering the above and once the assessments of the technical capacity of the EEs are available, the iTAP considers the country ownership as medium to high.

2.6 Efficiency and effectiveness

Scale: Medium to High

48. The design of this project responds to several existing barriers in Madagascar, including the limited financial access to invest in climate-resilient technologies and, as mentioned before, it includes some innovative approaches (see sections 1.1 and 1.2).

49. The total budget of the proposal is USD 150,847,444 with a GCF contribution of USD 53,861,003 distributed in the form of a grant of USD 41,045,298 (74 per cent) and a loan of USD 12,815,705 (26 per cent). According to the term sheet, key terms and conditions of the GCF loan to IFAD include a 40-year tenor, a 10-year grace period, zero (0.00) per cent interest rate per annum and a commitment fee of 0.5 per cent per annum. The co-funding by IFAD is USD 98 million, which means a co-financing ratio of 1:1.82 (GCF:IFAD).

50. The funding proposal includes a Letter of Intent from IFAD dated 12 December 2023 (annex 13) and a corresponding temporary annex for co-financing policy requirements (annex 14). According to these annexes, IFAD will contribute to the project with a total co-funding of USD 98 million distributed in the form of a grant of USD 37.4 million and senior loans amounting to USD 59.6 million. According to the annexes the amount has been approved by the IFAD Executive Board and the final approval is subject to the legal, budgetary and policy procedures of IFAD. The iTAP noted that the term sheet submitted on 15 December 2023 includes a condition on confirmation of the co-funding prior to the execution of the funded activity agreement.

51. The funding proposal includes an Economic and Financial Analysis (annex 3). Following the recommendations by the iTAP made during the previous assessment, the current analysis

⁶ Available at <https://unfccc.int/sites/default/files/resource/PNA-Madagascar.pdf>.

includes investment in infrastructure as well as in agriculture and landscape management and was carried out for investment types (ITs): Improved drainage networks; Irrigation schemes; De-silting systems; Landscape restoration; Water tanks; Irrigation parameters; Ridge ditches; Runoff collection; Road improved drainage and reinforcement during rehabilitation; Road storage infrastructure.

52. The economic financial analysis records an economic internal rate of return (E-IRR) of 24 per cent with a net present value (E-NPV) of USD 255.5 million. Although the second submission includes a far more complete narrative for the analysis, the documentation of the input data is incomplete. For this reason, the iTAP conducted a plausibility analysis of the calculations and found two sources of potential overestimation: (i) unclear distribution of the economic benefits per investment type per year; and (ii) implausible consideration of the economic benefits from year one regarding crops that are not currently being cultivated and would need some years before harvesting (e.g. vanilla). The second potential source of overestimation was confirmed during discussions with the AE. That said, this is understandable because the final decision about which crops will be planted, when and where, will only be taken during the inception phase. A clearer picture will be available once the Project Implementation Manual has been presented to the GCF Secretariat prior to the first disbursement.

53. A further concern is the long-term financial viability of the irrigation and road infrastructure. Considering the low payment capacity of the final beneficiaries of this funding proposal, a financial viability based on the payment of fees (for roads and/or access to irrigation) is either unrealistic or, in the best case, insufficient. The iTAP is aware that the funding proposal has a strategy regarding payments for irrigation costs by the Water User Associations which could resolve some of the irrigation costs, and is also aware that this is a structural problem that cannot be solved solely by this project. This is especially the case with regard to securing finance for road maintenance. However, the iTAP encourages the AE to search for innovative and context-appropriate options, especially because IFAD is well positioned in Madagascar.

54. Further, Madagascar is one of the countries with the highest financial risk rating in the global south, which is also reflected in the high interest rate (approx. 12 per cent annually). Madagascar would have to pay in order to hedge any foreign currency loans for local currency loans. Given the high volatility of macroeconomic conditions globally, and higher interest rates of the United States dollar leading to huge devaluations of emerging market currencies, Madagascar may have to do so in order to protect itself from a sovereign default.

55. Although the proposed project would be highly impactful to Madagascar in terms of the resilience to climate change of smallholder farmers, it could potentially harm Madagascar's ability to repay the loan. After all, the loan will be paid back from the Government of Madagascar, which has access only to taxpayers' money in local currency. Its ability to earn foreign currency is limited to very few exports and, given it is an island, it needs to import basic foods and fuels.

56. Considering the above, the iTAP assesses the efficiency and effectiveness of this funding proposal as medium to high.

III. Overall remarks from the independent Technical Advisory Panel

57. The iTAP notes that IFAD addressed most of the recommendations made for the first submission and highlights the many improvements in the funding proposal in this second submission.

58. The iTAP makes the following recommendation to be considered during the implementation of the project:

- (a) IFAD and the EEs are recommended to participate in a dedicated learning and exchange platform that facilitates sharing the lessons regarding the GR4W and long-term financial viability of the road maintenance actions with the actors responsible for road building and maintenance in Madagascar, especially with the Ministry of Public Works and Transport, the Ministry of Agriculture and other key players. If such a platform is not yet in place in Madagascar the iTAP recommends that IFAD consider facilitating such exchange.

59. Considering the above, the iTAP endorses this funding proposal for approval by the Board, subject to the following conditions.

3.6.1. Condition precedent to the effectiveness of the funded activity agreement:

60. Delivery to the GCF by the AE of written confirmation of the completion of the financial management and technical competence assessment of the EEs, in line with the AE's own policies and procedures, in a form and substance satisfactory to the GCF Secretariat and consistent with the requirements of clause 10.02(a)(i) of the accreditation master agreement, as well as a copy of such capacity assessment if requested by the GCF Secretariat;

3.6.2. Condition precedent to first disbursement of GCF proceeds under the funded activity agreement:

61. Delivery to GCF by the accredited entity, in a form and substance satisfactory to the GCF Secretariat, of a copy of the final version of the Project Implementation Manual, which shall include:

- (a) The definition of sub-activities, targets and means of verification aimed at contributing to improve land tenure and land-use rights for small-scale farmers in the two project regions with special attention paid to women and youth;
- (b) The description of the participatory processes for designing the site-specific climate smart agriculture (CSA) practices and the techniques and management activities regarding the application of the Green Roads for Water (GR4W) approach;
- (c) An outline of intended engagements with the relevant national and regional government agencies for applying the lessons learned on climate smart agriculture (CSA) and Green Roads for Water (GR4W) approach, including on investment in maintenance of new infrastructure, through relevant policymaking and budgeting; and
- (d) Clarification of the activities aimed at collaborating with micro-finance institutions with special consideration of how such collaboration will assist beneficiaries.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP227)

Proposal name:	Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme (DEFIS+)
Accredited entity:	International Fund for Agricultural Development (IFAD)
Country:	Madagascar
Project/programme size:	Medium

Impact potential

ITAP's concerns regarding the detailed planning of the landscape rehabilitation, site-specific climate-smart agriculture (CSA) interventions, and the long-term financial viability of the infrastructure are well noted. The Project Implementation Manual (PIM) to be prepared after project approval and expected to be finalized within Q2 2024, will include processes and steps to be followed for detailed landscape rehabilitation planning. The PIM will also provide options for CSA interventions based on site-specific conditions, from which beneficiaries will be able to select appropriate CSA measures following training and capacity building. Innovative ways of financing the infrastructure operation and maintenance will also be explored.

Paradigm shift potential

Regarding the comments on addressing the rights of women and youth, steps towards defining and implementing measures on specific land tenure and land-use rights in the project areas will be included in the PIM. Using existing platforms, IFAD and the Ministry of Agriculture and Livestock will work closely with the Ministry of Public Works and Transport to facilitate a learning mechanism to share lessons on Green Roads for Water (GR4W) with the various actors responsible for road construction and maintenance. Agricultural extension, Farmer Field Schools and other learning opportunities, are included in Component 3.

Sustainable development potential

To enhance the potential contribution to sustainable development in the south-central and south-east zones, concrete steps on access to land and land tenure security for women and youth will be included in the PIM. Gender-related activities will be funded using the resources allocated for capacity building and technology dissemination through the application of the Gender Action Learning System (GALS) during implementation. A Gender and Social Inclusion Specialist will be recruited to ensure that the targets in the Gender Action Plan are achieved. The participation of DEFIS+ smallholders in the entrepreneurship-related activities of the wider ongoing DEFIS programme will be ensured.

Needs of the recipient

As iTAP has pointed out, Madagascar is projected to face increased climate shocks and decreased financial stability, exacerbated with the factors mentioned by iTAP, such as low

income, food insecurity and lack of infrastructure. This project aims to help mitigate these extreme events through effective adaptation actions and sustainable financial measures.

Country ownership

A more recent no objection letter was provided by the NDA in July 2023. To improve country ownership, information on the participatory processes for designing the CSA practices and the techniques and management activities regarding the application of the GR4W approach will be included in the PIM and necessary budget provided. In addition, the assessments of the technical capacity of the Ministry of Economy and Finance and the Ministry of Agriculture and Livestock will be undertaken in line with IFAD policies and procedures, in a form and substance satisfactory to the GCF Secretariat and consistent with the requirements of the Funded Activity Agreement (FAA).

Efficiency and effectiveness

The sources of overestimation in the Economic and Financial Analysis (EFA) are recognized. Smallholders are expected to make informed decisions about what crops to plant, when and where, once their capacities have been enhanced. The EFA will be reviewed at mid-term and principles of adaptive management applied. Recognizing the low capacities of the beneficiaries to pay for the infrastructure operations and maintenance, particularly roads, IFAD, in collaboration with service providers that will engage in project implementation, will seek innovative and context-appropriate options to ensure the long-term sustainability of the infrastructure. This also applies to the irrigation systems and the pastures.

Overall remarks from the independent Technical Advisory Panel:

We appreciate the earlier feedback provided by iTAP, which enabled us to improve the Funding Proposal package. We also appreciate the positive feedback from the second review, which will improve the implementation of the project and contribute to the achievement of the DEFIS+ objective. We are grateful for iTAP's endorsement of the DEFIS+ proposal to the GCF Board.

As recommended by iTAP, IFAD and the Ministry of Agriculture and Livestock will work closely with the Ministry of Works and Transport on a dedicated learning and exchange platform to facilitate the sharing of lessons on GR4W and the long-term financial viability of the road maintenance interventions.

The condition precedent to the effectiveness of the FAA is accepted. IFAD will confirm the completion of the financial management and technical competence assessment of the EEs, in line with the IFAD's policies and procedures, in a form and substance satisfactory to the GCF Secretariat and consistent with the requirements of the AMA.

The condition precedent to first disbursement of GCF proceeds under the FAA is accepted. IFAD will provide a PIM that will include:

- sub-activities, targets and means of verification to improve land tenure and land-use rights for women and youth;
- participatory processes for designing site-specific CSA practices and applying the GR4W approach;
- intended engagements with the relevant government agencies to apply the lessons learned on CSA and the GR4W approach, including investment in maintenance of new infrastructure; and



- activities aimed at collaborating with micro-finance institutions focusing on assisting beneficiaries.

Annex 8 - Gender Analysis/Assessment Guide and Gender Action Plan

Part I: Gender Analysis/Assessment: Guide (Project/Programme Level)

The principle of equal rights of women and men is enshrined in the Malagasy Constitution and the Government of Madagascar has adopted a National Policy on the Advancement of Women (PNPF), which reflects the country's political commitment to eradicating disparities between men and women. Changes have been made to legal and regulatory texts to take gender into consideration, such as the alterations to legislation on marriage to establish the same legal marriageable age for both sexes (in 2007) and the texts on land tenure (in 2009), among others. However, cultural habits and customary practices remain strong and hinder the recognition of women's rights. In certain parts of the country, they limit or even prevent women from having access to land ownership or real estate assets; they do not allow women to inherit land or buildings in the event of the death of a family member or when a couple separates.

The possibility of getting involved and joining an association represents an opportunity for women to improve their status. At the national level, 43 per cent of women declare that they are members of a religious association and 5.5 per cent in a political and civic association. As for their place in the organization, only 4.6 per cent of the women who are members of an association hold the position of president or vice-president; 4.7 per cent, the position of secretary, and 5.1 per cent, other positions. It should be noted that nearly all (85 per cent) women between 15 and 49 years of age who are affiliated to an association are ordinary members (ENSOMD: 2012-2013).

According to the Madagascar Millennium Development Goals National Monitoring Survey (ENSOMD) 2012-2013, it seems that the majority of women participate, together with their husband, in decisions on major household expenses: this was the response in 65 per cent of the cases. In relation to the decisions on the use of women's income over a 12-month period, only one third (30 per cent) of women had this prerogative. In 58 per cent of the cases, this decision had been made together with their husband or partner.

Poverty and sex of the head of the household: At the national level, more than three quarters of households are headed by men (78 per cent). Female headed households (22 per cent) are more commonly found in urban areas (27 per cent) than in rural ones (21 per cent). The average age of women heads of household is higher than that of their male counterparts (46 years and 41 years, respectively). Agriculture is the main economic activity of the majority of heads of household (64 per cent); another 13 per cent are small business entrepreneurs in non-agricultural sectors; approximately 8 per cent are labourers or unskilled workers; and only 2 per cent hold middle- or upper-level management positions. The average number of dependents per household is 2.7. Approximately 30 per cent of heads of household have no schooling (INSTAT/ENSOMD 2013). The same survey shows that 57.2 per cent of female headed households are poor, in comparison to 62.8 per cent of the rest. Generally, women headed households are smaller in size than those headed by men: 3.5 individuals. In terms of the overall population, the poverty gap is shrinking, but is still in women's favour: 69.8 per cent, in comparison to 72.0 per cent.

Gender, education and literacy: In regard to education, the differences between girls and boys in Madagascar are minor at the primary and secondary levels of education in comparison to other sub-Saharan African countries. While there is parity between girls and boys at the primary school

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level (ratio of girls to boys' enrolment is 1.05 in girls' favour), at the secondary and superior education levels, girls are at a disadvantage in comparison to boys; this gap grows as the level of education increases. In Madagascar, illiteracy is still a matter of concern. The adult literacy rate is 64.5 per cent (2008-2012). Among youth (15-24 years of age), the literacy rate is 65.9 per cent for young men and 64 per cent for young women (2008-2012) (UNICEF). Differences in this rate are also found when place of residence is compared: literacy rates are higher in urban areas. Regional disparities also exist.

Gender and employment: On the labour market, there are fewer women wage earners in non-agricultural sectors than men (38 per cent and 62 per cent, respectively). The National Survey of Employment and the Informal Sector (ENEM PSI 2012)¹ noted that the development of the informal sector contributed to more than 24 per cent of GDP (non-farm business market) and to the creation of more than 93 per cent of jobs (of which 76 per cent were in the agricultural sector). This raises the levels of job dissatisfaction, precarity and instability among workers. Inadequate employment, which is a form of under-employment², affects 81 per cent of workers, especially women in rural areas. Among youth from 15 to 29 years of age who are employed, close to 56 per cent are not satisfied with their current status and wish to obtain a new job by changing companies.

Gender and time-use. Many studies show that women spend more than half of their daily time to perform domestic tasks (World Bank, 2014; AfDB, 2016; USAID, 2016) including water fetching, cooking, laundry washing and childcare. Time-use surveys from a USAID gender analysis in the east, southeast, and southwest found that women spend approximately four times as many hours on household activities compared men (USAID, 2016). Men very rarely perform domestic tasks, which are viewed as women's duties according to traditional stereotypes and spend a major part of their time on activities that provide income for the household's subsistence (AfDB, 2016). Women are also performing "productive activities" such as crop farming, gardening, and livestock care and participate in addition in community activities. Coping with these competing roles is a real challenge for women, mostly for those heads of households (FHHs), and led them to a situation of time poverty (USAID, 2016).

Gender and decision-making. Women's household decision making and access to resources is inequitable in Madagascar (USAID, 2013). According to the 2008-2009 DHS, only 33% of married and working women had primary control over their earnings (NSTAT and ICF Macro 2010). Men mostly have highest control over expenditure for all expenses beyond small routine household expenses. However, women tend to have more control over income that they earn from petty trade, poultry sales, or gardening, although income from livestock and agriculture (cash crops and food crops, particularly irrigated) falls more under the control of men (USAID, 2013). Men have also highest control over land and have the final says regarding their use and the choice of crops and agricultural technologies to apply (JICA, 2015). At the community level, women rarely participate in community-level decision-making meetings (CI & CONFORME, 2014). However, in some cases where women do not attend or participate in community meetings, they are still actively influencing men "behind the scenes" (PSI/Madagascar, 2015). Their rate of membership in community structures such as producer groups and local forest management structures remains low. Also, while they can be members of such structures, they have less access to

¹ First type 1 and type 2 mixed survey carried out in Madagascar by INSTAT, with the technical and financial support of UNDP, the International Labor Office and IRD/DIAL.

² Inadequate employment situation refers to workers whose wage is lower than the legal minimum wage (the minimum wage at the time of the survey was MGA 100,000/month) and who are looking for other jobs and are ready to accept new offers.

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executive positions and have limited access to decision-making spheres (CI & CONFORME, 2014).

Gender and agriculture: According to the agricultural census (2004-2005), the farm population is predominantly female, as women represent approximately 51.8 per cent of the total farm population and men, 48.2 per cent. The farm population is quite young: individuals under 15 represent 48.6 per cent of the total; people from 15 to 59 years of age, 47.2 per cent, and people 60 and over, only 4.2 per cent.

Both men and women in the households are involved in agricultural activities while tasks that need strong physical strengths are carried out by men. Women and youth participate actively in farm work in the rice fields and/or other family fields managed by the head of household. It is rare for a woman to manage a rice field autonomously or on her own. However, women who are part of structured groups farm on fields or perimeters rented or allocated to them by private individuals (often their husbands, a relative of one of the women of the group or a village chief). The women, especially the ones from the poorest farms, work as labourers for other rice growers or farmers in the area. In the case of rice, a fundamental crop of family farms, women are present in the various stages of the productive process: they play an important role in seeding, nurseries, transplanting, weeding and harvest. Men are more involved in the ploughing and preparing of rice fields.

Women tend to care for household gardens and subsistence agriculture, while men are more likely to engage in for-profit cash crops. Women can be involved in agricultural produce selling but men tend to have the highest control on the earned incomes. Men have also the final say about land-use and choice of agricultural technologies and own major of farming tools. Despite their work in agriculture, women's participation in decisions on, for example, the distribution of land and the choice of plots to develop is limited, as men have greater access to farm supplies, means of production and technology. The income women earn is used to cover their family's needs and their children's education.

In regard to livestock, women mainly raise animal species with short life cycles (poultry and pigs), which represent both an income-generating activity and a way to increase their capital using income earned from other crops or economic activities. Several experiences based on solidarity chains among women's groups at the village level both reinforced the small animal breeding sector and contributed to the economic empowerment of women.

The challenges facing the agriculture sector policy relate to: mobilizing all stakeholders, with the adoption of a gender approach, around the sector's objectives, while giving importance to the decentralization and through a program approach; providing the support needed to boost the competitiveness and productivity of businesses, including local small and medium-sized enterprises headed by women.

Women represent 53% of the rural agricultural population. However, they face a number of constraints in the family, community and professional spheres. They are poorly represented and have very little access to decision-making in economic bodies. Sub-programme 4.4 of the Agriculture, Livestock and Fisheries Sector Program concerns Capacity building for women in terms of agribusiness, agricultural entrepreneurship and value chain development. This sub-program aims to improve the economic empowerment of rural women, by strengthening their skills in agribusiness, agricultural entrepreneurship and value chain development.

According to this policy, by 2025:

- The number of rural women leaders emerging and strengthened will be 750,

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- The number of generating activities managed by women will be 120,000.
- 500,000 women will be trained and equipped in new agricultural technologies
- 180 exchange spaces for rural women will be created and equipped;
- 50 training courses will be provided for women on improving product quality;

The disruption of climatic conditions associated with changes in precipitation and temperatures should lead to changes in the types of crops and/or cropping methods and/or cropping patterns adopted in climate change adaptation strategies in agriculture. As women practice less mixed farming than men - on average per female-headed household versus four per male-headed household, and respectively male-headed households, and respectively 24% and 15% of female- and male-headed households engaged in monoculture - and with smaller areas under cultivation, women face greater risks of losing their agricultural sources of income and livelihoods. Their capacity to adapt to change is also reduced with the illiteracy rate, the lack of agricultural equipment and the weight of tradition more important among women. (Country gender profile Republic of Madagascar, 2021).

Inequality in access to land and production factors: In relation to access to factors of production, it is difficult for women to become landowners despite their strong presence in the agricultural sector. Civil laws grant equal rights to men and women, but customary laws remain the most applied in land management, mostly in rural areas (OECD, 2019). Customs regarding to land vary across regions and ethnic groups, but in most cases, women tend to have only secondary rights (Archambault & Zoomers, 2017). In case of decease of a landowner, often land is divided among the male siblings only, or land is given back to families once a woman leaves her village (FAO, nd). In addition, in the event of a divorce or death, women are only afforded 1/3 of the common property, and in some cases a wife may not receive anything at all (Archambault & Zoomers, 2017). Hence, the average area owned by FHHs (1 hectare) is almost the half of that MHHs possess (1.9 hectares) (INSTAT, 2011), and FHHs own generally marginalized lands. Women gain access to land as members of women's groups who work on plots that are lent, allocated or leased to them, as mentioned earlier, but they are unable to invest (e.g. plant trees). Women also have more limited access than men to factors of production, namely agricultural supplies, new technologies, financing, training and information.

Gender, forest and natural resources: Madagascar's forests are a valuable resource for rural populations. They provide essential materials for housing, fuelwood for cooking, medicines for adequate healthcare and a source of food for diverse foods (Randrianarivony et al. 2017; Borgerson et al. 2018). Collection of wood for cooking and timber for construction is men's responsibility, while the collection of traditional medicinal plants is predominantly a female activity (Golden et al. 2016, Borgerson et al., 2018). Men may in addition practice opportunistic hunting (using hunting pursuit methods and passive snare trap (Reuter et al., 2016; Borgerson et al., 2018). However, although people benefit from forest resources, deforestation remains a serious problem and constitutes the main cause of climate change effects exacerbation in the country (AfDB, 2016). Rural populations are involved in forest and natural resources management through local forest management structures (VOI/COBA, KASTI). However, women's membership in these structures remains low and executive positions are generally occupied by men (CI and Conforme, 2014).

Gender and climate change. Both men and women among smallholder farmer households in Madagascar are seriously affected by the effects of climate change. However, impacts on women,

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especially female household heads, seems to be worse than those on men. Women's work burden has increased due to water scarcity, food shortage and low availability of fuelwood (collecting water and often collecting firewood are tasks assigned to women and girls). Also, cases of gender-based violence (of various forms) perpetrated by men on women tend to increase in the face of harsh living conditions. Moreover, female-headed households are more exposed to risks of food insecurity because they produce less due to their poor access to land, labour, agricultural technologies, and capital. Statistics related to impacts of climatic hazards on the beneficiaries' crops show in addition that damages observed on women's farms is significantly greater than that reported by men. Studies reveal that this is due in part to the poor diversification of crops on women's farms (the average number of crops cultivated by FHHs is lower than by MHHs) and an unequal distribution of land (women often only have access to marginal land).

Gender-based violence. A third of women (32%) interviewed during an USAID's survey thought that a husband is justified in beating his wife for at least one of the following reasons: she burns the food, she argues with him, she goes out without telling him, she neglects the children, or she refuses to have sexual intercourse with him (USAID, 2013). According to a survey conducted by INSTAT in 2014, 30% of women across Madagascar, or 3 in 10 women, said they had experienced at least one of the four types of violence while 8% of them suffered from 2 types of violence.

Table 1
Social indicators for the country

Total labour force (thousands), 2014	12,077
Female labour force as % of total, 2014	49.38
Crude birth rate (per thousand people), 2014	34.46
Crude death rate (per thousand people), 2014	6.67
Infant mortality rate (per thousand live births), 2015	35.9
Life expectancy at birth (years), 2015	65.1
Total labour force (thousands), 2014	12,077
Female labour force as % of total, 2014	49.38

Source: World Bank. 2016. World Development Indicators.

Legal and administrative framework. Madagascar has a legal and administrative framework conducive to the promotion of gender equality, which explains the country's good rankings according to international indexes, despite the persistence of significant gender gaps as mentioned in the above section. In addition to adhering to most of the international conventions, the country has adopted several laws promoting women's rights and has a functional institutional mechanism for the promotion of gender.

The country has ratified or signed many international and regional conventions or protocols regarding women's rights and gender equity, namely:

- The Convention on the Elimination of All Forms of Discrimination against Women (1988),
- The Beijing Declaration, and its action plan (1995),
- The Southern African Development Community Protocol on Gender and Development,

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- The United Nations (UN) Security Council Resolution 1325 on Women, Peace and Security (2000).

More recently, the country adhered to the global agendas on gender equality and women empowerment in all spheres of life, through:

- The Political Declaration of the Commission on the Status of Women at its 59th session on Beijing + 20 and that of the International Organization of La Francophonie (March 2015),
- The Sustainable Development Goals where these questions are essential aspects for the vision of the 2030 Agenda (September 2015),
- The Agenda 2063 of the African Union which recognizes the role of women, girls, and young people in the achievement of the related objectives while inviting Africa to act in favour of women and girls and,
- The Gender Policy and Strategy of the Indian Ocean Commission (January 2016).

In 2016, the Declaration of Antananarivo at the end of the XVI Summit of the Heads of State of “La Francophonie”, materializes its previous commitments on women, girls and young people (especially young women) particularly concerning development, trafficking, human beings, education, training, the environment, entrepreneurship, employment and the representation of women in decision-making bodies, for “shared growth and [un] sustainable and responsible development”.

The country has a juridical arsenal that protect gender equality and promote women’s rights. The 2010 constitution reaffirms the equality of men and women before the law. Several national laws promote gender equality and protect women’s rights in access to land and assets (Law on Estate, wills, and donations, 1968; The new land policy, 2005; Law on marriage and matrimonial status, 2007). Some laws were also revised to be less discriminatory against women such as the Labour code which henceforth prohibits discrimination based on gender (Article 64, 2004), the Law on Marriage and Matrimonial Regimes which increase the legal age for marriage to 18 years old, allow both spouses to own individual property where they are entitled to administer without interference, and allow each spouse to administer joint property acquired during the marriage, which was before a right that was exclusively held by men. In addition, a new law related to the fight against Gender-Based Violence was adopted and provides severe penalties for perpetrators of GBV (Law n°2019-008).

Since the beginning of the 2000s, political efforts in favour of gender equality have been noted, in particular through revision of legislation and adoption of less discriminatory laws and the development of a National Policy for the Promotion of Women (“Politique Nationale pour la Promotion des Femmes”, PNPF). This policy aims to reduce disparities between men and women and aimed to: (i) Improve the income and economic status of women (especially rural women and women heads of household) (ii) Increase the level of education and training of women and girls, (iii) promote rights to health and reproductive rights, (iv) strengthen the participation of women in decision-making processes and (v) improve institutional mechanisms related to gender equality. Nevertheless, the recurrent socio-political crises in the country have limited the true integration of gender into institutional mechanisms and public policies and its implementation plan covers only the period 2004-2008.

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In 2016, the country launched a National Strategy for Fight against GBV (SNLVBG) for 2017-2021, with its action plan. The aim of this National Strategy is to contribute to reducing the prevalence of GBV and its general objective is to provide actors with a reference document to lead actions to prevent and respond to GBV in a coordinated and effective manner. However, although the country has policy and national programs related to Climate Change, those are not sufficiently considering gender in their orientation (AfDB, 2016). This neutrality is partially due to lack of studies and data that highlight differentiated impacts on men and women (World Bank, 2011).

The Ministry of Population, Social Protection and Promotion of Women (“Ministère de la Population, de la Protection Sociale et de la Promotion des Femmes”, MPPSPF) leads the implementation of policies, strategies, and national programs for the promotion of gender. Since 2015, the Ministry has dedicated gender mainstreaming to an attached structure, and the coordination of specific actions for women at the General Directorate for the Promotion of Women (DGPF). The Ministry has regional (directorates) and sub-regional (district level) branches which represent this structure at regional and districts level. In addition, with the support of its partners, the Ministry has set up “Trano Aro Zo” offices in some localities (mostly in urban and peripheral areas) which provide, among others, counselling and support to women victims of domestic and gender-based violence.

Civil society organizations (CSOs), including international and national NGOs and associations, are also having an increased role in advancing gender equality in Madagascar (AfFD, 2015). Through their interventions in various sectors (water and sanitation, governance, education, health, agriculture, food security, governance, etc.), they act to stimulate action and challenges related to gender equality / equity and collaborate with the Government as well as private sector. CSO's networking with sub-regional, regional, and global movements made remarkable progress and have positive impacts on gender equality in the country (AfDB, 2005).

It is important to recognize that women are a heterogeneous group (by age, wealth, socio-economic status, marital status), which results in significant implications for their resource endowments such as labour, access to land, access to finance and productive assets and their freedom to make independent decisions. In turn, this influences their information-seeking behaviour, accessing project services such as ability to attend training, develop new skills, adopt new practices and make decisions regarding income earned. Among those often most disadvantaged are women heading their own households as a result of never marrying, divorce or widowhood.

Summary of targeting approach and challenges in DEFIS areas: The overall targeting process favoured by DEFIS prioritizes vulnerable groups, including women, in particular women heads of households. The identification of investment areas is based, among other things, on the presence of a critical mass of beneficiaries, including the presence of a critical mass of women involved in the sectors. De facto, all the investments planned within the framework of all the components of DEFIS thus take into account the concern to reach more women as a priority group. Nevertheless, for the benefit of the management of activities and to strengthen the implementation of specific pro-gender actions, some components provide for specific activities for the benefit of women. In terms of indicators, the activities of DEFIS and DEFIS + are complementary and provide for capacity building of 2,400 women leaders in structuring

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organizations, leadership and management. Overall, out of the total direct beneficiaries of the Programme, the aim is to support at least 30% women, 50% of whom are young women.

On the whole, women face specific constraints related to access to land and physical capital and sometimes to access to leadership bodies and decision-making circles at the community level. They also sometimes do not have access to decision-making in the management of assets and capital (livestock) at the household level. Specifically, female heads of household are increasingly marginalized in decision-making at the group and community level. For lack of guaranty, they are also not credible towards financial institutions. They sometimes remain dependent on others.

With regards to infrastructure such as roads, which are largely deplorable in the DEFIS intervention areas, the population is forced to go on foot to market centers, schools, health centers and town/district headquarters to settle administrative matters. The transportation of agricultural products remains difficult. Pregnant women and young children are the most vulnerable to this situation. It should be noted that in terms of health, the geographical accessibility of health centers is a major issue influencing the attendance of health facilities by women and therefore affecting their state of health and that of their children. The densification of roads and rural tracks will be of definite benefit to women and men through the improvement in their living conditions. Jobs for women can also be created, directly linked to the project, in the construction of rural roads using the high labour intensity approach (HIMO).

In all cases, households headed by women derive less income from agricultural activities than those headed by men. Mechanization is limited by the smallness of the cultivated areas, but it also remains more and more inaccessible to women. Indeed, the acquisition and use of equipment more important than the spade remains the prerogative of men. The constraints and factors limiting the development of agricultural activities in general affect women even more. Various observations relate them, on the one hand, to the difficulty of access to seeds and fertilizers, the insufficient cultivated area and the low selling price, and on the other hand, to the low levels of human capital, the dilapidation of production and transport facilities (particularly rural roads), high exposure to climatic effects and poor irrigation infrastructure. The transmission by inheritance of land from parents is most often to the detriment of women.

DEFIS will give priority to women heads of household and young girls in targeting the AFEs benefiting from input support, the development of irrigated areas and the provision of production and processing equipment. Micro-subsidies are financed by mobilising the mechanism targeting the vulnerable in order to enable women to develop income-generating activities and strengthen their economic autonomy.

Proposed solutions for gender empowerment in DEFIS implementation: To strengthen the inclusion of women, the Program implements specific actions for the benefit of women, including the mechanism targeting the vulnerable group, the promotion of pro-gender activities as well as the establishment of small village savings and credit groups. By promoting income-generating activities, these mechanisms allow them to quickly generate more income and exceed the poverty line. Their gradual integration into remunerative sectors remains the guarantor of their real exit from poverty and in the long term.

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The taking into account of the gender dimension is rather institutionalized at the level of each institution including the steering committee of the Program (CNP at the national level and CROS at the regional level). The national federation of rural women is specifically represented in the national umbrella structure of producers (TTM). The establishment of producer organizations respects the quota system at the level of the governing body (over 30% women).

Promoting gender equality in agriculture requires support in various areas, including i) freer access for women to land ownership, requiring awareness-raising targeting the guardians of traditions on inheritance practices and on the rights of couples united by traditional/customary marriage; information and education for women and couples on land certification rights and procedures; (ii) greater access for women to improved production techniques (seeds, fertilizers, inputs, materials, processing, conservation, marketing, management, etc.), requiring training and financing facilities; (iii) a greater ability of women to regulate and sell their products, referring to the need for networking in the markets; (iv) greater access for women to their income, which would require actions to raise awareness among couples.

The legislative texts and regulatory framework in Madagascar do not exclude the rights of women and young people in terms of access to land. DEFIS support, such as technical assistance from FAO, respects the regulatory frameworks in force and provides for the promotion of the issuance of land certificates that include women. This guarantees the allocation of land to women and young people even after the completion of the Program.

In the areas of the DEFIS programme, problems with hygiene and water-related diseases do exist. Limited access to potable water and poor sanitation and hygiene practices are of particular concern for some areas. Awareness-raising activities on water-related diseases will be carried out during the implementation of the project, especially in the areas where irrigated perimeters have been set up, because households sometimes use non-potable sources. This is also part of the activities where beneficiaries will be sensitized on water-borne diseases.

Women intervene within local institutions and POs to defend their interests within the group, particularly in terms of access to inputs, financing, agricultural and processing equipment and above all access to capacity building and training modules. the autonomy of women. The quota system at the level of leading members can be strengthened and the activities supported can provide for the specific interests of women through the project.

Support for the emergence of young leaders and specific funding for youth projects will be provided through capacity building such as the gender action learning system.

Part II: Gender equality action plan: Model (at the project/programme level)

Gender strategy

DEFIS adopted a **gender strategy** based on the need to address inequalities related to sex, age and status when implementing the programme's activities. The programme's gender strategy is founded on Madagascar's national policy, IFAD's Gender Policy and the experiences and best practices developed in the framework of earlier projects in Madagascar. The objective is to: (i) eliminate the constraints that limit the inclusion, active participation and empowerment of women and youth (men and women) in the different links of agricultural chains; (ii) create opportunities to

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improve their living conditions; (iii) strengthen their productive capacities to increase the productivity and profitability of their economic activities by facilitating the access of women and youth to resources and factors of production (land, water, inputs, financing/credit, professional training, technology, support, etc.) and (iv) promote women's leadership and participation in decision-making at home, in farmers' organisations (grassroots peasant organisations, regional umbrella organisations and umbrella organisations – whose acronyms are OPB, OPR and OPF, respectively – as well as farmers' unions), the consultation forums of value chains and local communities in general.

The strategy was informed by consultations that took place during the realization project design through a series of public consultation (South East Region, South Region and Center Region) carried out that significantly involved the participation of women's groups and young people even the participation of women's groups since the grassroots (women's cooperatives, grassroots groups at village level). During the public consultations, specific requests were made for stakeholders to propose concrete actions for women that could be financed by the DEFIS program.

The strategy also takes into account that women are also active as agents of change, building climate resilience and finding innovative ways to adapt to climate change impacts on their communities and environments,³ for example by switching to drought-tolerant seeds, employing low impact or organic soil management techniques, or leading community-based reforestation and restoration efforts.⁴ These activities will be built on further during the implementation of DEFIS+.

The **strategy for the inclusion and empowerment of women** in the cultural and socioeconomic context of DEFIS' eight intervention regions will take into account the primary constraints that women face when engaging in agricultural activities. The biggest constraints on the five dimensions of women's empowerment are: (i) production: limited participation in decisions on the management of the family farm and limited autonomy to farm on small plots of land and/or land farmed by groups, and/or in certain post-harvest activities (processing); (ii) resources: restricted access and decision-making power over resources and factors of production (land, farm supplies, new production and processing techniques, training/information; financing/credit; etc.); (iii) income: limited access to family farm income and the fact that decisions on the use of these resources are made by the men; (iv) leadership: limited capacity for public speaking and to express their needs in the home, in communities, farmers' organizations (FOs) and other associations of which they are members; and (v) time: farming and domestic workloads that occupy women's entire day and the unequal division of labour between men and women.

DEFIS will support: (i) viable, sustainable and climate-resilient production activities that contribute to the economic empowerment of women; and (ii) capacity-building activities that increase women's voice, representation and participation in decision-making such as in farmer groups and cooperatives. (iii) Reduce workloads for women related to production and post-harvest activities (processing, sales, etc.) (ii) facilitating their access to factors of production, including land, water, seeds and financing; (iii) facilitating their access to efficient techniques and energy and labour saving technologies and and (iv) ensuring joint control of benefits over the management of the

³ WIFCAG, Oliver Wyman Forum, 2X Global. 2023. [Applying a Gender Lens to Climate Investing](https://www.oliverwymanforum.com/content/dam/oliver-wyman/ow-forum/climate/2023/WIFCAG_Gender_and_Climate_Framework_vF.pdf)
https://www.oliverwymanforum.com/content/dam/oliver-wyman/ow-forum/climate/2023/WIFCAG_Gender_and_Climate_Framework_vF.pdf

⁴ Schalatek, L. 2022. Gender and Climate Finance, Climate Funds Update, ODI and HBF https://climatefundsupdate.org/wp-content/uploads/2022/03/CFF10-Gender-and-CF_ENG-2021.pdf

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income generated by their activities. Access to more efficient technologies and techniques (combine harvesters, rice mills, machines to shell corn, etc.) and investments on water infrastructure will help reduce women's workload and the time spent on farm work.

To increase voice, representation and decision-making of women,, DEFIS' strategy includes actions that aim to: (i) improve the structure and governance of women's groups/associations and strengthen their leadership (training, awareness-building and information); (ii) increase knowledge on and respect for women's rights by building the awareness of all actors from the local communities on this issue, including men, women, youth, administrative, traditional and religious authorities and all individuals who influence the social and economic life of the community and (iii) facilitate access to business management tools and tools to improve their understanding of information provided, namely literacy training.

Women's influence on the choice of energy sources and equipment to be used remains limited by the level of the costs of this choice that the resource providers (boss; spouse) are willing to pay. Small-scale experiments with solar or wind-powered electrification of remote villages have shown that women have benefited both economically. Women's ability to adapt to change and their lack of access to climate-resilient farming systems and agroclimatic information are further reduced by illiteracy rates, a lack of farming equipment and the greater weight of tradition. Problems that may arise include husbands not accepting women's involvement in activities. Awareness-raising and information campaigns will be organized at village level for the entire population, including men, traditional authorities and all those with influence over the socio-economic dynamics of the villages, to encourage the involvement of young people and women.

Leadership training is also planned to reinforce women's active participation in POs, in the communities to which they belong and in the management of their organizations. This will enable women to make a real commitment to their communities. For all the activities planned by the DEFIS Program, eligibility criteria will promote the inclusion of the most vulnerable groups, FBOs, youth and women in project activities. The use of a quota system (achieving a minimum of 30% women and 30% youth) will reinforce these criteria.

Maintenance, rehabilitation and construction works are generally male dominated due to the manual nature of the work and labour efforts required. However, considerations for use of labour-saving equipment for women or allocation of lighter tasks related to rehabilitation and construction will be made. Therefore, the GAP includes an activity that guarantees women's participation in maintenance, rehabilitation and reconstruction work. This is also a result of DEFIS' targeting criteria that state women must be represented and participate in local decision-making bodies. During construction work, companies will be sensitized to ensure women's involvement in construction – light activities.

The difficulties encountered by women in participating in the Programme's activities are linked to: (i) their limited decision-making capacity. Women, especially married women, are very dependent on their husbands for decisions on the use of household production factors, (ii) their poor access to productive resources, (iii) women's workload at farm and household level. Women's participation in the various activities listed is likely to disrupt the relationship between men and women and give rise to conflicts within the household. Awareness-raising as well as the gender action learning that will be adopted and the mutual involvement of couples in the Programme's activities will help to eliminate this tension. The introduction of technologies to reduce the workload will also allow women to have more time to devote to Programme activities.

Constant monitoring and follow-up on these actions are necessary to ensure their sustainability and their appropriation by the women, women's groups, FOs and local communities. The

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programme will mobilize actors in the inter-regional/regional coordination units and/or service providers who will receive training on DEFIS' strategies and tools for implementing, monitoring and evaluating the gender approach and the achievement of its objectives. Furthermore, leaders (men and women) from the FOs and/or communities who are recognized as such by the population will be identified and trained on gender equality and women's rights so that they can then facilitate awareness-building/information sessions with local communities. This details of process will be refined once the sites/villages/poles of development for the programme's interventions have been selected. In this selection process, specific sociocultural elements of the local context (e.g. customs of ethnic groups) will be taken into account to avoid any conflicts in the future.

The implementation of the gender strategy

Operational measures have been identified to guarantee the effective mainstreaming of the gender dimension. They are to make sure that while executing all programme activities, special care is systematically taken to ensure that women and youth from the most vulnerable FFs effectively engage. This exercise, which began during the programme's design phase, will continue to be carried out at programme start and throughout its implementation, as it will be incorporated into the successive AWP/Bs. The entire programme team will be involved in the execution of AWP/Bs.

As in the case of the targeting strategy, the responsibility for executing the gender strategy lies with the entire programme coordination team, as well as the different partners and institutions in charge of programme implementation, who will receive training on gender issues. The PCU shall: (i) promote and monitor how gender issues are taken into account in the programme; (ii) guarantee that the programme's gender strategy is appropriate given its approach and the context and sociocultural reality in the intervention zone, which continue to have considerable influence on the role of women in the home and the FFs; (iii) ensure that aspects related to the integration of gender and the inclusion of youth are systematically taken into account and that M&E indicators are disaggregated by sex and age. Practical measures will be taken to: (i) introduce the gender dimension into the selection criteria of the recruitment of programme personnel; and (ii) build the awareness and capacity of the programme team and implementing partners on gender issues and respect for the gender principle during the operationalization of the programme.

Youth empowerment. The project will (i) increase youth and other vulnerable groups to acquire jobs through wage employment and entrepreneurship or income-generating activities. These actions include: (i) Increase access to land allocation for youth by reserving plots of land for them in the newly-developed perimeters; (ii) positioning youth including younger women as rural entrepreneurs in the services sector in areas linked to the poles of development; and (iii) ensuring male and female participate in capacity-building activities and access to financing for farmers. The goal is to ensure that at least 50 per cent of all beneficiaries are women and 30% of all beneficiaries are youth (men and women)⁵.

DEFIS will engage with service providers (national and international NGOs and research institutes) who have expertise and proven experience in the field of women's rights and youth empowerment. Priority will be given to collaborating with rural women's federations and associations to strengthen the women farmers' network and organize exchanges on issues that are common across the value chains in which the women are evolved. There are plans for

⁵ To arrive at these totals, the calculations are as follows: 15 per cent young men + 15 per cent young women + 15 per cent women in other age categories. See DEFIS DCP.

collaboration with the Ministry of Population, Social Protection and Women and its regional directorates in view of its mission to promote the advancement of women and gender equality

DEFIS will monitored youth targets through the Program monitoring and evaluation team (at the national and regional levels). The close collaboration with the FORMAPROD Program (an IFAD project, which deals with the integration of young people and rural training intervening in almost the regions where DEFIS is implemented) is developed at the field level: involvement of young people trained in agricultural advisory services, promotion of the Gender Action Learning System (GALS) methodology to integrate gender interventions and support and financing of income-generating activities.

The project will recruit a Gender and Social inclusion officer to enhance the capacity of the project and ensure the effective implementation of gender mainstreaming. The officer will support the monitoring of the gender results indicators assigned at the level of each activity will reinforce the involvement and the strong commitment of the implementing partners of the activities in the field. The Officer will coordinate the gender mainstreaming and also collaborate with the Technicians at different levels that dedicate permanent time to the management of their daily activities while being concerned with the gender dimension. At the level of the Steering Committee, rural women are represented in the TTM and participate in the validation of the AWPBs and the annual monitoring of implementation. Like the CNP, the CROS operate in the same way at regional level.

Gender-based violence and Sexual Exploitation, Abuse and Harassment

Gender-based violence and Sexual Exploitation, Abuse and Harassment (SEAH) undermines the security and well-being of individuals and entire communities, and threatens efforts to end poverty and stimulate shared prosperity. To address this, recommendations linked to the elimination of gender-based violence will include the following actions:

- Strengthen awareness of GBV and SEAH issues among staff and beneficiaries through trainings and continuous learning activities.
- Develop or create awareness on prevention, handling and referral of GBV protocol working with MOH in order to anticipate and, if necessary, deal with related incidents.
- Strengthen operational /screening processes for activities deemed to be at high risk of GBV and SEAH, including signing of codes of conduct for public works contractors prohibiting all forms of GBV and SEAH- specifically against sexual activity with anyone under the age of 18. Contractor contracts and collaboration agreements signed with the project will contain relevant clauses.
- Develop and create awareness on a complaints and grievance redress mechanism for the project
- Information and awareness-raising on GBV and SEAH issues for those involved and beneficiaries.

The various project stakeholders and communities will be informed of the existence of the project's complaint management mechanism, which is one of the entry points for identifying and recording cases of GBV and SEAH in activities linked to project implementation.

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Reporting procedures for gender-based violence, SEAH and violence against children

All staff, volunteers, consultants and contractors are encouraged and required to report suspected or actual cases of GBV as they are responsible for complying with the company's commitments and hold their direct reports accountable for compliance with the individual Code of Conduct.

The project will provide information to employees and the community on how to report cases of violations of the GBV Codes of Conduct through the grievance resolution mechanism (GRM). The project will follow up on cases of GBV and Code of Conduct violations reported through the grievance mechanism.

Service provider: In handling any cases of GBV and SEAH, DEFIS task team will work through a Service Provider. The service provider is a local organization that has the experience and capacity to support Victims of GBV. The client, contractor and consultant must establish a working relationship with the service provider so that GBV cases can be safely referred to it. The service provider will also provide support and advice to the GBV focal points as required.

GBV and SEAH focal point: The project must confirm that all complaints related to GBV and SEAH have been forwarded to IFAD. The project must consider all GBV complaints and agree on a resolution plan. The Project will advise on resolution, including referral to the police if necessary. They will be assisted by the service provider where appropriate. All reports must be kept confidential and returned immediately to the service provider. In cases of GBV and SEAH warranting police action, the project must refer the complaint appropriately to: (i) the authorities; (ii) the service provider; and (iii) management for further action. The client and IFAD must be informed immediately.

Transparent measures: All reports of GBV and SEAH must be treated confidentially in order to protect the rights of all those involved. The client, contractor and consultant must maintain the confidentiality of employees who report any act or threat of violence and of employees accused of committing acts or threats of violence (unless a violation of confidentiality is required to protect harmed persons or property or when required by law). To ensure that Victims feel confident in disclosing their experience of GBV, they can report cases of GBV and SEAH through various channels: (i) online, (ii) by telephone, (iii) in person, (iv) the local service provider, (v) the manager(s), (vi) village councils; or, (vii) the police. To ensure confidentiality, only the service provider will have access to information about the Victim. The project will be the main point of contact for information and follow-up concerning the perpetrator.

Monitoring and evaluation: The project shall monitor the follow-up of reported cases and maintain all reported cases in a confidential and secure location. Monitoring should collect the number of cases that have been reported and the proportion of those managed by the police, NGOs, etc. These statistics should be reported for inclusion in the project's annual report. These statistics should be reported for inclusion in their reports. For all cases of GBV and ECV warranting police action, the project must be informed immediately.

Response protocol: The project will be responsible for developing a written response protocol to meet project requirements, in accordance with national laws and protocols. The response protocol must include mechanisms for reporting and responding to aggressors in the workplace with a competent and confidential response to disclosures of GBV and SEAH.

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Gender Action Plan

Results	Activities	Indicators and targets	Timeline	Responsibilities	Budget (000 USD)
Objective: Strengthen the resilience of agricultural production systems and vulnerable marginal communities to the impacts of climate change while contributing to enhance mitigation co-benefits in six regions in Madagascar through gender mainstreaming and technical and financial assistance to gender-specific activities					
Outcome 1 Strengthened climate resilience of agricultural production systems through enhancement and maintenance of ecosystem services					
Output 1.1: Improved water management for sustainable agriculture	Increase women's voice, representation and decision-making in WUAs, ensuring a quota of at least 30%; providing leadership training, etc.) and the development of water resource preservation and management plans	50 % of women in WUA Overall target 1400 (700 females) 30 % of women in leadership positions in WUA Target 210 females	Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	40
	Construction of water tanks for drip-irrigation and for small livestock to reduce women's workloads	50 % of women amongst those benefiting from water tanks Overall target 150 (75 females)	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	3 461
	Support access to energy and labour-saving equipment for women and youth e.g. cooking stoves, small farm equipment, post harvest technologies	30 % of women and youth with labour saving equipment Target 80, 496 # of technologies supported and disseminated	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	100
	Establishment of input shops and workshops for the manufacture and repair of	# of youth managed inputs shops established	Annually	Monitoring and evaluation manager and Gender and	40

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	agricultural equipment for youth beneficiaries.	# of youth managed workshops for manufacture of agricultural equipment		social Inclusion officer	
	Increase access to land for women and youth through allocations e.g. through community sessions with traditional authorities and leaders etc	30 % of women and youth with new land allocations Target 80 496	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	50
Output 1.2: Enhanced agro-climatic information systems new technologies and initiatives ⁶	Training on climate information in capacity building of decentralized government (SRM & DRAEP) to ensure improved access to women by integrating their gendered needs	# of people (F,M) trained on gendered dimension of access and preferences regarding climate information Target 89 440	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	492
	Support roll out and adoption of renewable energy and energy-efficient technologies at households	50 % of women amongst those benefiting from renewable energy and energy-efficient technologies Target 134 160 females 30 % of Female Headed households benefitting from energy efficient technologies		Monitoring and evaluation manager and Gender and social Inclusion officer	871

⁶ Women play important roles in the preparation of agricultural activities and in relation to the promotion of agricultural activities in the off-season to develop income-generating activities and for nutrition (case of market gardening). Agro-meteorological information is essential for women in relation to these roles (planning of input supply, adjustment of the agricultural calendar, management of agricultural labor, etc.)

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	Training in Gender Action Learning System (GALS) for the dissemination of Climate Smart Agriculture practices and addressing gender issues	# of farmers (F,M, youth) trained on GALS # of farmers (trained in GALS including FHHs and vulnerable	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	291
Outcome 2: Increased income generation through improved market accessibility					
Output 2.1: Reinforced rural access roads	Increased voice for women through, for example, focus group discussions to ensure their priorities are considered for site selection, management committees	# of women in WUAs leadership Target 210	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	47
	Increased involvement of women in less intensive maintenance and rehabilitation/reconstruction works (e.g. requiring mandatory recruitment procedures in minor works contracts, preceded by conscience-raising activities targeting both men (to encourage them to allow female family members to participate) and women (to inform them of employment opportunities); ensuring equal pay for equal work and require contractors to submit weekly time sheets, broken out by	% of women involved in light maintenance and rehabilitation/reconstruction works Target 89 440 Number of time sheets submitted disaggregated by gender	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	47

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	gender; stimulating the development of women-owned construction enterprises; adapting work programmes to allow for women's considerable domestic workload and travel restrictions;...).				
Output 2.2: Improved capacity to build and maintain rural infrastructure	Training of women on construction standards related to climate change	20 % of women amongst those trained on construction standards related to climate change Target 35 776 women	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	47
	Training of youth on construction standards related to climate change	30 % of youth amongst those trained on construction standards related to climate change Target 53, 664 youth	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	45
Outcome 3: Improved food and nutrition security through managing knowledge on food production systems.					
Output 3.1: Knowledge on climate change adaptation and mitigation	Training of women and rural development actors on climate change adaptation and mitigation of food production systems for enhanced nutrition e.g. using participant quota of at least 50%; using female trainers, etc.)	50 % of women amongst those trained on climate change adaptation and mitigation of food production systems for enhanced nutrition Target 223 600 females	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	123

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generated and shared ⁷					
	increased participation of women in exchange visits on the adaptation of food systems to climate change and on the carbon sequestration potential of improved farming practices (e.g. using participant quota of at least 50%; organising women-only exchange visits and planning visits to successful female producers)	30 % of women and youth amongst those participating in exchange visits Target 120 women and youth	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	141
	Capitalization of best practices and knowledge on climate change adaptation and mitigation measures including special attention to gender dimensions	% of knowledge on climate change adaptation and mitigation measures including special attention to gender dimensions	Quarterly, Half yearly, annual	Monitoring and evaluation manager and Gender and social Inclusion officer	198
	Strengthen awareness of GBV and SEAH issues among staff and beneficiaries by developing orientation, training and continuous learning activities.	# of sessions held # of team members trained including at regional level	Annually	Monitoring and evaluation manager and Gender and social Inclusion officer	30

⁷ In terms of capitalizing on good practices, in practice, it is the women who keep the farm monitoring notebooks and the follow-up of the details in the management and conduct of the farm. It is rather relevant and effective to encourage their participation in the dissemination of resilient agricultural practices. Also, the assessment of the effects of climate change is rather quickly perceived at the level of daily management within the household (case of the management of food stocks, management of income attributed to agricultural activities, etc.). Women are always the vectors of knowledge sharing on climate change.

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	Develop an information/awareness protocol with a specialized entity in order to anticipate and, if necessary, deal with related incidents.	# of incidents reported and addressed	Annually	Monitoring and evaluation manager and Gender and social Inclusion officer	10
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The resources for the GAP are included in the overall project costs particularly under the capacity building activities and investment activities.
